ENS 301 Environmental Pollution and Management (1+1)



Theory Notes

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ENS 301 Environmental Pollution and Management 1+1

Scope:

It deals with the scientific study of environmental system (air, water, soil land), the inherent or induced changes on organisms and the environmental damages incurred as a result of human interaction with the environment.

Objectives:

- Imparting basic knowledge about the environment and its allied problems.
- Developing an attitude of concern for the environment.
- Motivating the students to participate in environment protection and environment
- improvement.
- Acquiring skills to help the concerned individuals in identifying and solving environmental problems and striving to attain harmony with nature.

Theory

Unit-I

Pollution in Environment-Introduction-Pollution- Pollutants – Contaminants – Source and types of pollution inSoil-Water-Air-Impact on environment-Pollution Status in India

Unit– II

Waste water Management: Waste water – Different types of waste water-pollutants and contaminants-Impact of waste water on ecosystem –Eutrophication – Biomagnification – Water borne diseases –Wastewater treatment methods – Physical, chemical and Biological – General water treatments-Wastewater recycling – Constructed wetlands-Reed bed system - Legislation and standards

Unit-III

Management of polluted soils: Soil pollutants – Sources – Urban and Industrial – Heavy metal – Pesticides – PAH's and PCB's-E-Waste-Fate of pollutants in Soil - Management of soil pollution – Bio and phyto remediation of polluted soil

Unit-IV

Air Pollution and its Management: Air pollutants from industrial anddomestic sources – Fate of air pollutants-Air pollution indicators - Monitoring and Control measures – Role of plants in controlling air pollutants-Legislation and Air quality standards - – Noise Pollution – Sources, Effect and Control Measures-Indoor air pollution and control measures

Unit-V

Solid waste management: Solid waste –Sources – Sludge from Industry and farm waste-Characteristics – Environmental problems – Management of sludge and farm wastes – Disposal methods – Sanitary land fills – Incineration – Pyrolysis - Recycling –Energy recovery –Composting – Vermicomposting – Maturity indices assessment-Standards and Legislation

Unit-VI

Environmental standards, Regulation and EIA - Environmental standards-CPCB Normsfor discharging industrial effluents to public sewers- CDM and Carbon foot print-Environmental Impact Assessment:Stages of EIA -Monitoring and Auditing – Environmental clearance procedure in India

Lecture Schedule

1. Introduction-Pollution- Pollutants – Contaminants – Source and types of pollution inSoil-Water-Air-Impact on environment-Pollution Status in India
2. Waste water – Different types of waste water-pollutants and contaminants-Impact of waste water on ecosystem –Eutrophication – Biomagnification – Water borne diseases
3. Wastewater treatment methods – Physical, chemical and Biological – General water treatments-

4. Wastewater recycling – Constructed wetlands-Reed bed system -Legislation and standards
5. Soil pollutants – Sources – Urban and Industrial – Heavy metal – Pesticides – PAH's and PCB's-E-Waste
6. Fate of pollutants in Soil - Management of soil pollution – Bio and phyto remediation of polluted soil
7. Air pollutants from industrial anddomestic sources – Fate of air pollutants-Air pollution indicators – Air pollution episodes-Monitoring and Control measures–
8. Role of plants in controlling air pollutants- Legislation and Air quality standards,
9. Mid Semester Examination
10. Noise Pollution, Sources, Effect and Control Measures, Indoor air pollutants and control measures
11. Solid waste –Sources – Sludge from Industry and farm waste-Characteristics – Environmental problems
12. Management of solid waste, Disposal methods, Sanitary land fills, Incineration, Pyrolysis
13. Recycling –Energy recovery –Composting – Vermicomposting – Maturity indices assessment-Standards and Legislation
14. Environmental standards-CPCB Normsfor discharging industrial effluents to public sewers
15. Environment Impact Assessment,Introduction,Stages of EIA, -Monitoring and Auditing
16. CDM and Carbon foot print
17. Environmental clearance procedure in India

Practical Schedule

1. Sample collection and preservation from contaminated sites.
2. Waste water treatment by physical (column study with vermiculite and activated charcoal) and chemical (Alum treatment)
3. Waste water treatment through constructed wetland system and characterization
4. Estimation of Chlorides, Phosphates in waste water
5. Analysis of Nitrogen in industrial effluent and sludge
6. Collection of PAH's contaminated soils and analysis by GC-MS
7. Biosorption of heavymetal (Cr) by using Water hyacinth and analysis through AAS
8. Pesticide Residue analysis in contaminated water
9. Analysis of SPM in air, Methane and CO_2 in Municipal dumping site
10. Assessing the efficiency of plants to control Indoor air pollutants
11. Analysis of Organic carbon in Sludge and Organic manure
12. Composting and Vermicomposting of farm wastes
13. Energy recovery from wastes
14. Maturity indices of compost- C:N ratio and Phytotoxicity test
15. Maturity indices of compost: starch iodine test and sulphide test
16. Visit to water treatment plant
17. Final practical examination

Reference:

1. Tyler Miller and Scot Spoolman. 2009. Living in the Environment (Concepts, Connections, and Solutions). Brooks/cole, Cengage learning publication, Belmont, USA
2. P.D. Sharma, 2009, Ecology and Environment, Rastogi Publications, Meerat, India
3. E-Books: Chiras D.D., 2016. Environmental Science, Tenth Edition. Jones & Bartlett Learning, Burlington, MA. ISBN: 978-1-284-05705-8, 708 Pages

LECTURE 1

Introduction –Pollution-Pollutants-Contaminants-Source and Types of pollution in Soil-Water-Air-impact on environment-pollution status in India.

INTRODUCTION

There are two types of the cause of pollution, natural and man-made. Natural pollution occurs naturally and won't cause excessive harm to our lives due to its regeneration ability. While the man-made pollution is caused by human activities, and hard to get rid of. The backbones of man-made pollution are human population and technology. Pollution is a growing pain. Pollution is not a problem that came suddenly from the sky; it's our fault and has been a part of our life through many years. We must be wise in managing our resources, and take positive action towards preventing any forms of pollution to the environment. Make the world a better place to live.

POLLUTION

Pollution is the introduction of contaminants into an environment that causes instability, disorder, harm or discomfort to the ecosystem i.e. physical systems or living organisms. Pollution can take the form of chemical substances, or energy, such as noise, heat, or light energy. Pollutants, the elements of pollution, can be foreign substances or energies, or naturally occurring; when naturally occurring, they are considered contaminants when they exceed natural levels. Pollution is often classed as point source or nonpoint source pollution.

POLLUTANT

Pollutants include solid, liquid or gaseous substances present in greater than natural abundance produced due to human activity, which have a detrimental effect on our environment. The nature and concentration of a pollutant determines the severity of detrimental effects on human health.

From an ecological perspective pollutants can be classified as follows:

Degradable or non-persistent pollutants:Thesecan be rapidly broken down by natural processes.Eg: domestic sewage, discarded vegetables,etc.

Slowly degradable or persistent pollutants: Pollutants that remain in the environment for many years in an unchanged condition and take decades or longer to degrade. Eg: DDT and most plastics.

Non-degradable pollutants: These cannot be degraded by natural processes. Once they are released into the environment they are difficultto eradicate and continue to accumulate.

Eg: toxic elements like lead or mercury.

CONTAMINANTS

Biological, Chemical, physical or radiological substance (normally absent in the environment) which, in sufficient concentration, can adversely affect living organisms through air, water, soil or food.

Types Of Pollution

[1] Soil pollution
[2] Water pollution
[3] Air Pollution
[4] Land Pollution
[5] Noise Pollution
[6] Radioactive Pollution
[7] Thermal Pollution

SOIL POLLUTIONSOURCE AND TYPES OF POLLUTION IN SOIL

Soil pollution, involves the following mechanism:

a) Deposition of solid waste
b) Accumulation of non-biodegradable materials
c) Toxification of chemicals into poisons

d) Alteration of soil chemical composition (imbalance of chemical equilibrium to soil medium)
e) By as much, land pollution of this has amass globally, everyday threatening the very foundation and mechanical support of every matter on earth.

Statistically, it has been shown that:

a) Loss of 6 million hectares of land per year
b) Loss of 24 billion tons of topsoil per year
c) Loss of minimum 15 million acres prime agricultural land to overuse and mismanagement
d) Desertification of land results in the lost of 16 million per square miles of world's land surface
e) The causes for such devastation are generally due to 2 (two) forms of malpractices

Unhealthy soil management methods;

- Improper tillage of soil in which excessive tillage result in the deterioration of soil structure
- Non-maintenance of a proper supply of organic matter in the soil from the imbalance composition of the reserves of organic matter especially nitrogen, phosphorus and sulfur unplenished supply after cultivation of vegetation, living the soil prone to soil infertility, unable to stabilize the soil physicality which ultimately let to desertification
- Irregular maintenance of a proper nutrient supply of trace elements gives rise to the use of excessive synthetic fertilizers, which are non biodegradable and accumulate in the soil system which eventually destroys useful organisms such as bacteria, fungi and other organisms
- Improper maintenance of the correct soil acidity which ultimately disrupt the adaptation of various crops and native vegetation of different soils as the solubility of minerals present will be affected. In a more acidic soil, minerals tend to be more soluble and washed away during rainfall while alkaline soil, minerals are more insoluble which form complex minerals unable to be absorbed into the flora system physiological usage.
- **Improper irrigation practices;**
- Poorly drained soil result in salt deposits leading to high soil salinity that inhibit plant growth and may lead to crop failure
- Unirrigated land giving rise to stagnation of agriculture waste products which accumulates and increases land toxicity and also decreasing
- Irregular irrigation leads to decreasing moisturization of land for soil medium and replenishments of solvents for minerals

SOURCES AND METHODS

We can classify major sources that lead to land pollution to the following categories:

- ✓ Agriculture
- ✓ Mining and quarrying
- ✓ Sewage sludge
- ✓ Dredged spoils
- ✓ Household
- ✓ Demolitions and constructions
- ✓ Industrial

Soil Pollution impact on environment

- ➢ Causes cancers including leukaemia
- ➢ Lead in soil is especially hazardous for young children causing developmental damage to the brain

- Mercury can increase the risk of kidney damage; cyclodienes can lead to liver toxicity
- Causes neuromuscular blockage as well as depression of the central nervous system
- Also causes headaches, nausea, fatigue, eye irritation and skin rash
- Contact with contaminated soil may be direct (from using parks, schools etc) or indirect (by inhaling soil contaminants which have vaporized)
- Soil pollution may also result from secondary contamination of water supplies and from deposition of air contaminants (for example, via acid rain)
- Contamination of crops grown in polluted soil brings up problems with food security

Effects of Soil Pollution on Animals

- Can alter metabolism of microorganisms and arthropods in a given soil environment; this may destroy some layers of the primary food chain, and thus have a negative effect on predator animal species
- Small life forms may consume harmful chemicals which may then be passed up the food chain to larger animals; this may lead to increased mortality rates and even animal extinction

Soil Pollution Effects on Trees and Plants

- May alter plant metabolism and reduce crop yields
- Trees and plants may absorb soil contaminants and pass them up the food chain

Salination of Soil

Increase in salt concentration of soil affects soil productivity and degrades its quality. Inadequate drainage especially in flood ravaged and well irrigated areas leads toaccumulation of salt at the soil surface. During summer season salts from lower layers move up by capillary action and get aggravated on soil surface. In India intensive farming with poor drainage has increased soil salinity. Total saline land in our country-6 million hectares. In Punjab, 6000-8000 hectares. Nearly 1/6th of arid and semi-arid lands of the world have high salinity.

SOIL POLLUTANTS AND THEIR EFFECTS

1. FLUORIDES.

They combine with Mg_{2+} of chlorophyll and hence inhibit photosynthesis, cause leaf abscission and of fruit. Maize is the sensitive indicator of fluoride pollution. In human beings mottling of teeth (fluorosis) is an indication of fluorination. Bone fluorosis results in weak bones, boat shaped posture and knocking of knee.

2. NITROGEN FERTILIZATION (NITRATES + NITRITE)

Toxic concentration in leaves and fruits enters into food chain. In alimentary canal, activity of bacteria changes nitrates into nitries. The latter entres blood and combine with haemoglobin to form met6-haemoglobin so oxygen transport is reduced. It give rise to disease called methanaemoglobinaemia. In infants it cause cyanosis (blue babies due to bluish tint of skin). Nitrate poisoning is fatal unless methylene blue is injected (in infants) while in adult it produces breathlessness.

3. WEEDICIDES

They are usually metabolic inhibitors which stop photosynthesis and other metabolic activities killing the plant. Some causes death due to proliferation of phloem cells to block transport of organic food.

Water pollution

Water pollution is contamination of water by foreign matter that deteriorates the quality of the water. Water pollution covers pollutions in liquid forms like ocean pollution and river pollution. As the term applies, liquid pollution occurs in the oceans, lakes, streams, rivers, underground water and bays, in short liquid-containing areas. It involves the release of toxic substances, pathogenic germs, substances that require much oxygen to decompose, easy-soluble substances, radioactivity, etc. that becomes deposited upon the bottom and their accumulations will interfere with the condition of aquatic ecosystems. For example, the

eutrophication: lack of oxygen in a water body caused by excessive algae growths because of enrichment of pollutants.

Sources of pollution

- Petroleum Products
- Synthetic Agricultural Chemicals
- Heavy Metals
- Hazardous Wastes
- Excess Organic Matter
- Sediment
- Infectious Organisms
- Air Pollution
- Thermal Pollution

Water Pollution Effects on humans

Waterborne diseases caused by polluted drinking water:

- Typhoid
- Amoebiasis
- Giardiasis
- Ascariasis
- Hookworm
- Waterborne diseases caused by polluted beach water:
- Rashes, ear ache, pink eye
- Respiratory infections
- Hepatitis, encephalitis, gastroenteritis, diarrhoea, vomiting, and stomach aches
- Conditions related to water polluted by chemicals (such as pesticides, hydrocarbons, persistent organic pollutants, heavy metals etc):

Water Pollution effects

- Cancer, incl. prostate cancer and non-Hodgkin's lymphoma
- Hormonal problems that can disrupt reproductive and developmental processes
- Damage to the nervous system
- Liver and kidney damage
- Damage to the DNA
- Exposure to mercury (heavy metal):
- ***In the womb:*** may cause neurological problems including slower reflexes, learning deficits, delayed or incomplete mental development, autism and brain damage
- ***In adults:*** Parkinson's disease, multiple sclerosis, Alzheimer's disease, heart disease, and even death
- **Effects of Water Pollution on Animals**
- *Nutrient pollution* (nitrogen, phosphates etc) causes overgrowth of toxic algae eaten by other aquatic animals, and may cause death; nutrient pollution can also cause outbreaks of fish diseases
- **Oil Coated Duck**
- *Chemical contamination* can cause declines in frog biodiversity and tadpole mass
- *Oil pollution* (as part of chemical contamination) can negatively affect development of marine organisms, increase susceptibility to disease and affect
- reproductive processes; can also cause gastrointestinal irritation, liver and kidney damage, and damage to the nervous system
- *Mercury* in water can cause abnormal behavior, slower growth and development, reduced reproduction, and death
- *Persistent organic pollutants* (POPs) may cause declines, deformities and death of fish life

- Too much sodium chloride (ordinary salt) in water may kill animals (7)
- **Effect of Water Pollution on trees and plants**
- May disrupt photosynthesis in aquatic plants and thus affecting ecosystems that depend on these plants
- Terrestrial and aquatic plants may absorb pollutants from water (as their main nutrient source) and pass them up the food chain to consumer animals and humans
- Plants may be killed by too much sodium chloride (ordinary slat) in water
- Plants may be killed by mud from construction sites as well as bits of wood and leaves, clay and other similar materials
- Plants may be killed by herbicides in water; herbicides are chemicals which are most harmful to plants

AIR POLLUTION is indication of disturbances to the composition of compounds in the atmosphere, as it may be summarized as shown:

- Excess emission of gases/vapors into atmosphere.
- Saturation of chemical compounds/particulates.
- Rate of dissipation < (smaller than) rate of absorption through various cycles (i.e. carbon and nitrogen cycle).
- Emergence of new chemical reactions of reactive and non-biodegradable compounds.
- Global warming, acid rain, smog, ozone depletion are some effects of air pollution.

In relation to this, we may observe the cycle which involves in our daily lives: carbon and nitrogen cycle. These 2 cycles are the most important of all, regulating the composition of carbon and nitrogen of Earth.

Sources and Methods

We can classify major sources that lead to air pollution to the following categories:

- Motor Vehicle Exhaust
- Heat And Power Generation Facilities
- Industrial Processes
- Auto Manufacturing
- Fertilizers Plants
- Building Demolition
- Solid Waste Disposal
- Solvent Evaporation
- Volcanic Eruption
- Fuel Production
- Roadway Construction
- Electrical Components Manufacturing
- Extraction Of Metals
- Forest Fires
- Agriculture.

Air Pollution Effects on Humans

- Pollution causes not only physical disabilities but also psychological and behavioral disorders in people.
- Reduced lung functioning.
- Irritation of eyes, nose, mouth and throat.
- Asthma attacks.
- Respiratory symptoms such as coughing and wheezing.

Air Pollution

- Increased respiratory disease such as bronchitis
- Reduced energy levels
- Headaches and dizziness

- Disruption of endocrine, reproductive and immune systems
- Neurobehavioral disorders
- Cardiovascular problems
- Cancer
- Premature death.

Effects of Air Pollution on Animals

- Acid rain (formed in the air) destroys fish life in lakes and streams.
- Excessive ultraviolet radiation coming from the sun through the ozone layer in the upper atmosphere which is eroded by some air pollutants, may cause skin cancer in wildlife.
- Ozone in the lower atmosphere may damage lung tissues of animals.

Air Pollution Effects on Trees and Plants

- Acid rain can kill trees, destroy the leaves of plants, can infiltrate soil by making it unsuitable for purposes of nutrition and habitation.
- Ozone holes in the upper atmosphere can allow excessive ultraviolet radiation from the sun to enter the Earth causing damage to trees and plants.
- Ozone in the lower atmosphere can prevent plant respiration by blocking stomata (openings in leaves) and negatively affecting plants' photosynthesis rates which will stunt plant growth; ozone can also decay plant cells directly by entering stomata

AIR POLLUTANTS

Gaseous Air Pollutants

Main types of gaseous air pollutants:

- Sulfur dioxide (SO_2)
- Oxides of nitrogen ($NO_x = NO + NO_2$)
- Ozone (O_3)
- Sulfur dioxide and nitric oxide (NO) are the primary air pollutants, and ozone is a secondary pollutant (though there are negligible direct emissions of the gas itself).
- Nitrogen dioxide (NO2) is both a primary and secondary air pollutant.
- **Other important gaseous pollutants are:** ammonia, carbon monoxide, volatile organic compounds (VOCs) and persistent organic pollutants (POPs)

Sources of SO2

- Energy Production
- Electric power generation
- Petroleum refining
- Other combustion
- Commercial and residential use
- Combustion for industry use
- Production processes
- Extraction and distribution of fossil fuels
- Transport
- Road transport
- Other Transport (such as aviation, ships, trains).
- Fuel itself can produce some nitrogen (for example, oil and coal contain around 0.5 – 1.5% of nitrogen, and natural gas contains less than that , most of nitrogen oxides' production comes from the reaction of atmospheric nitrogen and oxygen within the combustion chamber.

Nitrogen Oxides

- The two main nitrogen oxides are nitric oxide (NO), or nitrogen monoxide, and nitrogen dioxide (NO_2) the sum of which is equal to NO_x.

- Nitric oxide (NO) is a colorless gas. Nitrogen dioxide (NO_2) is a gas of reddish-brown color with a distinct sharp, biting odor.

Sources

- Combustion of fuels always produces both NO_2 and NO. But almost 90% of the NOX combustion product is in the form of NO which is then oxidized to nitrogen dioxide (NO_2) in the air. Therefore, only a small percentage of NO_2 found in the atmosphere is directly emitted there in this form. The rest has been formed as a result of chemical reactions in the atmosphere itself.
- **Road transport** contributed nearly half of all nitrogen emissions, followed by contributions from electric power generating plants which only contributed around 20% of total nitrogen emissions.
- Road transport
- Other Transport

3. Energy Production
   Electric power generation
   Petroleum refining
   Other combustion
4. Combustion for industry use
5. Production processes
6. Extraction and distribution of fossil fuels.

Agriculture is by far the biggest source of ammonia emissions. Livestock farming and animal waste account for the biggest percentage of total ammonia emissions which are due to the decomposition of urea from large animal wastes and uric acid from poultry wastes.

Major sources:

- Livestock – contributes more than 50% of all emissions
- Fertilizer application
- Oceans
- Vegetation
- Biomass burning

Carbon monoxide (CO)

- The combustion of carbon-based fuels produces carbon dioxide (CO_2).
- But not all such combustion is complete, and this leads to the production of carbon monoxide (CO).
- Motor vehicles and industry are among the largest anthropogenic sources of carbon monoxide emissions.

Volatile Organic Compounds (VOCs)

Volatile organic compounds (VOCs) are defined as organic compounds which easily evaporate and enter the atmosphere. VOCs may include a wide range of organic air pollutants, from pure hydrocarbons to partially oxidized hydrocarbons to organic compounds containing
chlorine, sulphur or nitrogen.

Major sources:

- Painting (evaporation of solvents)
- Oil production (flaring and venting of gas)
- Oil refining (flaring and fugitive emissions)
- Distribution of oil or refinery products (evaporation from storage, displacement losses when venting tanks)
- Dry cleaning (final drying of clothes)
- Production of alcoholic drinks (breweries and distilleries)

- Arable farming (crop growing, silage manufacture, sludge spreading)

Natural Sources of Volatile Organic Compounds

Forests are the primary natural sources of VOC emissions. Tropical forests are estimated to produce about half of all global natural non-methane VOC emissions. Plants synthesize many organic molecules and release some VOCs (including a range of terpenes) into the atmosphere.

OZONE (O_3)

Ozone can be found in:

- The stratosphere, one of the upper layers of the atmosphere, where it occurs naturally,
- The troposphere, the lowest layer of the atmosphere, where it occurs both naturally and as a result of human-generated emissions.
- The natural stratospheric ozone is considered to be of beneficial nature – it keeps harmful excessive ultraviolet sunlight from reaching the surface of the Earth.
- Ozone which is formed in the troposphere as a result of anthropogenic emissions of primary pollutants, has negative effects on humans and the natural environment. And from this point of view it is an air pollutant.
- This human-caused ozone in the troposphere is a secondary pollutant because it is produced by the reaction of primary pollutants, nitrogen oxides and hydrocarbons [including VOCs], in the presence of sunlight.
- The tropospheric ozone is the main component of the photochemical smog.

Persistent Organic Pollutants (POPs)

- Persistent organic pollutants are compounds which are resistant to degradation and persistent in the environment, with half lives of years in the soil or sediment and days in the atmosphere.
- Such compounds may include dioxins, furans, polychlorinated biphenyls (PCBs) and organochlorine pesticides such as DDT.
- They enter the food chains via the process of biomagnifications accumulated in human and animal tissue, and are capable of long range transport through being attached to airborne particles.

Sources of Persistent Organic Pollutants:

Some POPs are used as pesticides. Others are used in industrial processes as well as in the production of goods such as solvents, polyvinyl chloride and medicines.

POLLUTION STATUS IN INDIA

Air pollution in India

The World health Organization (WHO) which rates only mega cities of the world has rated Delhi the fourth most polluted city ion the world. However compared to other cities in India, Delhi is not at the top of the list of polluted cities. Our country has several pollution hotspots. The recent release from the Central Pollution ControlBoard (CPCB), Parivesh, January 2003 states that Ahmedabad's air is most noxious flowed by Kanpur, Solapur and Lucknow with small particulate levels (PM10) 3-4 times the standard of 60 microgram per cubic meter (mg/m3).

The reporthas ranked 29 cities according to Respirable Particulate Matter (RSPM) levels recorded during the year 2000.. Nitrogen dioxide levels in most major cities are generally close to the acceptable annual standard of 60 mg/m3. However sharp increases have been noticed in a few cities with heavy vehicular traffic and density as in a few locations in Kolkata and Delhi indicating stronger impact of traffic. d. The Supreme Court's order of April 5, 2002 has directed the Central Government for an action plan for other pollutedcities. The Supreme Court also played a vital role protecting the Taj Mahal. Being exposed to sulphur dioxide and suspended particulate matter, the Taj had contracted 'marble cancer', a

fungal growth that corroded its surface giving it a yellowish tinge. The SPM deposits blackened it. Shri MC Mehta an environmental lawyer filed a publicinterest litigation in 1984 expressing concern over the havoc the polluting units in Agra were wreaking on the Taj Mahal. Twelve years later the Supreme Court ordered 292 industries in the vicinity to either adopt pollution control measures or shut down.

The state of India's rivers

. Urbanization, industrialization, excess withdrawal of water, agricultural run-off, improper agricultural practices and various religious and social practices all contribute to river pollution in India. Every single river in India be it the Ganga, Yamuna, Cauvery or the Krishna have their own share of problems due to pollution. Waters from the Ganga and the Yamuna are drawn for irrigation through the network of canals as soon as these rivers reachthe plains reducing the amount of water that flows downstream..

Sewage and municipaleffluents account for 75% of the pollution load in rivers while the remaining 25% is from industrial effluents and non-point pollution sources. In 1985, India launched the Ganga Action plan (GAP) the largest ever river clean-up operationin the country. The plan has been criticized for, overspending and slow progress. The GAP Phase II in 1991 included cleaning operations for thetributaries of the Ganga, ie; the Yamuna, Gomti and the Damodar. Thus the Yamuna Action Plan (YAP), Gomti Action Plan and the Damodar Actionplan were added. In 1995 the National River Conservation plan was launched.

A CASE STUDY OF GROUNDWATER POLLUTION ININDIA

An example of groundwater pollution caused by excessive extraction is that fluoride contamination. Fluorisis is not a localized problem. It has spread across 19 states and across a variety of ecological regions ranging from the Thar desert, the Gangetic plains and the Deccan plateau. Each of these regions are distinct in terms of rainfall, soil type, groundwater recharge regime, climatic conditions and hydrology.. According to a report of the Rajiv Gandhi National Drinking Water mission, the bedrock of the Indian peninsula consists of a number of fluoride bearing minerals. When the bedrock weathers the fluoride leaches into water and the soil.. This is related to the over extraction of groundwater which has resulted in the tapping of aquifers with high fluoride concentrations.

LECTURE 2

Wastewater-different types of wastewater-pollutants and contaminants-impact of wastewater on ecosystem-eutrophication-biomagnification-waterborn diseases

WASTEWATER

Wastewater is liquid waste discharged by domestic residences, commercial properties, industry, agriculture, which often contains some contaminants that result from the mixing of wastewater from different sources.Based on its origin wastewater can be classed as sanitary, commercial, industrial, agricultural or surface runoff. Term wastewater need to be separated from the term sewage, sewage is subset of wastewater that is contaminated with feces or urine though many people use term sewage referring to any waste water.

Wastewater mostly consists of pure water (more than 95%), and there are numerous processes that can be used to clean up waste waters depending on the type and extent of contamination. Treated wastewater can then be reused as the drinking water after it has been cleared of contaminants.The treatment of wastewater is not only important for our own health but also to keep our environment clean and healthy. Without the proper wastewater treatment many ecosystems would be severely damaged once the treated water gets recharged back into the environment.

DIFFERENT TYPES OF WASTEWATER

Domestic Wastewater

Although domestic wastewater originates from domestic household activities, it can also include water that is discharged from commercial and business buildings and institutions, along with ground water. Water that collects from a storm can also be present in domestic wastewater. The source of domestic wastewater typically consists of liquid discharge from sanitary facilities, bathing, laundry and cooking. This type of water can be treated due to its characteristics.

Industrial Wastewater

Industrial wastewater, which originates from the manufacturing process, can be more difficult to treat due to the examination that must take place at an industry-based level. Industrial sources of wastewater contain contaminants such as oils, pharmaceuticals, pesticides, silt, chemicals and other byproducts.

Composition

Wastewater, which is constantly changing, is classified as having set particles, or colloids, which are materials that do not settle readily, along with solids that are in a dissolved state. It also contains large numbers of microscopic organisms that are mostly bacteria, which are capable of consuming the organic components of fats, proteins and carbohydrates that force changes to the wastewater.

Collection

Efficient and effective treatment for wastewater requires that it pass through a network of pipes, pumps and pump stations within a specific amount of time. The collection of water needs to be done with a velocity of at least two feet per second to ensure that solids are not settling and clogging up the pipes, causing odors. Manholes at every 300-500 feet allow inspection and cleaning of the sewer. In low-land areas, a pump station is typically installed to lift the wastewater to a higher elevation to ensure a flow that works with gravity.

Different types of water pollutants can be classified in to following major categories:

(1) Organic pollutants,
(2) Pathogens,
(3) Nutrients and agriculture runoff,
(4) Suspended solids and sediments (organic and inorganic),
(5) Inorganic pollutants (salts and metals),
(6) Thermal Pollution

(7) Radioactive pollutants.

ORGANIC POLLUTANTS

Organic pollutants can be further divided into following categories:

1.Oxygen Demanding Wastes:

The wastewaters such as, domestic and municipal sewage, wastewater from food processing industries, canning industries, slaughter houses, paper and pulp mills, tanneries, breweries, distilleries, etc. have considerable concentration of biodegradable organic compounds either in suspended, colloidal or dissolved form. These wastes undergo degradation and decomposition by bacterial activity. The dissolved oxygen available in the water body will be consumed for aerobic oxidation of organic matter present in the wastewater. Hence, depletion of the DO will be a serious problem adversely affecting aquatic life, if the DO falls below 4.0 mg/L. This decrease of DO is an index of pollution.

2.Synthetic Organic Compounds

Synthetic organic compounds are also likely to enter the ecosystem through various manmade activities such as production of these compounds, spillage during transportation, and their uses in different applications. These include synthetic pesticides, synthetic detergents, food additives, pharmaceuticals, insecticides, paints, synthetic fibers, plastics, solvents and volatile organic compounds (VOCs). Most of these compounds are toxic and biorefractory organics i.e., they are resistant to microbial degradation. Even concentration of some of these in traces may make water unfit for different uses. The detergents can form foams and volatile substances may cause explosion in sewers. Polychlorinated biphenyls (PCBs) are used in the industries since 1930s which are complex mixtures of chlorobiphenyls. Being a fat soluble they move readily through the environment and within the tissues or cells. Once introduced into environment, these compounds are exceedingly persistent and their stability to chemical reagents is also high.

3.Oil

Oil is a natural product which results from the plant remains fossilized over millions of years, under marine conditions. It is a complex mixture of hydrocarbons and degradable under bacterial action, the biodegradation rate is different for different oils, tars being one of the slowest. Oil enters in to water through oil spills, leak from oil pipes, and wastewater from production and refineries. Being lighter than water it spreads over the surface of water, separating the contact of water with air, hence resulting in reduction of DO. This pollutant is also responsible for endangering water birds and coastal plants due to coating of oils and adversely affecting the normal activities. It also results in reduction of light transmission through surface waters, thereby reducing the photosynthetic activity of the aquatic plants. Oil includes polycyclic aromatic hydrocarbons (PAH), some of which are known to be carcinogenic.

Pathogens

The pathogenic microorganisms enter in to water body through sewage discharge as a major source or through the wastewater from industries like slaughterhouses. Viruses and bacteria can cause water borne diseases, such as cholera, typhoid, dysentery, polio and infectious hepatitis in human.

4.Nutrients

The agriculture run-off, wastewater from fertilizer industry and sewage contains substantial concentration of nutrients like nitrogen and phosphorous. These waters supply nutrients to the plants and may stimulate the growth of algae and other aquatic weeds in receiving waters. Thus, the value of the water body is degraded. In long run, water body reduces DO, leads to eutrophication and ends up as a dead pool of water. People swimming in eutrophic waters containing blue-green algae can have skin and eye irritation, gastroenteritis and vomiting.

High nitrogen levels in the water supply, causes a potential risk, especially to infants under six months. This is when the**methemoglobin** results in a decrease in the oxygen carrying capacity of the blood (blue baby disease) as nitrate ions in the blood readily oxidize ferrous ions in the hemoglobin. In freshwater systems, eutrophication is a process whereby water bodies receive excess inorganic nutrients, especially N and P, which stimulate excessive growth of plants and algae.

Eutrophication can happen naturally in the normal succession of some freshwater ecosystems. However, when the nutrient enrichment is due to the activities of humans, sometimes referred to as "cultural eutrophication", the rate of this natural process is greatly intensified. Two major nutrients, nitrogen (N) and phosphorus (P), occur in streams in various forms as ions or dissolved in solution. Aquatic plants convert dissolved inorganic forms of nitrogen (nitrate, nitrite, and ammonium) and phosphorus (orthophosphate) into organic or particulate forms for use in higher trophic production. The main effects caused by eutrophication can be summarized as follows:

1. Species diversity decreases and the dominant biota changes
2. Plant and animal biomass increase
3. Turbidity increases
4. Rate of sedimentation increases, shortening the lifespan of the lake, and
5. Anoxic conditions may develop.

5.Suspended Solids And Sediments

These comprise of silt, sand and minerals eroded from land. These appear in the water through the surface runoff during rainy season and through municipal sewers. This can lead to the siltation, reduces storage capacities of reservoirs. Presence of suspended solids can block the sunlight penetration in the water, which is required for the photosynthesis by bottom vegetation.

Deposition of the solids in the quiescent stretches of the stream or ocean bottom can impair the normal aquatic life and affect the diversity of the aquatic ecosystem. If the deposited solids are organic in nature, they will undergo decomposition leading to development of anaerobic conditions. Finer suspended solids such as silt and coal dust may injure the gills of fishes and cause asphyxiation.

6.Inorganic Pollutants

Apart from the organic matter discharged in the water body through sewage and industrial wastes, high concentration of heavy metals and other inorganic pollutants contaminate the water. These compounds are non-biodegradable and persist in the environment. These pollutants include mineral acids, inorganic salts, trace elements, metals, metals compounds, complexes of metals with organic compounds, cyanides, sulphates, etc. ¬

The accumulation of heavy metals may have adverse effect on aquatic flora and fauna and may constitute a public health problem where contaminated organisms are used for food.Algal growth due to nitrogen and phosphorous compounds can be observed. ¬ Metals in high concentration can be toxic to biota e.g. Hg, Cu, Cd, Pb, As, and Se. Copper greater than 0.1 mg/L is toxic to microbes.

7.Thermal Pollution

Considerable thermal pollution results due to discharge of hot water from thermal power plants, nuclear power plants, and industries where water is used as coolant. As a result of hot water discharge, the temperature of water body increases. Rise in temperature reduces the DO content of the water, affecting adversely the aquatic life. This alters the spectrum of organisms, which can adopt to live at that temperature and DO level. When organic matter is also present, the bacterial action increases due to rise in temperature; hence, resulting in rapid decrease of DO. The discharge of hot water leads to the thermal stratification in the water body, where hot water will remain on the top.

8.Radioactive Pollutants

Radioactive materials originate from the following: ¬ Mining and processing of ores, ¬ Use in research, agriculture, medical and industrial activities, such as I131, P32, Co60, Ca45, S35, C14, etc. ¬ Radioactive discharge from nuclear power plants and nuclear reactors, e.g., Sr90, Cesium Cs137, Plutonium Pu248, Uranium-238, Uranium-235, ¬ Uses and testing of nuclear weapons. These isotopes are toxic to the life forms; they accumulate in the bones, teeth and can cause serious disorders. The safe concentration for lifetime consumption is 1 x 10-7 microcuries per ml. The summary of various pollutants and their adverse effect on the environment is presented in Table 10.1. The major impacts have been described, however there are additional adverse effects of release of these pollutants in the environment.

IMPACT OF WASTEWATER ON ECOSYSTEM

Effect of Water Pollution:

i. Disorders:

Some pollutants like sodium can cause cardiovascular diseases, while mercury and lead cause nervous disorders.

ii. Toxic Substances:

DDT is toxic material which can cause chromosomal changes. Some of these substances like pesticides, methyl mercury etc., move into the bodies of organisms from the medium in which these organisms live. These substances tend to accumulate in the organism's body from the medium food. This process is called bioaccumulation or bio concentration. The concentration of these toxic substances builds up at successive levels of food chain. This process is called bio magnifications.

iii. Water Pollution:

Fluoride pollution causes defects in teeth and bones, a disease called fluorosis while arsenic can cause significant damage to the liver and the nervous system. In addition to all these, organic compounds present in the polluted water facilitate the growth of algae and other weeds, which in turn use more oxygen dissolved in the water. This reduces the amount of oxygen dissolved in the water and the consequent shortage of oxygen for other aquatic life.

iv. Asbestos:

This pollutant is a serious health hazard and carcinogenic. Asbestos fibers can be inhaled and cause illnesses such as asbestosis, mesothelioma, lung cancer, intestinal cancer and liver cancer.

v. Mercury:

This is a metallic element and can cause health and environmental problems. It is a non-biodegradable substance so is hard to clean up once the environment is contaminated. Mercury is also harmful to animal health as it can cause illness through mercury poisoning.

vi. Phosphates:

The increased use of fertilizers means that phosphates are more often being washed from the soil and into rivers and lakes. This can cause eutrophication, which can be very problematic to marine environments.

vii. Oils:

Oil does not dissolve in water; instead it forms a thick layer on the water surface. This can stop marine plants receiving enough light for photosynthesis. It is also harmful for fish and marine birds.

viii. Petrochemicals:

This is formed from gas or petrol and can be toxic to marine life.

ix. Organic matter

which reaches water bodies is decomposed by micro-organisms present in water. For this degradation, oxygen dissolved in water is consumed. Dissolved oxygen (DO) is the

amount of oxygen dissolved in a given quantity of water at a particular temperature and atmospheric pressure.

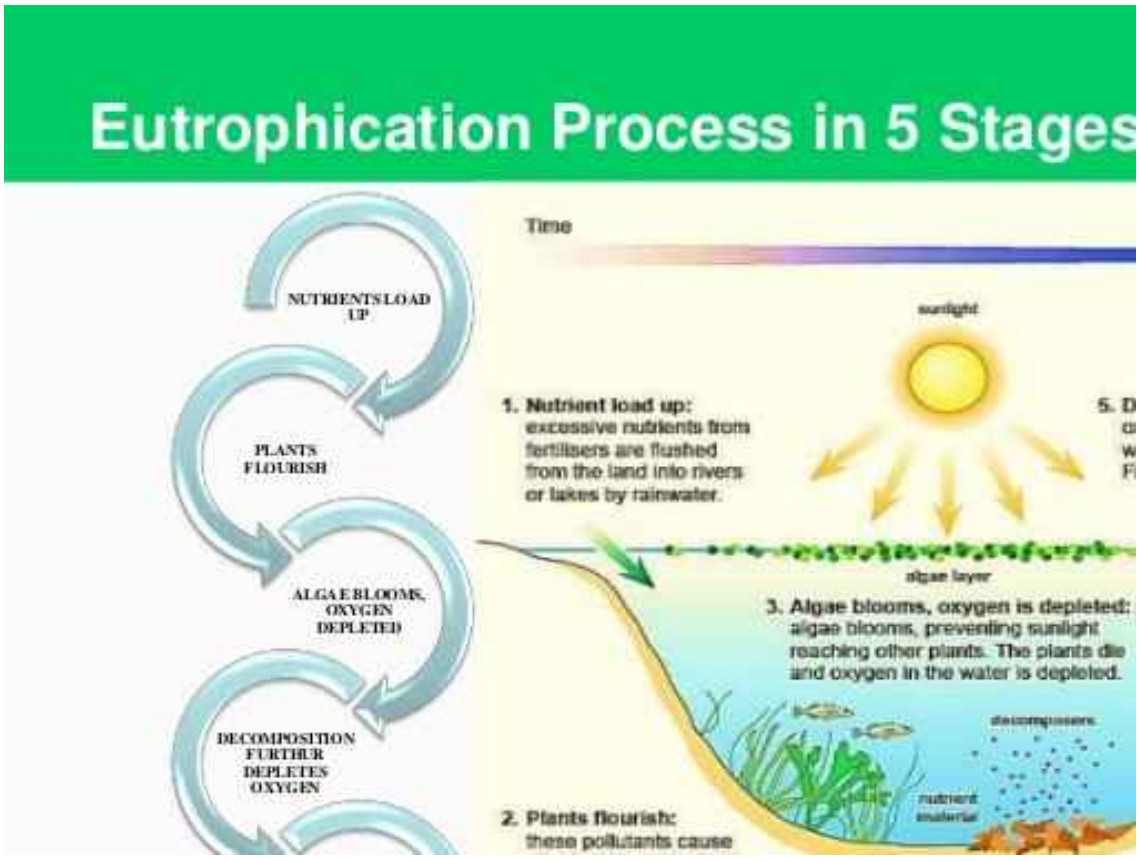
Amount of dissolved oxygen depends on aeration, photosynthetic activity in water, respiration of animals and plants and ambient temperature.The saturation value of DO varies from 8-15 mg/L. For active fish species (trout and Salmon) 5-8 mg/L of DO is required whereas less desirable species like carp can survive at 3.0 mg/L of DO.Lower DO may be harmful to animals especially fish population. Oxygen depletion (deoxygenating) helps in release of phosphates from bottom sediments and causes eutrophication.

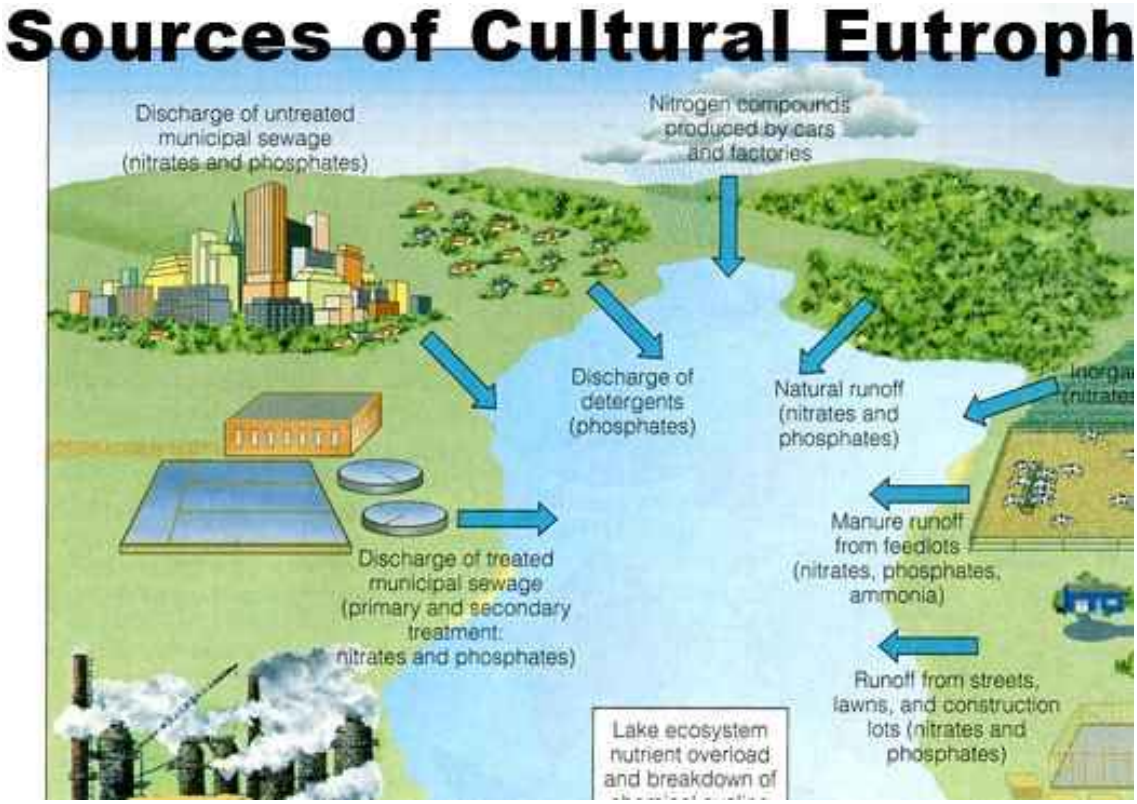
x. Lead in water may be released from water pipes as lead is used in plumbing. Lead poisoning affects kidneys reproductive system, liver, brain and central nervous system. It also causes anaemia and mental retardation in children.

xi. Nitrate ions present in the water are harmful to human health. From nitrogen fertilizers, nitrate ions seep into water bodies from where these may bioaccumulate in the bodies of the consumers. In the stomach nitrate is reduced to nitrite and is responsible for blue baby syndrome and stomach cancer.

EUTROBICATION

All living things need specific nutrients to survive. Usually, nature does a pretty good job of providing just the right amount of nutrients, because too many or too few can cause problems. This is especially true in aquatic ecosystems because they are so dynamic. When too few nutrients are present, the water is **oligotrophic**. It makes sense that when there is not enough nutrition available for the variety of organisms living in an aquatic environment, serious problems will arise.





However, problems can also arise when the aquatic system has an overabundance of nutrients. When this happens we get **eutrophication**. A **eutrophic** stream, river or lake occurs when too many nutrients, like nitrogen and phosphorous, are present, usually as a result of runoff from the surrounding land. Algae, plankton and other microorganisms love these types of nutrients, and when they are plentiful, these aquatic organisms can take over. When a lake, river or other aquatic system becomes eutrophic, it can have serious negative effects on other organisms like fish, birds and even people. But first, let's look at what causes eutrophication.

CAUSES

Eutrophication is most often the result of human activity. Farms, golf courses, lawns and other fields tend to be heavily fertilized by people. These fertilizers are the perfect type of nutrients to feed hungry algae and plankton, and when it rains, these fertilizers run off into lakes, streams, rivers and oceans. Concentrated animal feeding operations (CAFOs) are also a major source of polluting nutrients.

Eutrophication can also come from natural events. If a stream, river or lake floods, it may wash away any excess nutrients off the land and into the water. However, eutrophication is less likely to occur in areas that are not surrounded by fertilized lands.

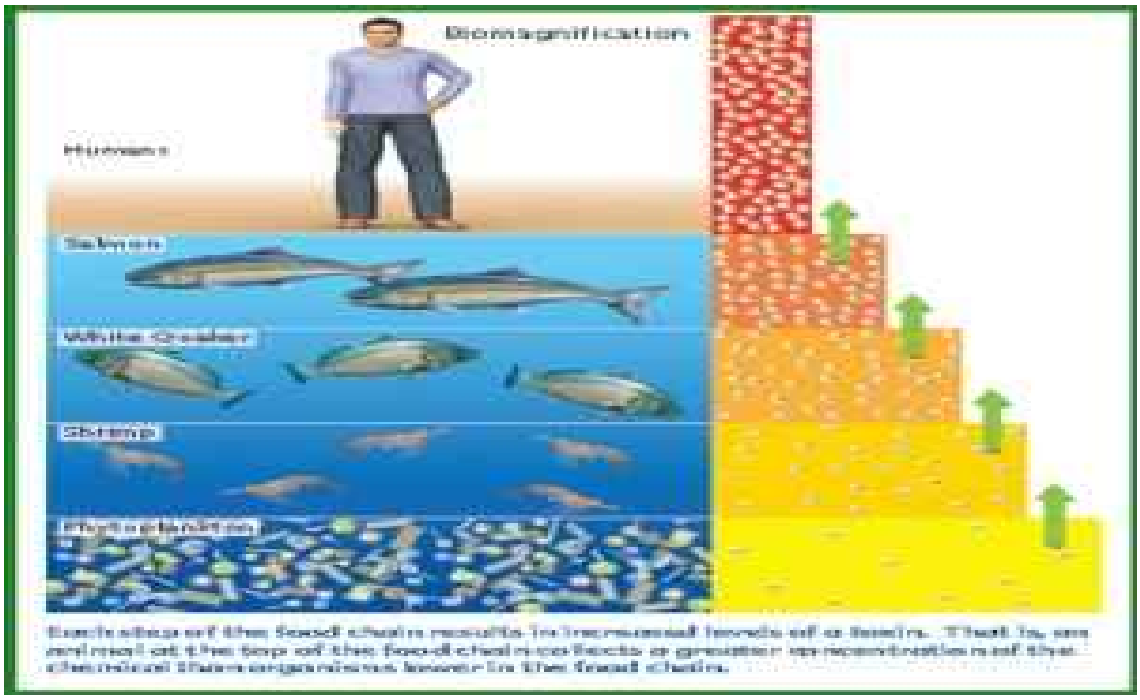
EFFECTS

Eutrophication can have serious, long-term effects. The most notable effect of eutrophication is **algal blooms**. When a bloom occurs, the stream, river, lake or ocean becomes covered with algae, which is usually bright green. In addition to looking pretty ugly, it also blocks light from reaching the water. This prevents the aquatic plants from photosynthesizing, a process which provides oxygen in the water to animals that need it, like fish and crabs. If an algal bloom is so bad that it causes wide-spread death in the water, the organisms that die will all sink to the bottom and start to decompose. The microbes that break

down these dead organisms use oxygen to do their work. So, in addition to the lack of oxygen from photosynthesis, there is also now a lack of oxygen from the decomposition of dead organisms.

BIOMAGNIFICATION

Biomagnification is a cumulative increase in the concentrations of a persistent substance (e.g. pesticides, metals, etc.) as it moves up the food chain. This occurs when agricultural, industrial, or human waste pours into the ocean directly or via rivers, sewage, etc. Some of the most dangerous of these toxins settle in the sediment of the sea floor and are consumed by bottom feeders.



With each step up the food chain, there is an increase in the concentration of these chemicals. This affects humans because a large percentage of the fish we consume are higher on the food chain and therefore likely to carry a high concentration of potentially harmful chemicals.The contaminants include heavy metals namely mercury, arsenic, pesticides such as DDT, and polychlorinated biphenyls (PCBs) compounds, which are then taken up by organisms because of the food they consume or the intoxication of their environment.

PROCESS OF BIOMAGNIFICATION

Release of toxic chemicals and pollutants into the environment

The entire process starts with the release of toxic chemicals and pollutants into the environment, which eventually work their way into soils, rivers or lakes, and the seas. The concentration of these toxic chemicals and pollutants seem to be very low when released in different environments. They are very minute even in terms of weight.

Phytoplankton

Phytoplankton are small plants that float in the seas that normally absorb these toxins. Once absorbed, the toxins stay in their tissues without being excreted or broken down. With time, the toxins accumulate to high concentrations (up to 200 parts a trillion), which represents a toxin accumulation increase by about four-fold.

Zooplankton

Zooplankton refers to small marine animals that float in the seas. They consume the phytoplankton and thus take in the toxins, which stay locked in the organism's tissue without being excreted or broken down. Over time, the toxin concentration increases up to two parts per billion, which represents about a ten-fold increase in the previous concentration.

Small fish consume the zooplankton

Whenever the small fish feed on the zooplanktons, they take up the toxins, which get absorbed in their fatty tissues. As a result, accumulation occurs and the concentrations build up to about 20 parts per billion, which is another ten-fold increase.

Large fish graze on the smaller fish

Again, when the large fish graze on the smaller fish for food, they consume the toxins that accumulate in their fatty tissues. The concentrations become higher up to ranges of 80 to 100 parts per billion. This is about four to five-fold increase in the toxic levels.

The top food chain organisms consume the fish

The organisms at the top of the food chain including the marine mammals such as dolphin, sea birds, and humans gradually build up the toxins in their tissues such as their liver when they consume the large fish. The levels of concentrations here increase to the highest ranges of 10,000 to 15,000 parts per billion. The results impact the animal's fertility and make them vulnerable to diseases as they interfere with the normal functioning of vital organs.

EFFECTS OF BIOMAGNIFICATION

Human Health

Humans become more susceptible to cancers, liver and kidney failure, respiratory disorders, birth defects in pregnant women, brain damage, and heart diseases are a result of mercury, cadmium, lead, cobalt, chromium and other chemical poisonings. For instance, diseases like hepatitis and cancer have been attributed to consuming seafood that has been poisoned by mercury and polycyclic aromatic hydrocarbons (PAHs).

Marine Creatures

The various toxic elements get accumulated in the vital organs of the various aquatic creatures, thus affecting their reproduction and development. Selenium and heavy metals such as mercury also affect the reproduction of aquatic creatures such as fish as it destroys their reproductive organs. Besides, PCBs (polychlorinated biphenyls) also biomagnify and impair reproduction and are considerably high in aquatic systems.

Coral Reefs

The coral reefs are destroyed by cyanide, which is used in leaching gold and in fishing. The reefs provide for spawning, feeding, and dwelling grounds for numerous sea creatures. When destroyed, the survival of aquatic creatures is highly compromised.

Food Chain

Many sea creatures depend on the natural food chain for survival. When chemicals and other toxins are carried into the soils, rivers, lakes or oceans and taken up by various organisms, it disrupts the interconnected relationships within the food chain. The creatures or plants intoxicated with substances such as mercury, copper, chromium, selenium and cobalt may also be consumed by humans and top animals in the food chain leading to susceptibility to diseases, reproductive disorders, and even deaths. Biomagnification affects humans THE MOST because we are at the top of the food chain, thus the accumulated toxins and chemicals are in their largest concentration amongst us.

WATER BORN DISEASE IN INDIA

Cholera is a water-related disease, and is diarrhoeal in nature.

It can kill in hours if left unattended.

Cholera strikes when one ingests water that is infested with the Vibrio Cholerae bacterium.

2) Diarrhoea

Diarrhoeal infection is spread through food and drinking water that has been contaminated.

A diarrhoeal attack can last up to 2 weeks and leave the person completely dehydrated.

3) Malaria

Malarial fever is spread by the Plasmodium parasite mosquito that breeds in water bodies like lakes, paddy fish and stagnant water.

Malaria can kill a child who does not have the immunity against the disease.

4) Typhoid

Fluctuating high fever, exhaustion, sleepiness, diarrhoea etc are signs of typhoid.

The infection spreads through contaminated food and water or through close contact with an infected person.

5) Filariasis

Filariasis is a parasitic disease and affects people who live near unsanitary water bodies or sewages. Filariasis is spread by mosquitoes that breeds in fresh and stagnant water bodies and is the host of the filarial nematode worm. This worm affects humans and leads to elephantitis.

Lecture 3

Wastewater treatment methods – Physical, chemical and Biological – General water treatments-

Wastewater treatment consists of applying known technology to improve or upgrade the quality of a wastewater. Usually wastewater treatment will involve collecting the wastewater in a central, segregated location (the Wastewater Treatment Plant) and subjecting the wastewater to various treatment processes. Most often, since large volumes of wastewater are involved, treatment processes are carried out on continuously flowing wastewaters (continuous flow or "open" systems) rather than as "batch" or a series of periodic treatment processes in which treatment is carried out on parcels or "batches" of wastewaters. While most wastewater treatment processes are continuous flow, certain operations, such as vacuum filtration, involving as it does, storage of sludge, the addition of chemicals, filtration and removal or disposal of the treated sludge, are routinely handled as periodic batch operations.

Wastewater treatment, however, can also be organized or categorized by the nature of the treatment process operation being used; for example, physical, chemical or biological.

Physical, Chemical and Biological Wastewater Treatment Methods

Physical

Sedimentation (Clarification) : This water treatment process used to settle out suspended solids in water under the influence of gravity.

Screening: This water treatment process used to trap and remove the floating matter such as paper, wood etc. by using automated mechanically raked bar screens.

Aeration: This Water aeration process is used for water bodies that suffer from anoxic conditions, usually caused by adjacent human activities such as sewage discharges, agricultural run-off, or over-baiting a fishing lake. Aeration can be achieved through the infusion of air into the bottom of the lake, lagoon or pond or by surface agitation from a fountain or spray-like device to allow for oxygen exchange at the surface and the release of noxious gasses such as carbon dioxide, methane or hydrogen sulfide.

Filtration: This water treatment process is used at the end to remove remaining suspended particles and unsettled floc.

Chemical

Chlorination: This is a method of treatment which has been employed for many purposes to disinfect or destruct the pathogenic organisms and to prevent water from decomposition.

Ozonation: In this process O_3 is used, Which is an unstable molecule gives up one atom of Oxygen providing a powerful oxidizing agent which is toxic to most waterborne organisms.

Neutralization: The neutralization process can be performed for the industrial wastewater containing acidic substances (acidic) or base (alkaline) that need to be neutralized before discharge of water into the waste water treatment process.

Coagulation: Coagulation is one of the most important physio-chemical reactions used in water treatment. The precipitation of ions (heavy metals) and colloids (organic and inorganic) are mostly held in solution by electrical charges. By the addition of ions with opposite charges, these colloids can be destabilized; coagulation can be achieved by chemical or electrical methods. The coagulant is added in the form of suitable chemical substances. Alum [$Al_2(SO_4)_3 \cdot 18H_2O$] is such a chemical substance, which has been widely used for ages for wastewater treatment.

Adsorption: This is the physical adhesion of chemicals on to the surface of the solid. The effectiveness of the adsorbent is directly related to the amount of surface area available to attract the particles of contaminant. The most commonly used adsorbent is a very porous matrix of granular activated carbon.

Ion Exchange: This technique has been used extensively to remove hardness, and iron and manganese salts in drinking water supplies. It has also been used selectively to remove specific impurities and to recover valuable trace metals like chromium,nickel, copper, lead and cadmium from industrial waste discharges. The process takes advantage of the ability of certain natural and synthetic materials to exchange one of their ions.

Biological

Aerobic: Under aerobic (O_2) conditions bacteria rapidly consume organic matter and convert into CO_2 .The following treatment methods come under this method.

- Activated Sludge Treatment Methods
- Trickling Filtration
- Oxidation Ponds
- Lagoons
- Aerobic Digestion

Anaerobic: This is a bacterial process of domestic sewage in septic tanks which normally retain the sewage from one day to two days reducing B.O.D by about 35 to 40 percentage. The following treatment methods come under this method.

- Anaerobic Digestion
- Septic Tanks
- Lagoons

Physical treatment methods

Include processes where no gross chemical or biological changes are carried out and strictly physical phenomena are used to improve or treat the wastewater.

Examples would be coarse screening to remove larger entrained objects and sedimentation (or clarification). In the process of sedimentation, physical phenomena relating to the settling of solids by gravity are allowed to operate. Usually this consists of simply holding a wastewater for a short period of time in a tank under quiescent conditions, allowing the heavier solids to settle, and removing the "clarified" effluent. Sedimentation for solids separation is a very common process operation and is routinely employed at the beginning and end of wastewater treatment operations. While sedimentation is one of the most common physical treatment processes that is used to achieve treatment, another physical treatment process consists of aeration -- that is, physically adding air, usually to provide oxygen to the wastewater. Still other physical phenomena used in treatment consist of filtration. Here wastewater is passed through a filter medium to separate solids. An example would be the use of sand filters to further remove entrained solids from a treated wastewater. Certain phenomena will occur during the sedimentation process and can be advantageously used to further improve water quality. Permitting greases or oils, for example, to float to the surface and skimming or physically removing them from the wastewaters is often carried out as part of the overall treatment process.

In certain industrial wastewater treatment processes strong or undesirable wastes are sometimes produced over short periods of time. Since such "slugs" or periodic inputs of such wastes would damage a biological treatment process, these wastes are sometimes held, mixed with other wastewaters, and gradually released, thus eliminating "shocks" to the treatment plant. This is call equalization. Another type of "equalization" can be used to even out wide variations in flow rates. For example, the wet well of a pump station can receive widely varying amounts of wastewater and, in turn, pump the wastes onward at more uniform rates.

Chemical treatment Methods

These methods consist of using some chemical reaction or reactions to improve the water quality. Probably the most commonly used chemical process is chlorination. Chlorine, a strong oxidizing chemical, is used to kill bacteria and to slow down the rate of decomposition of the wastewater. Bacterial kill is achieved when vital biological processes

are affected by the chlorine. Another strong oxidizing agent that has also been used as an oxidizing disinfectant is ozone.

A chemical process commonly used in many industrial wastewater treatment operations is neutralization. Neutralization consists of the addition of acid or base to adjust pH levels back to neutrality. Since lime is a base it is sometimes used in the neutralization of acid wastes.

Coagulation consists of the addition of a chemical that, through a chemical reaction, forms an insoluble end product that serves to remove substances from the wastewater. Polyvalent metals are commonly used as coagulating chemicals in wastewater treatment and typical coagulants would include lime (that can also be used in neutralization), certain iron containing compounds (such as ferric chloride or ferric sulfate) and alum (aluminum sulfate).

Certain processes may actually be physical and chemical in nature. The use of activated carbon to "adsorb" or remove organics, for example, involves both chemical and physical processes. Processes such as ion exchange, which involves exchanging certain ions for others, are not used to any great extent in wastewater treatment.

Biological treatment methods

These methods use microorganisms, mostly bacteria, in the biochemical decomposition of wastewaters to stable end products. More microorganisms, or sludges, are formed and a portion of the waste is converted to carbon dioxide, water and other end products. Generally, biological treatment methods can be divided into aerobic and anaerobic methods, based on availability of dissolved oxygen. The purpose of wastewater treatment is generally to remove from the wastewater enough solids to permit the remainder to be discharged to a receiving water without interfering with its best or proper use. The solids which are removed are primarily organic but may also include inorganic solids. Treatment must also be provided for the solids and liquids which are removed as sludge. Finally, treatment to control odors, to retard biological activity, or destroy pathogenic organisms may also be needed.

While the devices used in wastewater treatment are numerous and will probably combine physical, chemical and biological methods, they may all be generally grouped under six methods:

1.Preliminary Treatment
2. Primary Treatment
3. Secondary Treatment
4. Disinfection
5.Sludge Treatment
6.Tertiary Treatment

Degrees of treatment are sometimes indicated by use of the terms primary, secondary and tertiary treatment. Tertiary treatment, properly, would be any treatment added onto or following secondary treatment.

Preliminary Treatment

At most plants preliminary treatment is used to protect pumping equipment and facilitate subsequent treatment processes. Preliminary devices are designed to remove or cut up the larger suspended and floating solids, to remove the heavy inorganic solids, and to remove excessive amounts of oils or greases.

To effect the objectives of preliminary treatment, the following devices are commonly used:

- Screens -- rack, bar or fine
- Comminuting devices -- grinders, cutters, shredders
- Grit chambers
- Pre-aeration tanks

In addition to the above, chlorination may be used in preliminary treatment. Since chlorination may be used at all stages in treatment, it is considered to be a method by itself. Preliminary treatment devices require careful design and operation.

Primary Treatment

In this treatment, most of the settleable solids are separated or removed from the wastewater by the physical process of sedimentation. When certain chemicals are used with primary sedimentation tanks, some of the colloidal solids are also removed. Biological activity of the wastewater in primary treatment is of negligible importance.

The purpose of primary treatment is to reduce the velocity of the wastewater sufficiently to permit solids to settle and floatable material to surface. Therefore, primary devices may consist of settling tanks, clarifiers or sedimentation tanks. Because of variations in design, operation, and application, settling tanks can be divided into four general groups:

1. Septic tanks
2. Two story tanks -- Imhoff and several proprietary or patented units
3. Plain sedimentation tank with mechanical sludge removal
4. Upward flow clarifiers with mechanical sludge removal

When chemicals are used, other auxiliary units are employed. These are:

1. Chemical feed units
2. Mixing devices
3. Flocculators

The results obtained by primary treatment, together with anaerobic sludge digestion as described later, are such that they can be compared with the zone of degradation in stream self-purification. The use of chlorine with primary treatment is discussed under the section on Preliminary Treatment.

Secondary Treatment

Secondary treatment depends primarily upon aerobic organisms which biochemically decompose the organic solids to inorganic or stable organic solids. It is comparable to the zone of recovery in the self-purification of a stream.

The devices used in secondary treatment may be divided into four groups:

1. Trickling filters with secondary settling tanks
2. Activated sludge and modifications with final settling tanks
3. Intermittent sand filters
4. Stabilization ponds

The use of chlorine with secondary treatment is discussed under the section on Secondary Treatment

Chlorination

This is a method of treatment which has been employed for many purposes in all stages in wastewater treatment, and even prior to preliminary treatment. It involves the application of chlorine to the wastewater for the following purposes:

1. Disinfection or destruction of pathogenic organisms
2. Prevention of wastewater decomposition --

(a)odor control, and
(b) protection of plant structures

1. Aid in plant operation --

(a) Sedimentation,
(b)trickling filters,
(c) activated sludge bulking

2. Reduction or delay of biochemical oxygen demand (BOD)

While chlorination has been commonly used over the years, especially for disinfection, other methods to achieve disinfection as well as to achieve similar treatment

ends are also used. Among the most common is the use of ozone. In view of the toxicity of chlorine and chlorinated compounds for fish as well as other living forms, ozonation may be more commonly used in the future. This process will be more fully discussed in the section on disinfection.

Sludge Treatment

The solids removed from wastewater in both primary and secondary treatment units, together with the water removed with them, constitute wastewater sludge. It is generally necessary to subject sludge to some treatment to prepare or condition it for ultimate disposal. Such treatment has two objectives -- the removal of part or all of the water in the sludge to reduce its volume, and the decomposition of the organic solids to mineral solids or to relatively stable organic solids. This is accomplished by a combination of two or more of the following methods:

3. Thickening
4. Digestion with or without heat
5. Drying on sand bed -- open or covered
6. Conditioning with chemicals
7. Elutriation
8. Vacuum filtration
9. Heat drying
10. Incineration
11. Wet oxidation
12. Centrifuging

Tertiary and Advanced Wastewater Treatment

Tertiary treatment is the next wastewater treatment process after secondary treatment. This step removes stubborn contaminants that secondary treatment was not able to clean up. Wastewater effluent becomes even cleaner in this treatment process through the use of stronger and more advanced treatment systems. Tertiary treatment technologies can be extensions of conventional secondary biological treatment to further stabilize oxygen-demanding substances in the wastewater, or to remove nitrogen and phosphorus. Tertiary treatment may also involve physical-chemical separation techniques such as carbon adsorption,
flocculation/precipitation, membranes for advanced filtration, ion exchange, dechlorination and reverse osmosis.

LECTURE 4

Wastewater recycling –Constructed wetlands –Reed bed system-Legislation and standards

WATER RECYCLING

Water recycling is reusing treated wastewater for beneficial purposes such as agricultural and landscape irrigation, industrial processes, toilet flushing.

Another type of recycled water is "gray water". Gray water, or gray water, is reusable wastewater from residential, commercial and industrial bathroom sinks, bath tub shower drains, and clothes washing equipment drains. National Science Foundation (NSF) International has established a wastewater treatment task group on onsite residential and commercial gray water treatment systems. They have developed a draft new standard – *NSF 350 – Onsite Residential and Commercial Reuse Treatment Systems*.

RECYCLED WATER BENEFIT

Recycled water can satisfy most water demands, as long as it is adequately treated to ensure water quality appropriate for the use. EPA regulates many aspects of wastewater treatment and drinking water quality, and the majority of states in the US have established criteria or guidelines for the beneficial use of recycled water.

USES FOR RECYCLED WATER

- Agriculture
- Landscape
- Public parks
- Golf course irrigation
- Cooling water for power plants and oil refineries
- Processing water for mills, plants
- Toilet flushing
- Dust control,
- Construction activities
- Concrete mixing
- Artificial lakes

ENVIRONMENTAL BENEFITS OF WATER RECYCLING

In addition to providing a dependable, locally-controlled water supply, water recycling provides tremendous environmental benefits. By providing an additional source of water, water recycling can help us find ways to decrease the diversion of water from sensitive ecosystems. Other benefits include decreasing wastewater discharges and reducing and preventing pollution. Recycled water can also be used to create or enhance wetlands and riparian habitats.

WATER RECYCLING CAN DECREASE DIVERSION OF FRESHWATER FROM SENSITIVE ECOSYSTEMS

Plants, wildlife, and fish depend on sufficient water flows to their habitats to live and reproduce. The lack of adequate flow, as a result of diversion for agricultural, urban, and industrial purposes, can cause deterioration of water quality and ecosystem health. People who reuse water can supplement their demands by using a reliable source of recycled water, which can free considerable amounts of water for the environment and increase flows to vital ecosystems.

RECYCLED WATER MAY BE USED TO CREATE OR ENHANCE WETLANDS AND RIPARIAN (STREAM) HABITATS.

Wetlands provide many benefits, which include wildlife and wildfowl habitat, water quality improvement, flood diminishment, and fisheries breeding grounds. For streams that

have been impaired or dried from water diversion, water flow can be augmented with recycled water to sustain and improve the aquatic and wildlife habitat.

WATER RECYCLING CAN REDUCE AND PREVENT POLLUTION

When pollutant discharges to oceans, rivers, and other water bodies are curtailed, the pollutant loadings to these bodies are decreased. Moreover, in some cases, substances that can be pollutants when discharged to a body of water can be beneficially reused for irrigation. For example, recycled water may contain higher levels of nutrients, such as nitrogen, than potable water. Application of recycled water for agricultural and landscape irrigation can provide an additional source of nutrients and lessen the need to apply synthetic fertilizers.

RECYCLING WATER CAN SAVE ENERGY

As the demand for water grows, more water is extracted, treated, and transported sometimes over great distances which can require a lot of energy. If the local source of water is ground water, the level of ground water becomes lower as more water is removed and this increases the energy required to pump the water to the surface. Recycling water on site or nearby reduces the energy needed to move water longer distances or pump water from deep within an aquifer. Tailoring water quality to a specific water use also reduces the energy needed to treat water. The water quality required to flush a toilet is less stringent than the water quality needed for drinking water and requires less energy to achieve. Using recycled water that is of lower quality for uses that don't require high quality water saves energy and money by reducing treatment requirements.

WATER REUSE IN DIFFERENT INDUSTRIES

1.Water reuse in the paper industry

In the paper industry water can be recycled by means of membrane filtration and membrane bio reactors. (MBR)

2. Water reuse in the textile industry

For water recycling in the textile industry membrane filtration is a preferable option compared to other wastewater treatment techniques.

3. Water reuse in the poultry industry

By means of filtration and ozone disinfection, cooling water can be recycled.

4. Water reuse in the food and beverage industry

Water recycling using UV-disinfection and membrane filtration in the food and beverage industry.

5. Water disinfection and reuse in greenhouse horticulture

Water recycling and disinfection by means of heat treatment, sand filtration and ozone- and UV treatment.

6. Water treatment and reuse of oily wastewater

By means of membrane filtration and ozone/UV treatment, oil can be removed from wastewater

7. Water savings and reuse for cooling processes in which surface and ground water is used.

Cooling water savings by means of open recirculating cooling towers.

8. Water reuse for agricultural irrigation

The use of municipal wastewater effluent for agricultural applications.

WETLANDS

The term "wetlands" encompasses a broad range of wet environments, including marshes, bogs, swamps,wet meadows, tidal wetlands, floodplains, and ribbon (riparian) wetlands along stream channels.

WETLAND FUNCTIONS AND VALUES

Wetlands provide a number of functions and values. All wetlands provide all functions and values. constructed wetlands can provide: water quality improvement flood storage and the desynchronization of storm rainfall and surface runoff cycling of nutrients

and other materials habitat for fish- and wildlife passive recreation, such as bird watching and photography active recreation, such as hunting education and research aesthetics and landscape enhancemerit.'

CONSTRUCTED WETLANDS

A constructed wetland is a shallow basin filled with some sort of substrate, usually soil or gravel, and planted with vegetation tolerant of saturated conditions. Water is introduced at one end and flows over the surface or through the substrate, and is discharged at the other end through a weir or other structure which controls the depth of the water in the wetland.

COMPONENTS OF CONSTRUCTED WETLANDS

A constructed wetland consists of a properly-designed basin that contains water, a substrate, and, most commonly, vascular plants. These components can be manipulated in constructing a wetland. Other important components of wetlands, such as the communities of microbes and aquatic invertebrates, develop naturally.

W A T E R

Wetlands are likely to form where landforms direct surface water to shallow basins and where a relatively impermeable subsurface layer prevents the surface water from seeping into the ground. These conditions can be created to construct a wetland. A wetland can be built almost anywhere in the land-scape by shaping the land surface to collect surface water and by sealing the basin to retain the water.

Hydrology is the most important design factor in constructed wetlands because it links all of the functions in a wetland and because it is often the primary factor in the success or failure of a con-structed wetland. While the hydrology of con-structed wetlands is not greatly different than that of other surface and near-surface waters, it does differ in several important respects:

1. small changes in hydrology can have fairly signifi-cant effects on a wetland and its treatment effec-tiveness .because of the large surface area of the water and its shallow depth, a wetland system interacts strongly with the atmosphere through rainfall and evapotranspiration (the combined loss of water by evaporation from the water surface and loss through transpiration by plants)
2. The density of vegetation of a wetland strongly affects its hydrology, first, by obstructing flow paths as the water finds its sinuous way through the network of stems, leaves, roots, and rhizomes and, second, by blocking exposure to wind and sun.
3. They support many of the living organisms in wetlands.
4. Substrate permeability affects the movement of water through the wetland.
5. Many chemical and biological (especially microbial) transformations take place within the substrates.

   Substrates provide storage for many contaminants

   The accumulation of litter increases the amount of organic matter in the wetland.

Organic matter provides sites for material exchange and micro-bial attachment, and is a source of carbon, the energy source that drives some of the important biological reactions in wetlands. The physical and chemical characteristics of soils and other substrates are altered when they are flooded. In a saturated substrate, water replaces the atmospheric gases in the pore spaces and microbial metabolism consumes the available oxygen. Since oxygen is consumed more rapidly than it can be replaced by diffusion from the atmosphere, substrates become anoxic (without oxygen). This reducing environment is important in the removal of pollutants such as nitrogen and metals.

V E G ET A T I O N

Both vascular plants (the higher plants) and non-vascular plants (algae) are important in constructed wetlands. Photosynthesis by algae increases the dissolved oxygen content of the water which in turn affects nutrient and metal

Constructed wetlands attract waterfowl and wading birds, including mallards, green-winged teal, wood ducks, moorhens, green and great blue herons, and bitterns. Snipe, red-winged blackbirds, marsh wrens, bank swallows, redtailed hawks, and Northern harriers feed and/or nest in wetlands.

M I C R O O R G A N I S M S

A fundamental characteristic of wetlands is that their functions are largely regulated by microorganisms and their metabolism (Wetzel 1993). Microorganisms include bacteria, yeasts, fungi, protozoa, rind algae. The microbial biomass is a major sink for organic carbon and many nutrients.

Microbial activity:

Transforms a great number of organic and inorganic substances into innocuous or Some microbial transformations are aerobic (that is, they require free oxygen) while others are anaerobic (they take place in the absence of free oxygen). Many bacterial species are faculta-tive anaerobes, that is, they are capable of func-tioning under both aerobic and anaerobic condi-tions in response to changing environmental conditions.

Microbial populations adjust to changes in the water delivered to them. Populations of microbes can expand quickly when presented with suitable energy-containing materials. When environmental conditions are no longer suitable, many microorganisms become dormant and can remain dormant for years (Hilton 1993). The microbial community of a constructed wetland can be affected by toxic substances, such as pesticides and heavy metals, and care must be taken to prevent such chemicals from being introduced at damaging concentrations.

A N I M A L S

Constructed wetlands provide habitat for a rich diversity of invertebrates and vertebrates. Invertebrate animals, such as insects and worms. contribute to the treatment process by fragment-ing detritus and consuming organic matter. The larvae of many insects are aquatic and consume significant amounts of material during their larval stages, which may last for several years. Invertebrates also fill a number of ecological roles; for instance, dragonfly nymphs are impor-tant predators of mosquito larvae. Although invertebrates are the most important animals as far as water quality im-provement is concerned, constructed wetlands also attract a variety of amphibians, turtles.

WETLANDS IMPROVE WATER QUALITY

A wetland is a complex assemblage of water, substrate, plants (vascular and algae), litter (primarily fallen plant material), invertebrates (mostly insect larvae and worms). and an array of microorganisms (most importantly bacteria). The mechanisms that are available to improve water quality are therefore numerous and often interrelated.

These mechanisms include:

1.Settling of suspended particulate matter

2.Filtration and chemical precipitation through contact of the water with the substrate and litter

3.Chemical transformation

4.Adsorption and ion exchange on the surfaces of plants, substrate, sediment, and litter

5.Breakdown and transformation of pollutants by microorganisms and plants

6.Uptake and transformation of nutrients

ADVANTAGES OF CONSTRUCTED WETLANDS

Constructed wetlands are a cost-effective and technically feasible approach to treating waste-water and runoff for several reasons:

1. 1.wetlands can be less expensive to build than other treatment options
2. 2.Operation and maintenance expenses (energy and supplies) are low
3. 3.Operation and maintenance require only periodic, rather than continuous, on-site labor
4. wetlands are able to tolerate fluctuations in f l o w
5. 5.they facilitate water reuse and recycling.

In addition:

They provide habitat for many wetland organisms

They can be built to fit harmoniously into the landscape

They provide numerous benefits in addition to water quality improvement, such as wildlife habitat and the aesthetic enhancement of open spaces

They are an environmentally-sensitive approach that is viewed with favor by the general public.

LIMITATIONS OF C O N S T R U C T E D W E T L A N D S

There are limitations associated with the use of constructed wetlands:

1.They generally require larger land areas than do conventional wastewater treatment sys-terns. 2.Wetland treatment may be economical relative to other options only where land is available and affordable.

3.Performance may be less consistent than in conventional treatment.

Wetland treatment efficiencies may vary ‘seasonally in response to changing environmental conditions, including rainfall and drought. While the average perfor-mance over the year may be acceptable, wetland treatment cannot be relied upon if effluent quality must meet stringent discharge standards at all times.The biological components are sensitive to toxic chemicals, such as ammonia and pesticides . Flushes of pollutants or surges in water flow may temporarily reduce treatment effectiveness.

TYPES OF CONSTRUCTED WETLANDS

There are several types of constructed wetlands: surface flow wetlands, subsurface flow wetlands, and hybrid systems that incorporate surface and subsurface flow wetlands. Constructed wetland systems can also be combined with conventional treatment technologies.

The types of constructed wetlands appropriate for domestic wastewater, agricultural wastewater, coal mine drainage, and stormwater runoff .Water level is above the ground surface; vegetation is rooted and emerges above the water surface: waterflow is primarily above ground. Water level is below ground; water flow is through a sand or gravel penetrate to the bottom of the bed

REED BED

A reed bed is essentially a channel, lined with an impermeable membrane, that is filled with gravel and planted with macrophytes i.e. reeds, rushes and used to treat wastewater. Wastewater, black or grey, is passed through the root zone of the reeds where it undergoes treatment. Inlet and outlet pipes are positioned below the gravel surface, so that the water always remains below the gravel surface, thus excluding human exposure to the wastewater, mosquito breeding and unpleasant odours.

Reed beds work

Primary treated effluent from the house is initially filtered prior to entering the reed bed through an effluent filter fitted to the greywater collection tank or septic tank outlet pipe. After filtration of these large solids/floatables the wastewater undergoes many processes as it passes through the reed bed.

Reed beds are generally designed to detain the wastewater for a period of 5 to 7 days. This residence time aids with the treatment by allowing sufficient time for the settling and filtering of suspended solids, nitrification/denitrification to occur, fixation onto the substrate, breakdown of organic matter and nutrient removal via micro-organisms and plant uptake. Residence time is generally governed by the surface area and depth of the reed bed.

PRINCIPLE

The reed bed technology is an high functional efficiency eco-friendly wastewater treatment plant that has a lower cost and lower carbon footprint compare to conventional wastewater treatment methods. Reed Bed use waste to produce new resources in order to contribute to the natural environment cycle regeneration. The main principle consists of filtering and aerating waste water in different stage of sand filters basins planted with green plants, in order to produce treated water. We incorporate and combine this eco-friendly principle in our wide range design of wastewater treatment plant.

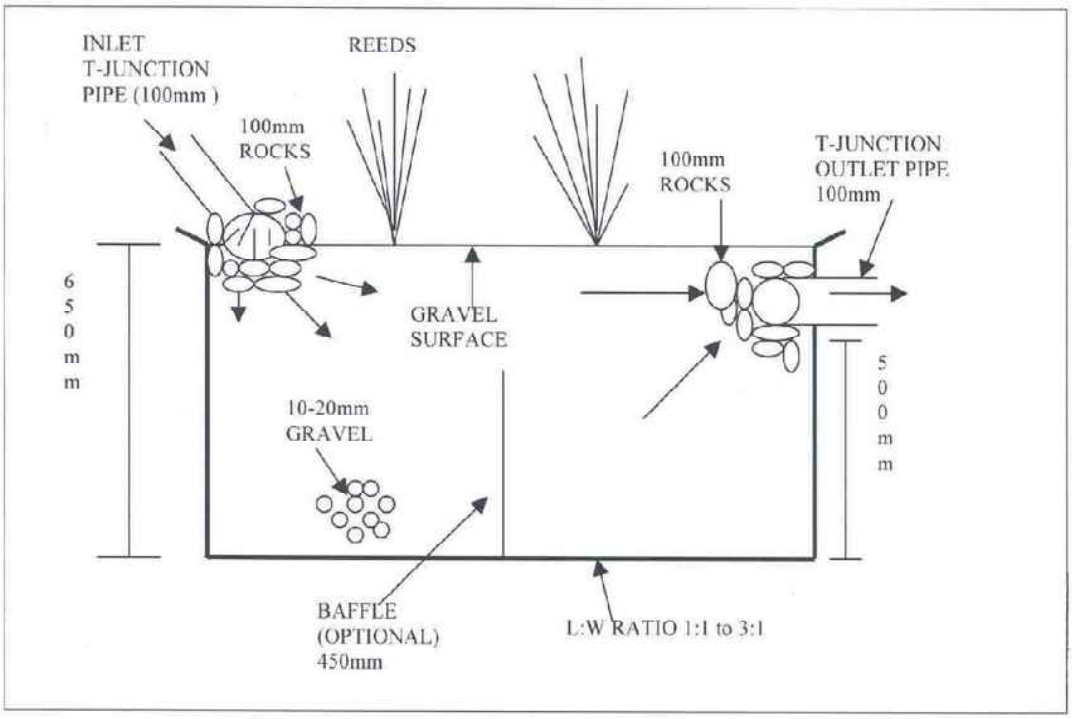


Figure : Basic Reed bed Design (Lateral View)

BASIC REED BED DESIGN PARAMETERS

Reed beds should be designed to have a wastewater residence time of 5 to 7 days and can range between 300-1000mm deep (Figure). Reed beds should have a length to width ratio between 3:1 and 1:1. If installing a circular reed bed it is advisable to install a central baffle and strongly consider installing inlet and outlet baffles to prevent short-circuiting of wastewater in the reed bed and a possible reduction of treatment performance.

Many reed beds contain 10-20mm gravel as a medium for the main body of the reed bed, although some reed beds do have a top layer of sand for planting the reeds in. Clogging due to solids in the influent can be minimised by installing an effluent filter on the outlet of the greywater/septic tank. Current research is indicating that once a reed bed has been established and the reeds have reached maturity that worms enter the reed bed and actually "transport" solids to the surface that would otherwise clog the reed bed.

Wastewater enters the reed bed via the inlet pipe positioned at a height greater than the outlet pipe, and disperses the wastewater as evenly as possible into the gravel. Some reed

beds use perforated T-junctions , made from 100mm PVC sewer grade pipe, whilst others use perforated 300mm capped, perforated stormwater pipe . It is important to prevent surfacing of effluent and the escape of odours. Therefore, the inlet pipe should be covered with aggregate. Large 50/100mm diameter rocks must be placed around the inlet and outlet pipes to allow the effluent to disperse easily and quickly, to minimise clogging and make checking for root intrusion easier.

Some reed beds contain baffles that minimise short circuiting of the wastewater flow and direct the wastewater up and down through the aerobic and anaerobic zones in the reed bed, creating unfavourable conditions for pathogens and assisting nutrient removal. The reed bed membrane can be made from ferro-concrete (concrete cattle troughs) or polyethylene (i.e. cattle troughs, water tanks cut in half, Duraplas moulds).

MACROPHYTES

Macrophytes i.e. sedges, reeds, play an important role in the treatment of water within a reed bed. They directly take-up nutrients, pump oxygen into the substrate and provide a food source for the micro-organisms responsible for breaking down pollutants (L.Davison&T.Headley, 2003).

Until recently*, Phragmitesaustralis*(the Common Reed) and *Tyhpaorientalis*were the main reeds used in reed beds on the NSW North coast. However, with the understanding of reed beds growing continuously many new plant species are being used, such as, *Lomandrahystrix, Baumeaarticulata*and *Schoenaplectusmucronatus*. Davison & Headley (2003) recommend that locally occurring native species that exhibit rapid and luxurious growth should be used.

Planting out the Reed bed

Macrophytes should be planted at a density of at least 4 to 5 clumps per square metre (approximately 30-40cm apart), with the rhizomes measuring approximately 15cm^3. Essentially, the denser they are planted, the better! All shoots/stems should be trimmed to a length of 20cm before placing the clump in the gravel. Reeds will grow best if the reed bed is situated in a sunny location

Macrophyte species suitable for use in reed beds and their maximum expected height. Species name	**Common name**	**Height (m)**
Baumeaarticulata	Jointed twigrush	2.5
Baumearubiginosa	-	1
Bolboschoenusfluviatilis	marsh clubrush	2.5
Eleocharissphacelata	Tall spikerush	2
Lepironiaarticulata	-	4
Phragmitesaustralis	Common reed	4
Schoenoplectusmucronatus	-	1
Schoenoplectusvalidus	River clubrush	3
Typhaorientalis	Bullrush or cumbungi	4

HARVESTING THE MACROPTHYES

Harvesting is undertaken to remove nutrients from the reed bed and promote macrophyte growth. Harvested vegetation can be used as mulch on the reed bed or composted. If mulching the reed bed you must ensure that the amount of mulched material placed on the reed bed does not prevent the re-growth of new shoots. Council advises people to use only 20% of the harvested material on the reed bed as a mulch until such a time as you the owner knows how much mulch can be used without limiting the growth of new shoots. Recent studies have shown that mulching the reed bed helps attract garden/compost worms

that help "transport" solids in the reed bed to the gravel surface that would otherwise clog the reed bed, and thus having a positive effect on the treatment performance of the reed bed.

LEGISLATION AND STANDARDS

There are three key policies relating to environmental protection in India.
They are:

- The National Forest Policy, 1988
- Policy statement for Abatement of Pollution, 1992
- National Conservation Strategy and Policy Statement on Environment and Development, 1992

According to Section 2(a) of the Environmental Protection Act, 1986, 'Environment' includes

a)Water,air and land

b) The inter-relationship which exists among and between,

i) water, air, land, and

ii) human beings, other living creatures, plants, microorganisms and property

The different statutes / legislations enacted in India exclusively for environment protection are

- The Water (Prevention and Control of Pollution) Act, 1974
- The Water (Prevention and Control of Pollution) Rules, 1975
- The Water (Prevention and Control of Pollution) Cess Act, 1977
- The Water (Prevention and Control of Pollution) Cess Rules, 1978
- The Air (Prevention and Control of Pollution) Act, 1981
- The Air (Prevention and Control of Pollution) Rules, 1982
- The Environment (Protection) Act, 1986
- The Environment (Protection) Rules, 1986
- Hazardous Wastes (Management and Handling) Rules, 1989
- Manufacture, Storage and Import of Hazardous Chemical Rules, 1989
- The Forest (Conservation) Act, 1980
- The Forest (Conservation) Rules, 1981
- The Wildlife Protection Act, 1972
- The Wildlife (Transactions and Taxidermy) Rules, 1973
- The Wildlife (Stock Declaration) Central Rules, 1973
- The Wildlife (Protection) Licensing (Additional Matters for Consideration) Rules, 1983
- The Wildlife (Protection) Rules, 1995
- The Wildlife (Specified Plants - Conditions for Possession by Licensee) Rules, 1995
- The Public Liability Insurance Act, 1991
- The Public Liability Insurance Rules, 1991
- The National Environment Tribunal Act, 1995
- The National Environment Appellate Authority Act, 1997

What is the difference between the laws enacted before and after independence with respect to Environmental protection in India?

There are about two hundred laws dealing with environmental protection both before and after independence in India. However, the pre-independence laws have not dealt with environmental protection exclusively. For example, the Indian Penal Code (IPC), 1860, had a chapter (chapter XIV) which dealt with offences affecting public health, safety and convenience, which covered aspects like water, air and noise pollution, whereas the post-independence laws mentioned above deal exclusively with environmental protection.

What are the provisions in the Indian Penal Code for Environmental protection ?

The Indian Penal Code has a chapter on offences affecting Public Health, Safety, Convenience . Sec. 268 provides that "a person is guilty of a public nuisance who does any act or is guilty of an illegal omission which causes any common injury, danger or annoyance to the public or to the people in general who dwell or occupy property in the vicinity, or which must necessarily cause injury, obstruction, danger, or annoyance to persons who may have occasion to use any public right." The section further explains that a common nuisance is not excusable on the ground that it causes some convenience or advantage. Other concerned provisions are: a "negligent act likely to spread infection or disease dangerous to life" (Sec. 269 IPC.), a "malignant act likely to spread infection or disease dangerous to life" (Sec. 270 IPC.), "making atmosphere noxious to health" (Sec. 278 IPC.).

But the essential requirement of the provision to punish a man is the guilty intention of the accused, i.e. either the act of the accused should be negligent, malignant or voluntary, which vitiates the atmosphere. In case of public nuisance, the Penal Code provides for fines up to Rs. 200/- by way of punishment (Sec. 290 IPC.) and for making the atmosphere noxious to health Rs. 500/- only (Sec.78 IPC.).The punishments are too meagre to meet the objectives. With these penal provisions, it is not possible to check environmental pollution.

Lecture 5

SOIL POLLUTANTS-SOURCES-URBAN AND INDUSTRIAL-HEAVYMETAL-PESTICIDES-PAHS AND PCBS-E-WASTE

TYPES OF SOIL POLLUTANTS

Soil pollution consists of pollutants and contaminants. The main pollutants of the soil are the biological agents and some of the human activities. Soil contaminants are all products of soil pollutants that contaminate the soil. Human activities that pollute the soil range from agricultural practices that infest the crops with pesticide chemicals to urban or industrial wastes or radioactive emissions that contaminate the soil with various toxic substances.

Short-term damage by changing the growth rate of plant or animal species, or by interfering with human amenities, comfort, health, or property values. Pollutants may be classified by various criteria: (1) By the origin: whether they are natural or man-made (synthetic). (2) By the effect: on an organ, specie, or an entire ecosystem. (3) By the properties: mobility, persistence, toxicity. (4) By the controllability: ease or difficulty of removal.

SOIL POLLUTION

Soil pollution is defined as the contamination of soil in a particular region. Soil pollution is a result of penetration of harmful pesticides and insecticides that deteriorate the soil quality making it unfit for later use. Soil contamination is caused by xenobiotic components or alteration of natural environment. This is usually caused by industrial activity, agricultural chemicals and improper disposal of waste. The most common pollutants are petroleum, polynuclear aromatic carbons, hydrocarbons, solvents, pesticides, lead and other heavy metals.

CAUSES OF SOIL POLLUTION

Soil Pollution is a result of many activities by mankind which contaminate the soil. Some the causes of soil pollution can be as follows:

- Industrial effluents like harmful gases and chemicals.
- Use of chemicals in agriculture like pesticides, fertilizers and insecticides.
- Improper or ineffective soil management system.
- Unfavorable irrigation practices.
- Improper management and maintenance of septic system.
- Sanitary waste leakage.
- Toxic fumes from industries get mixed with rains causing acid rains.
- Leakages of fuel fro automobiles are washed off due to rains and are deposited in the nearby soil.
- Unhealthy waste management techniques release sewage into dumping grounds and nearby water bodies.
- Use of pesticides in agriculture retains chemicals in the environment for a long time. These chemicals also effect beneficial organisms like earthworm in the soil and lead to poor soil quality.
- Absence of proper garbage disposal system leads to scattered garbage in the soil. These contaminants can block passage of water into the soil and affects its water holding capacity.
- Unscientific disposal of nuclear waste contaminate soil and can cause mutations.
- Night soil contamination due to improper sanitary system in villages can cause harmful diseases.

EFFECTS OF SOIL POLLUTION

Soil pollution causes huge disturbances in the ecological balance and health of living organisms at an alarming rate. Some the effects of soil pollution are:

- Reduced soil fertility hence decrease in soil yield.
- Loss of natural nutrients in soil.
- Reduced nitrogen fixation.
- Loss of soil and nutrients.
- Increased soil erosion.
- Imbalance in the flora and fauna of the soil.
- Increase in soil salinity, makes it unfit for cultivation.
- Creation of toxic dust.
- Foul odor due to industrial chemicals and gases.
- Alteration in soil structure can lead to death of organisms in it.

SOIL POLLUTION - TYPES, EFFECTS, SOURCES AND CONTROL OF SOIL POLLUTION

Soil pollution

Soil pollution is defined as, "contamination of soil by human and natural activities which may cause harmful effect on living organisms". Composition of soil is listed below:

COMPONENT (%)

Organic mineral matter (45)
Organic matter (05)
Soil water (25)
Soil air (25)

TYPES, EFFECTS AND SOURCES OF SOIL POLLUTION

Soil pollution mainly occurs due to the following:

1. Industrial wastes
2. Urban wastes
3. Agricultural practices
4. Radioactive pollutants
5. Biological agents

Industrial wastes –

Disposal of Industrial wastes is the major problem for soil pollution

Sources:

Industrial pollutants are mainly discharged from various origins such as pulp and paper mills, chemical fertilizers, oil refineries, sugar factories, tanneries, textiles, steel, distilleries, fertilizers, pesticides, coal and mineral mining industries, drugs, glass, cement, petroleum.

Effect:

These pollutants affect and alter the chemical and biological properties of soil. As a result, hazardous chemicals can enter into human food chain from the soil or water, disturb the biochemical process and finally lead to serious effects on living organisms.

Urban wastes

Urban wastes comprise of both commercial and domestic wastes consisting of dried sludge and sewage. All the urban solid wastes are commonly referred to as refuse. Constituents of urban refuse: This refuse consists of garbage and rubbish materials like plastics, glasses, metallic cans, fibres, paper, rubbers, street sweepings, fuel residues, leaves, containers, abandoned vehicles and other discarded manufactured products. Urban domestic wastes though disposed off separately from industrial wastes, can still be dangerous. This happens because they are not easily degraded.

Agricultural practices –Modern agricultural practices pollute the soil to a large extent. With the advancing agro-technology, huge quantities of fertilizers, pesticides, herbicides and weedicides are added to increase the crop yield. Apart from these farm wastes, manure, slurry, debris, soil erosion containing mostly inorganic chemicals are reported to cause soil pollution

Radioactive pollutants/ - Radioactive substances resulting from explosions of nuclear testing laboratories and industries giving rise to nuclear dust radioactive wastes, penetrate the soil and accumulate giving rise to land/soil pollution.

Ex: Radio nuclides of Radium, Thorium, Uranium, isotopes of Potassium (K-40) and Carbon (C-14) are commonly found in soil, rock, water and air.

Explosion of hydrogen weapons and cosmic radiations include neutron, proton reactions by which Nitrogen (N-15) produces C-14. This C-14 participates in Carbon metabolism of plants which is then into animals and human beings.

Radioactive waste contains several radio nuclides such as Strontium90, Iodine-129, Cesium-137 and isotopes of Iron which are most injurious. Strontium get deposited in bones and tissues instead of calcium.

Nuclear reactors produce waste containing Ruthenium-106, Iodine-131, Barium-140, Cesium-144 and Lanthanum-140 along with primary nuclides Sr-90 with a half life 28 years and Cs-137 with a half life 30 years. Rain water carries Sr-90 and Cs-137 to be deposited on the soil where they are held firmly with the soil particles by electrostatic forces. All the radio nuclides deposited on the soil emit gamma radiations.

Biological agents –

Soil gets a large amount of human, animal and bird excreta which constitute a major source of land pollution by biological agents.

Ex: 1. Heavy application of manures and digested sludge can cause serious damage to plants within a few years

CONTROL MEASURES OF SOIL POLLUTION:

1.Soil erosion can be controlled by a variety of forestry and farm practices.

Ex: Planting trees on barren slopes,Contour cultivation and strip cropping may be practiced instead of shifting cultivation.Terracing and building diversion channels may be undertaken.

2.**Proper dumping of unwanted materials:**

Excess wastes by man and animals pose a disposal problem. Open dumping is the most commonly practiced technique. Nowadays, controlled tipping is followed for solid waste disposal. The surface so obtained is used for housing or sports field.

3.Production of natural fertilizers:

Bio-pesticides should be used in place of toxic chemical pesticides. Organic fertilizers should be used in place of synthesized chemical fertilizers. Ex: Organic wastes in animal dung may be used to prepare compost manure instead of throwing them wastefully and polluting the soil.

4.**Proper hygienic condition**:

People should be trained regarding sanitary habits.Ex: Lavatories should be equipped with quick and effective disposal methods.

5.Public awareness:

Informal and formal public awareness programs should be imparted to educate people on health hazards by environmental education.

Ex: Mass media, Educational institutions and voluntary agencies can achieve this.

6.Recycling and Reuse of wastes:

To minimize soil pollution, the wastes such as paper, plastics, metals, glasses, organics, petroleum products and industrial effluents etc should be recycled and reused.

Ex: Industrial wastes should be properly treated at source. Integrated waste treatment methods should be adopted.

7.Ban on Toxic chemicals:

Ban should be imposed on chemicals and pesticides like DDT, BHC, etc which are fatal to plants and animals. Nuclear explosions and improper disposal of radioactive wastes should be banned.

PAH

Polycyclic aromatic hydrocarbons are primarily found in natural sources such as creosote. They can result from the incomplete combustion of organic matter. PAHs can also be produced geologically when organic sediments are chemically transformed into fossil fuels such as oil and coal.

PAHs are considered ubiquitous in the environment and can be formed from either natural or man-made combustion sources. The dominant sources of PAHs in the environment are thus from human activity: Wood-burning and combustion of other biofuels such as dung or crop residues contribute more than half of annual global PAH emissions, particularly due to biofuel use in India and China.

As of 2004, industrial processes and the extraction and use of fossil fuels made up slightly more than one quarter of global PAH emissions, dominating outputs in industrial countries such as the United States.

PAHs are typically found as complex mixtures. Lower-temperature combustion, such as tobacco smoking or wood-burning, tends to generate low molecular weight PAHs, whereas high-temperature industrial processes typically generate PAHs with higher molecular weights.

PAHs typically disperse from urban and suburban non-point sources through road run-off, sewage, and atmospheric circulation and subsequent deposition of particulate air pollution. Soil and river sediment near industrial sites such as creosote manufacturing facilities can be highly contaminated with PAHs. Oil spills, creosote, coal mining dust, and other fossil fuel sources can also distribute PAHs in the environment.

PAH IN URBAN SOILS

` Benzo[a]pyrene (BaP) the most toxic of the parent PAH is widely considered a key marker PAH for environmental assessments; the normal background concentration of BaP in the London urban sites was 6.9 mg/kg (dry wt soils). the overall distribution also suggested that the PAHs in London soils had undergone weathering and been modified by a variety of pre-and post-depositional processes such as volatilization and microbial biodegradation.

Volcanic eruptions may emit PAHs. Certain PAHs such as perylene can also be generated in anaerobic sediments from existing organic material, although it remains undetermined whether abiotic or microbial processes drive their production.

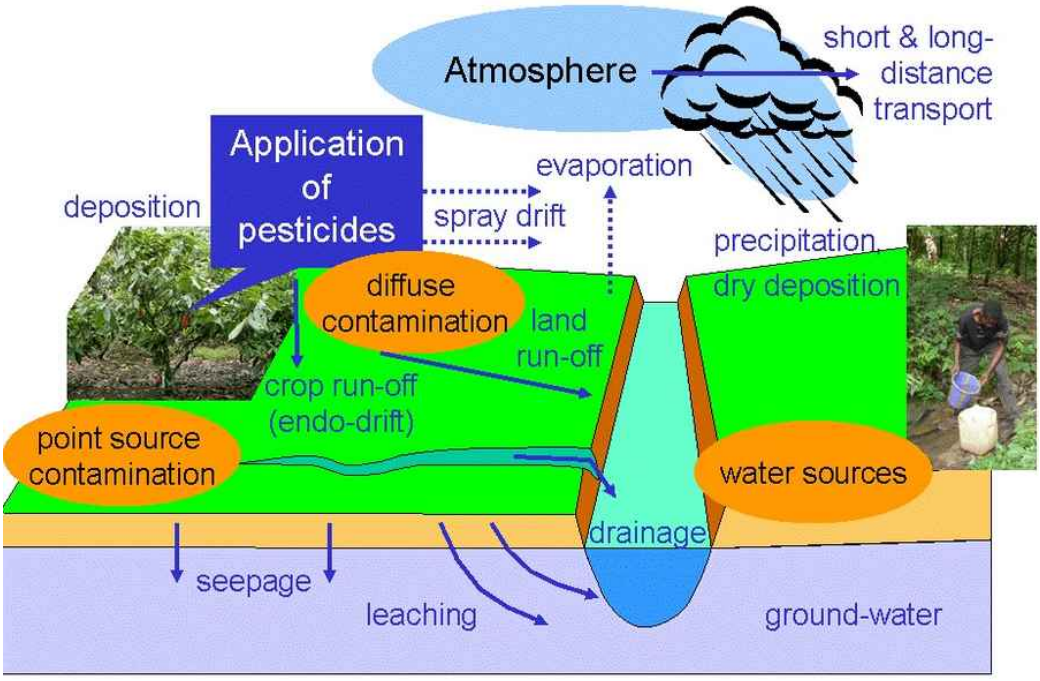
PESTICIDES

Pesticides, the most cost-effective means of pest and weed control, allow the maintenance of current yields and so contribute to economic viability. Concern about the environmental impact of repeated pesticide use has prompted research into the environmental fate of these agents, which can emigrate from treated fields to air, other land and waterbodies. How long the pesticide remains in the soil depends on how strongly it is bound by soil components and how readily it is degraded. It also depends on the environmental conditions at the time of application, *e.g.*, soil water content. Pesticide use must ensure public safety and environmental protection with regards to both the chemical itself and their potentially harmful metabolites.

Many of the chemicals used in pesticides are persistent soil contaminants, whose impact may endure for decades and adversely affect soil conservation.

The use of pesticides decreases the general biodiversity in the soil. Not using the chemicals results in higher soil quality, with the additional effect that more organic matter in the soil allows for higher water retention. This helps increase yields for farms in drought years, when organic farms have had yields 20-40% higher than their conventional counterparts. A smaller content of organic matter in the soil increases the amount of pesticide that will leave the area of application, because organic matter binds to and helps break down pesticides.

Degradation and sorption are both factors which influence the persistence of pesticides in soil. Depending on the chemical nature of the pesticide, such processes control directly the transportation from soil to water, and in turn to air and our food. Breaking down organic substances, degradation, involves interactions among microorganisms in the soil. Sorption affects bioaccumulation of pesticides which are dependent on organic matter in the soil. Weak organic acids have been shown to be weakly sorbed by soil, because of pH and mostly acidic structure. Sorbed chemicals have been shown to be less accessible to microorganisms. Aging mechanisms are poorly understood but as residence times in soil increase, pesticide residues become more resistant to degradation and extraction as they lose biological activity.



The **impact of pesticides** consists of the effects of pesticides on non-target species. Pesticides are chemical preparations used to kill fungal or animal pests. Over 98% of sprayed insecticides and 95% of herbicides reach a destination other than their target species, because they are sprayed or spread across entire agricultural fields. Runoff can carry pesticides into aquatic environments while wind can carry them to other fields, grazing areas, human settlements and undeveloped areas, potentially affecting other species. Other problems emerge from poor production, transport and storage practices. Over time, repeated application increases pest resistance, while its effects on other species can facilitate the pest's resurgence.

Each pesticide or pesticide class comes with a specific set of environmental concerns. Such undesirable effects have led many pesticides to be banned, while regulations have limited and/or reduced the use of others. Over time, pesticides have generally become less persistent and more species-specific, reducing their environmental footprint. In addition the

amounts of pesticides applied per hectare have declined, in some cases by 99%. However, the global spread of pesticide use, including

ELECTRONIC WASTE

e-waste describes discarded electrical or electronic devices. Used electronics which are destined for reuse, resale, salvage, recycling, or disposal are also considered e-waste. Informal processing of e-waste in developing countries can lead to adverse human health effects and environmental pollution.

Electronic scrap components, such as CPUs, contain potentially harmful components such as lead, cadmium, beryllium, or brominated flame retardants. Recycling and disposal of e-waste may involve significant risk to health of workers and communities in developed countries and great care must be taken to avoid unsafe exposure in recycling operations and leaking of materials such as heavy metals from landfills and incinerator ashes.

E-waste or electronic waste is created when an electronic product is discarded after the end of its useful life. The rapid expansion of technologymeans we are creating a very large amount of e-waste every passing minute.

"Electronic waste" or E-Waste may be defined as discarded computers, office electronic equipment, entertainment device electronics, mobile phones, television sets, and refrigerators. This includes used electronics which are destined for reuse, resale, salvage, recycling, or disposal. Others are re-usables (working and repairable electronics) and secondary scrap (copper, steel, plastic, etc.) to be "commodities", and reserve the term "waste" for residue or material which is dumped by the buyer rather than recycled, including residue from reuse and recycling operations, because loads of surplus electronics are frequently commingled (good, recyclable, and non-recyclable), several public policy advocates apply the term "e-waste" broadly to all surplus electronics. Cathode ray tubes (CRTs) are considered one of the hardest types to recycle.the use of older/obsolete pesticides that have been banned in some jurisdictions, has increased overall.

ENVIRONMENTAL IMPACT OF THE PROCESSING OF DIFFERENT ELECTRONIC WASTE COMPONENTS

E-Waste Component	Process Used	Potential Environmental Hazard
Cathode ray tubes (used in TVs, computer monitors, ATM, video cameras, and more)	Breaking and removal of yoke, then dumping	Lead, barium and other heavy metals leaching into the ground water and release of toxic phosphor
Printed circuit board (image behind table – a thin plate on which chips and other electronic components are placed)	De-soldering and removal of computer chips; open burning and acid baths to remove metals after chips are removed.	Air emissions and discharge into rivers of glass dust, tin, lead, brominated dioxin, beryllium cadmium, and mercury
Chips and other gold plated components	Chemical stripping using nitric and hydrochloric acid and burning of chips	PAHs, heavy metals, brominated flame retardants discharged directly into rivers acidifying fish and flora. Tin and lead contamination of surface and

		groundwater. Air emissions of brominated dioxins, heavy metals, and PAHs
Plastics from printers, keyboards, monitors, etc.	Shredding and low temp melting to be reused	Emissions of brominated dioxins, heavy metals, and hydrocarbons
Computer wires	Open burning and stripping to remove copper	PAHs released into air, water, and soil.

A **POLYCHLORINATED BIPHENYL** (**PCB**) is an organic chlorine compound with the formula $C_{12}H_{10-x}Cl_x$. Polychlorinated biphenyls were once widely deployed as dielectric and coolant fluids in electrical apparatus, carbonless copy paper and in heat transfer fluids. Because of their longevity, PCBs are still widely in use, even though their manufacture has declined drastically since the 1960s, when a host of problems were identified. Because of PCBs' environmental toxicity and classification as a persistent organic pollutant, PCB production was banned by the United States Congress in 1979 and by the Stockholm Convention on Persistent Organic Pollutants in 2001.

The International Agency for Research on Cancer (IARC), rendered PCBs as definite carcinogens in humans. According to the U.S. Environmental Protection Agency (EPA), PCBs cause cancer in animals and are probable human carcinogens.[4] Many rivers and buildings including schools, parks, and other sites are contaminated with PCBs, and there have been contaminations of food supplies with the substances.Some PCBs share a structural similarity and toxic mode of action with dioxins.Other toxic effects such as endocrine disruption (notably blocking of thyroid system functioning) and neurotoxicity are known.The maximum allowable contaminant level in drinking water in the United States is set at zero, but because of the limitations of water treatment technologies, a level of 0.5 parts per billion is the de facto level.

The bromine analogues of PCBs are polybrominated biphenyls (PBBs), which have analogous applications and environmental concerns. toxicity of PCBs varies considerably among congeners. The coplanar PCBs, known as nonortho PCBs because they are not substituted at the ring positions ortho to (next to) the other ring, (such as PCBs 77, 126 and 169), tend to have dioxin-like properties, and generally are among the most toxic congeners. Because PCBs are almost invariably found in complex mixtures, the concept of toxic equivalency factors (TEFs) has been developed to facilitate risk assessment and regulation, where more toxic PCB congeners are assigned higher TEF values on a scale from 0 to 1. One of the most toxic compounds known, 2,3,7,8-tetrachlorodibenzo[p]dioxin, a PCDD, is assigned a TEF of 1.[37]

EXPOSURE AND EXCRETION

In general, individuals are exposed to PCBs overwhelmingly through food, much less so by breathing contaminated air, and least by skin contact. Once exposed, some PCBs may change to other chemicals inside the body. These chemicals or unchanged PCBs can be excreted in feces or may remain in a person's body for years, with half lives estimated at 10–15 years. PCBs collect in body fat and milk fat. PCBs biomagnify up the food web and are present in fish and waterfowl of contaminated aquifers.[M]Infants are exposed to PCBs through breast milk or by intrauterine exposure through transplacental transfer of PCBs and are at the top of the food chain.

SIGNS AND SYMPTOMS

HUMANS

The most commonly observed health effects in people exposed to extremely high levels of PCBs are skin conditions, such as chloracne and rashes, but these were known to be symptoms of acute systemic poisoning dating back to 1922. Studies in workers exposed to PCBs have shown changes in blood and urine that may indicate liver damage. In Japan in 1968, 280 kg of PCB-contaminated rice bran oil was used as chicken feed, resulting in a mass poisoning, known as Yushō disease, in over 1800 people.Common symptoms included dermal and ocular lesions, irregular menstrual cycles and lowered immune responses. Other symptoms included fatigue, headaches, coughs, and unusual skin sores.Additionally, in children, there were reports of poor cognitive development. Women exposed to PCBs before or during pregnancy can give birth to children with lowered cognitive ability, immune compromise, and motor control problems.

There is evidence that crash dieters that have been exposed to PCBs have an elevated risk of health complications. Stored PCBs in the adipose tissue becomes mobilized into the blood when individuals begin to crash diet. PCBs have shown toxic and mutagenic effects by interfering with hormones in the body. PCBs, depending on the specific congener, have been shown to both inhibit and imitate estradiol, the main sex hormone in females. Imitation of the estrogen compound can feed estrogen-dependent breast cancer cells, and possibly cause other cancers, such as uterine or cervical. Inhibition of estradiol can lead to serious developmental problems for both males and females, including sexual, skeletal, and mental development issues. In a cross-sectional study, PCBs were found to be negatively associated with testosterone levels in adolescent boys.

High PCB levels in adults have been shown to result in reduced levels of the thyroid hormone triiodothyronine, which affects almost every physiological process in the body, including growth and development, metabolism, body temperature, and heart rate. It also resulted in reduced immunity and increased thyroid disorders.

ANIMALS

Animals that eat PCB-contaminated food even for short periods of time suffer liver damage and may die. In 1968 in Japan, 400,000 birds died after eating poultry feed that was contaminated with PCBs.Animals that ingest smaller amounts of PCBs in food over several weeks or months develop various health effects, including anemia; acne-like skin conditions (chloracne); liver, stomach, and thyroid gland injuries (including hepatocarcinoma),[citation needed], and thymocyte apoptosis.[38] Other effects of PCBs in animals include changes in the immune system, behavioral alterations, and impaired reproduction.PCBs that have dioxin-like activity are known to cause a variety of teratogeniceffects in animals. Exposure to PCBs causes hearing loss and symptoms similar to hypothyroidism in rats.[55]

CANCER

In 2013, the International Agency for Research on Cancer (IARC) classified dioxin-like PCBs as human carcinogens. According to the U.S. EPA, PCBs have been shown to cause cancer in animals and evidence supports a cancer-causing effect in humans.Per the EPA, studies have found increases in malignant melanoma and rare liver cancers in PCB workers.

In 2013, the International Association for Research on Cancer (IARC) determined that the evidence for PCBs causing non-Hodgkin lymphoma is "limited" and "not consistent". In contrast an association between elevated blood levels of PCBs and non-Hodgkin lymphoma had been previously accepted. PCBs may play a role in the development of cancers of the immune system because some tests of laboratory animals subjected to very high doses of PCBs have shown effects on the animals' immune system, and some studies of human

populations have reported an association between environmental levels of PCBs and immune response.

HEAVY METALS

Although heavy metals are naturally present in the soil, geologic and anthropogenic activities increase the concentration of these elements to amounts that are harmful to both plants and animals. Some of these activities include mining and smelting of metals, burning of fossil fuels, use of fertilizers and pesticides in agriculture, production of batteries and other metal products in industries, sewage sludge, and municipal waste disposal .

Growth reduction as a result of changes in physiological and biochemical processes in plants growing on heavy metal polluted soils has been recorded . Continued decline in plant growth reduces yield which eventually leads to food insecurity. Therefore, the remediation of heavy metal polluted soils cannot be overemphasized.

Various methods of remediating metal polluted soils exist; they range from physical and chemical methods to biological methods. Most physical and chemical methods (such as encapsulation, solidification, stabilization, electrokinetics, vitrification, vapour extraction, and soil washing and flushing) are expensive and do not make the soil suitable for plant growth . Biological approach (bioremediation) on the other hand encourages the establishment/reestablishment of plants on polluted soils.

It is an environmentally friendly approach because it is achieved via natural processes. Bioremediation is also an economical remediation technique compared with other remediation techniques. This paper discusses the nature and properties of soils polluted with heavy metals. Plant growth and performance on these soils were examined. Biological approaches employed for the remediation of heavy metal polluted soils were equally highlighted.

HEAVY METAL POLLUTED SOILS

Heavy metals are elements that exhibit metallic properties such as ductility, malleability, conductivity, cation stability, and ligand specificity. They are characterized by relatively high density and high relative atomic weight with an atomic number greater than 20 . Some heavy metals such as Co, Cu, Fe, Mn, Mo, Ni, V, and Zn are required in minute quantities by organisms.

However, excessive amounts of these elements can become harmful to organisms. Other heavy metals such as Pb, Cd, Hg, and As (a metalloid but generally referred to as a heavy metal) do not have any beneficial effect on organisms and are thus regarded as the “main threats” since they are very harmful to both plants and animals.

Metals exist either as separate entities or in combination with other soil components. These components may include exchangeable ions sorbed on the surfaces of inorganic solids, nonexchangeable ions and insoluble inorganic metal compounds such as carbonates and phosphates, soluble metal compound or free metal ions in the soil solution, metal complex of organic materials, and metals attached to silicate minerals .

LECTURE 6

Fate of pollutants in Soil - Management of soil pollution – Bio and phyto remediation of polluted soil

BIOREMEDIATION OF HEAVY METAL POLLUTED SOIL

Bioremediation is the use of organisms (microorganisms and/or plants) for the treatment of polluted soils. It is a widely accepted method of soil remediation because it is perceived to occur via natural processes. It is equally a cost effective method of soil remediation. Blaylock et al. reported 50% to 65% saving when bioremediation was used for the treatment of 1 acre of Pb polluted soil compared with the case when a conventional method (excavation and landfill) was used for the same purpose. Although bioremediation is a nondisruptive method of soil remediation, it is usually time consuming and its use for the treatment of heavy metal polluted soils is sometimes affected by the climatic and geological conditions of the site to be remediated

Heavy metals cannot be degraded during bioremediation but can only be transformed from one organic complex or oxidation state to another. Due to a change in their oxidation state, heavy metals can be transformed to become either less toxic, easily volatilized, more water soluble (and thus can be removed through leaching), less water soluble (which allows them to precipitate and become easily removed from the environment) or less bioavailable.Bioremediation of heavy metals can be achieved via the use of microorganisms, plants, or the combination of both organisms.

USING MICROBES FOR REMEDIATION OF HEAVY METAL POLLUTED SOIL

Several microorganisms especially bacteria (Bacillus subtilis, Pseudomonas putida, and Enterobacter cloacae) have been successfully used for the reduction of Cr (VI) to the less toxic Cr (III) . B.subtilis has also been reported to reduce nonmetallic elements. For instance, Garbisu et al. recorded that B. subtilis reduced the selenite to the less toxic elemental Se. Further, B. cereus and B. thuringiensis have been shown to increase extraction of Cd and Zn from Cd-rich soil and soil polluted with effluent from metal industry .

It is assumed that the production of siderophore (Fe complexing molecules) by bacteria may have facilitated the extraction of these metals from the soil; this is because heavy metals have been reported to simulate the production of siderophore and this consequently affects their bioavailability . For instance, siderophore production by Azotobactervinelandii was increased in the presence of Zn (II) . Hence, heavy metals influence the activities of siderophore-producing bacteria which in turn increases mobility and extraction of these metals in soil.

Bioremediation can also occur indirectly via bioprecipitation by sulphate reducing bacteria (Desulfovibriodesulfuricans) which converts sulphate to hydrogen sulphate which subsequently reacts with heavy metals such as Cd and Zn to form insoluble forms of these metal sulphides .

Most of the above microbe assisted remediation is carried out ex situ. However, a very important in situmicrobe assisted remediation is the microbial reduction of soluble mercuric ions Hg (II) to volatile metallic mercury and Hg (0) carried out by mercury resistant bacteria . The reduced Hg (0) can easily volatilize out of the environment and subsequently be diluted in the atmosphere .

Genetic engineering can be adopted in microbe assisted remediation of heavy metal polluted soils. For instance, Valls et al. reported that genetically engineered Ralstoniaeutropha can be used to sequester metals (such as Cd) in polluted soils. This is made possible by the introduction of metallothionein (cysteine rich metal binding protein) from mouse on the cell surface on this organism. Although the sequestered metals remain in the soil, they are made less bioavailable and hence less harmful. The controversies

surrounding genetically modified organisms and the fact that the heavy metal remains in the soil are major limitations to this approach to bioremediation.

Making the soil favourable for soil microbes is one strategy employed in bioremediation of polluted soils. This process known as biostimulation involves the addition of nutrients in the form of manure or other organic amendments which serve as C source for microorganisms present in the soil. The added nutrients increase the growth and activities of microorganisms involved in the remediation process and thus this increases the efficiency of bioremediation.

Although biostimulation is usually employed for the biodegradation of organic pollutants, it can equally be used for the remediation of heavy metal polluted soils. Since heavy metals cannot be biodegraded, biostimulation can indirectly enhance remediation of heavy metal polluted soil through alteration of soil pH. It is well known that the addition of organic materials reduces the pH of the soil ; this subsequently increases the solubility and hence bioavailability of heavy metals which can then be easily extracted from the soil .

Biochar is one organic material that is currently being exploited for its potential in the management of heavy metal polluted soils. Namgay et al. recorded a reduction in the availability of heavy metals when the polluted soil was amended with biochar; this in turn reduced plant absorption of the metals. The ability of biochar to increase soil pH unlike most other organic amendments may have increased sorption of these metals, thus reducing their bioavailability for plant uptake. It is important to note that, since the characteristics of biocharvary widely depending on its method of production and the feedstock used in its production, the effect different biochar amendments will have on the availability of heavy metals in soil will also differ. Further, more research is needed in order to understand the effect of biochar on soil microorganisms and how the interaction between biochar and soil microbes influences remediation of heavy metal polluted soils because such studies are rare in literature.

USING PLANTS FOR REMEDIATION OF HEAVY METAL POLLUTED SOIL

Phytoremediation is an aspect of bioremediation that uses plants for the treatment of polluted soils. It is suitable when the pollutants cover a wide area and when they are within the root zone of the plant. Phytoremediation of heavy metal polluted soils can be achieved via different mechanisms. These mechanisms include phytoextraction, phytostabilization, and phytovolatilization.

PHYTOEXTRACTION

This is the most common form of phytoremediation. It involves accumulation of heavy metals in the roots and shoots of phytoremediation plants. These plants are later harvested and incinerated. Plants used for phytoextraction usually possess the following characteristics: rapid growth rate, high biomass, extensive root system, and ability to tolerate high amounts of heavy metals. This ability to tolerate high concentration of heavy metals by these plants may lead to metal accumulation in the harvestable part; this may be problematic through contamination of the food chain .

There are two approaches to phytoextraction depending on the characteristics of the plants involved in the process. The first approach involves the use of natural hyperaccumulators, that is, plants with very high metal-accumulating ability, while the second approach involves the use of high biomass plants whose ability to accumulate metals is induced by the use of chelates, that is, soil amendments with metal mobilizing capacity .

Hyperaccumulators accumulate 10 to 500 times more metals than ordinary plant ; hence they are very suitable for phytoremediation. An important characteristic which makes hyperaccumulation possible is the tolerance of these plants to increasing concentrations of these metals (hypertolerance). This could be a result of exclusion of these metals from the plants or by compartmentalization of these metal ions; that is, the metals are retained in the

vacuolar compartments or cell walls and thus do not have access to cellular sites where vital functions such as respiration and cell division take place .

Generally, a plant can be called a hyperaccumulator if it meets the following criteria: (i) the concentration of metal in the shoot must be higher than 0.1% for Al, As, Co, Cr, Cu, Ni, and Se, higher than 0.01% for Cd, and higher than 1.0% for Zn ; (ii) the ratio of shoot to root concentration must be consistently higher than 1 ; this indicates the capability to transport metals from roots to shoot and the existence of hypertolerance ability : (iii) the ratio of shoot to root concentration must be higher than 1; this indicates the degree of plant metal uptake .

Reeves and Baker reported some examples of plants which have the ability to accumulate large amounts of heavy metals and hence can be used in remediation studies. Some of these plants include Haumaniastrumrobertii (Cohyperaccumulator); Aeollanthussubacaulis (Cuhyperaccumulator);Maytenusbureaviana (Mnhyperaccumulator); Minuartiaverna and Agrostistenuis (Pbhyperaccumulators);Dichapetalumgelonioides, Thlaspitatrense,and Thlaspicaerulescens (Zn hyperaccumulators); Psycotriavanhermanni and Streptanthuspolygaloides (Nihyperaccumulators); Lecythisollaria (Sehyperaccumulator). Pterisvittata is an example of a hyperaccumulator that can be used for the remediation of soils polluted with As . Some plants have the ability to accumulate more than one metal. For instance, Yang et al. observed that the Zn hyperaccumulator, Sedum alfredii, can equally hyperaccumulate Cd.

The possibility of contaminating the food chain through the use of hyperaccumulators is a major limitation in phytoextraction. However, many species of the Brassicaceae family which are known to be hyperaccumulators of heavy metals contain high amounts of thiocyanates which make them unpalatable to animals; thus this reduces the availability of these metals in the food chain .

Most hyperaccumulators are generally slow growers with low plant biomass; this reduces the efficiency of the remediation process . Thus, in order to increase the efficiency of phytoextraction, plants with high growth rate as well as high biomass (e.g., maize, sorghum, and alfalfa) are sometimes used together with metal chelating substances for soil remediation exercise. It is important to note that some hyperaccumulators such as certain species within the Brassica genus (Brassica napus, Brassica juncea, and Brassica rapa) are fast growers with high biomass .

In most cases, plants absorb metals that are readily available in the soil solution. Although some metals are present in soluble forms for plant uptake, others occur as insoluble precipitate and are thus unavailable for plant uptake. Addition of chelating substances prevents precipitation and metal sorption via the formation of metal chelate complexes; this subsequently increases the bioavailability of these metals . Further, the addition of chelates to the soil can transport more metals into the soil solution through the dissolution of precipitated compounds and desorption of sorbedspecies . Certain chelates are also able to translocate heavy metal into the shoots of plants .

Marques et al. documented examples of synthetic chelates which have successfully been used to extract heavy metals from polluted soils. Some of these chelates include EDTA (ethylenediaminetetraacetic acid), EDDS (SS-ethylenediaminedisuccinic acid), CDTA (trans-1,2-diaminocyclohexane-N,N,N′,N′-tetraacetic acid), EDDHA (ethylenediamine-di-o-hydroxyphenylacetic acid), DTPA (diethylenetriaminepentaacetic acid), and HEDTA (N-hydroxyethylenediaminetriacetic acid).

EDTA is a synthetic chelate that is widely used not only because it is the least expensive compared with other synthetic chelates but also because it has a high ability to successfully improve plant metal uptake . Organic chelates such as citric acid and malic acid can also be used to improve phytoextraction of heavy metals from polluted soils .

One major disadvantage of using chelates in phytoextraction is the possible contamination of groundwater via leaching of these heavy metals . This is because of the increased availability of heavy metals in the soil solution when these chelates are used. In addition, when chelates (especially synthetic chelates) are used in high concentrations, they can become toxic to plants and soil microbes . In general, solubility/availability of heavy metals for plant uptake and suitability of a site for phytoextraction are additional factors that should be considered (in addition to suitability of plants) before using phytoextraction for soil remediation .

PHYTOSTABILIZATION

Phytostabilization involves using plants to immobilize metals, thus reducing their bioavailability via erosion and leaching. It is mostly used when phytoextraction is not desirable or even possible . Marques et al. argued that this form of phytoremediation is best applied when the soil is so heavily polluted so that using plants for metal extraction would take a long time to be achieved and thus would not be adequate. Jadia and Fulekar .on the other hand showed that the growth of plants (used for phytostabilization) was adversely affected when the concentration of heavy metal in the soil was high.

Phytostabilization of heavy metals takes place as a result of precipitation, sorption, metal valence reduction, or complexation . The efficiency of phytostabilization depends on the plant and soil amendment used. Plants help in stabilizing the soil through their root systems; thus, they prevent erosion. Plant root systems equally prevent leaching via reduction of water percolation through the soil. In addition, plants prevent man's direct contact with pollutants and they equally provide surfaces for metal precipitation and sorption .

Based on the above factors, it is important that appropriate plants are selected for phytostabilization of heavy metals. Plants used for phytostabilization should have the following characteristics: dense rooting system, ability to tolerate soil conditions, ease of establishment and maintenance under field conditions, rapid growth to provide adequate ground coverage, and longevity and ability to self-propagate.

Soil amendments used in phytostabilization help to inactivate heavy metals; thus, they prevent plant metal uptake and reduce biological activity . Organic materials are mostly used as soil amendments in phytostabilization. Marques et al. showed that Zn percolation through the soil reduced by 80% after application of manure or compost to polluted soils on which Solanumnigrum was grown.

Other amendments that can be used for phytostabilization include phosphates, lime, biosolids, and litter . The best soil amendments are those that are easy to handle, safe to workers who apply them, easy to produce, and inexpensive and most importantly are not toxic to plants . Most of the times, organic amendments are used because of their low cost and the other benefits they provide such as provision of nutrients for plant growth and improvement of soil physical properties .

In general, phytostabilization is very useful when rapid immobilization of heavy metals is needed to prevent groundwater pollution. However, because the pollutants remain in the soil, constant monitoring of the environment is required and this may become a problem.

PHYTOVOLATILIZATION

In this form of phytoremediation, plants are used to take up pollutants from the soil; these pollutants are transformed into volatile forms and are subsequently transpired into the atmosphere .Phytovolatilization is mostly used for the remediation of soils polluted with Hg. The toxic form of Hg (mercuric ion) is transformed into the less toxic form (elemental Hg). The problem with this process is that the new product formed, that is, elemental Hg, may be redeposited into lakes and rivers after being recycled by precipitation; this in turn repeats the process of methyl-Hg production by anaerobic bacteria .

Raskin and Ensley reported the absence of plant species with Hg hyperaccumulating properties. Therefore, genetic engineered plants are mostly used in phytovolatilization. Examples of transgenic plants which have been used for phytovolatilization of Hg polluted soils are Nicotianatabacum, Arabidopsis thaliana,and Liriodendron tulipifera . These plants are usually genetically modified to include gene for mercuric reductase, that is, merA. Organomercuriallyase (merB) is another bacterial gene used for the detoxification of methyl-Hg. Both merA and merB can be inserted into plants used to detoxify methyl-Hg to elemental Hg . Use of plants modified with merA and merB is not acceptable from a regulatory perspective. However, plants altered with merB are more acceptable because the gene prevents the introduction of methyl-Hg into the food chain .

Phytovolatilization can also be employed for the remediation of soils polluted with Se . This involves the assimilation of inorganic Se into organic selenoamino acids (selenocysteine and selenomethionine). Selenomethionine is further biomethylated to dimethylselenide which is lost in the atmosphere via volatilization . Plants which have successfully been used for phytovolatilization of soils polluted with Se are Brassica juncea and Brassica napus .

TECHNOLOGIES FOR THE RECLAMATION OF POLLUTED SOILS

The cleaning of contaminated soils from heavy metals is the most difficult task, particularly on a large scale. The soil is composed of organic and inorganic solid constituents, water and mixture of different gases present in various proportions. The mineral components vary according to parent materials on which the soil had been developed under a particular set of climatic conditions. Therefore, soils vary enormously in physical, chemical and biological properties. Soil water movement is controlled by physical properties, such as soil structure and texture.

The soil moisture has great bearing in controlling solute movement, salt solubility, chemical reactions and microbiological activities and ultimately the bioavailability of the metal ions. A successful phytoremediation program, therefore, must take into consideration variations in soil properties of the specific site.Different approaches have been used or developed to mitigate/reclaim the heavy metal polluted soils and waters including the landfill/damping sites. These may be broadly classified into physicochemical and biological approaches.

The physicochemical approach includes excavation and burial of the soil at a hazardous waste site, fixation/inactivation (chemical processing of the soil to immobilize the metals), leaching by using acid solutions or proprietary leachants to desorb and leach the metals from soil followed by the return of clean soil residue to the site (Salt et al., 1995), precipitation or flocculation followed by sedimentation, ion exchange, reverse osmosis and microfiltration (Raskin et al., 1996). The physicochemical approaches are generally costly and have side effects (Raskin et al., 1997; McGrath et al., 2001).

Biological approaches of remediation include: (1) use of microorganisms to detoxify the metals by valence transformation, extracellular chemical precipitation, or volatilization [some microorganism can enzymatically reduce a variety of metals in metabolic processes that are not related to metal assimilation], and (2) use of special type of plants to decontaminate soil or water by inactivating metals in the rhizosphere or translocating them in the aerial parts. This approach is called phytoremediation, which is considered as a new and highly promising technology for the reclamation of polluted sites and cheaper than physicochemical approaches (Garbisu and Alkorta, 2001; McGrath et al., 2001; Raskin et al., 1997).

Phytoremediation, also referred as botanical bioremediation (Chaney et al., 1997), involves the use of green plants to decontaminate soils, water and air. It is an emerging technology that can be applied to both organic and inorganic pollutants present in the soil,

water or air (Salt et al., 1998). However, the ability to accumulate heavy metals varies significantly between species and among cultivars within species, as different mechanisms of ion uptake are operative in each species, based on their genetic, morphological, physiological and anatomical characteristics. There are different categories of phytoremediation, including phytoextraction, phytofiltration, phytostabilization, phytovolatization and phytodegradation, depending on the mechanisms of remediation.

Phytoextraction involves the use of plants to remove contaminants from soil. The metal ion accumulated in the aerial parts that can be removed to dispose or burnt to recover metals. Phytofiltration involves the plant roots or seedling for removal of metals from aqueous wastes. In phytostabilization, the plant roots absorb the pollutants from the soil and keep them in the rhizosphere, rendering them harmless by preventing them from leaching. Phytovolatization involves the use of plants to volatilize pollutants from their foliage such as Se and Hg. Phytodegradation means the use of plants and associated microorganisms to degrade organic pollutants (Garbisu and Alkorta, 2001). Some plants may have one function whereas others can involve two or more functions of phytoremediation.

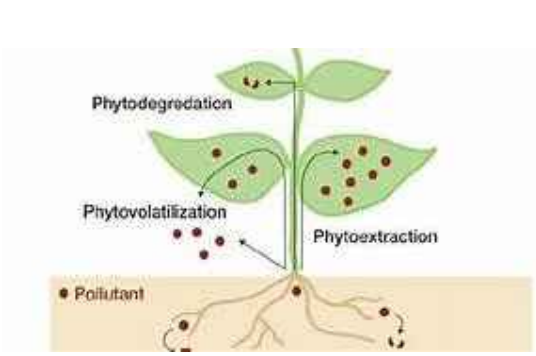
PLANT SPECIES FOR PHYTOREMEDIATION

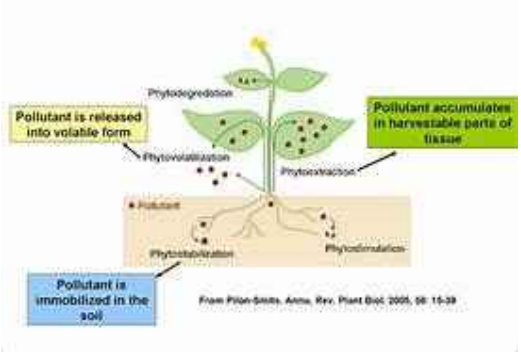
To identify plant populations with the ability to accumulate heavy metals, 300 accessions of 30 plant species were tested by Ebbs et al.(1997) in hydroponics for 4 weeks, having moderate levels of Cd, Cu and Zn. The results indicate that many Brasssica spp. such as B. juncea L., B. juncea L. Czern, B. napus L. and B. rapa L. exhibited moderately enhanced Zn and Cd accumulation.

They were also found to be most effective in removing Zn from the contaminated soils. To date, more than 400 plant species have been identified as metal hyperaccumulators, representing less than 0.2% of all angiosperms (Brooks, 1998; Baker et al., 2000). The plant species that have been identified for remediation of soil include either high biomass plants such as willow (Landberg and Greger, 1996) or those that have low biomass but high hyperaccumulating characteristics such as Thlaspi and Arabidopsis species. On worldwide basis, the number of species identified to have ability to accumulate one or more metals >1000 mg/kg dry weight is listed in Table (Reeves, 2003). The number of plant species that are reported to have hyperaccumulation traits (metal concentration >1000 mg/kg dry weight) (Reeves, 2003)

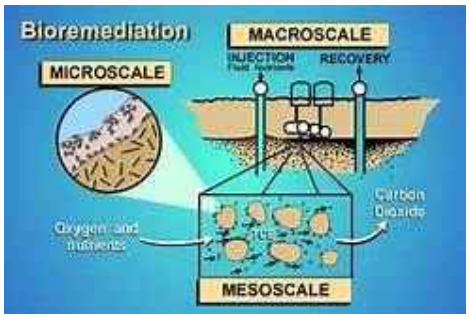
Metals	Number of species
As	04
Cd	01
Co	34
Cu	34
Pb	14
Ni	>320
Se	20

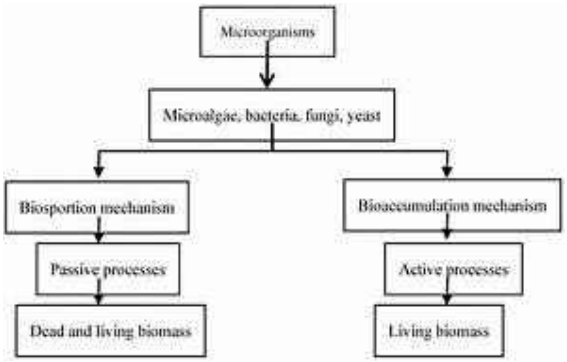
PHYTOREMEDDIATION





BIOREMEDIATION





Lecture 7

Air pollutants from industrial and domestic sources-fate of air pollutants-airpollution indicators-airpollution episodes -monitoring and control measures

History of air pollution:

The origin of air pollutionon the earth can be traced from the timeswhen man started using firewood as a meansof cooking and heating. Hippocrates has mentionedair pollution in 400 BC. In spite of this airpollution became a serious problem in Londonduring the industrial revolution due to the useof coal in industries. The earliest recorded majordisaster was the 'London Smog' that occurredin 1952 that resulted in more than 4000 deathsdue to the accumulation of air pollutants overthe city for five days.

Air pollution began to increase in the beginningof the twentieth century with the developmentof the transportation systems and large-scale useof petrol and diesel. The severe air quality problemsdue to the formation of photochemicalsmog from the combustion residues of dieseland petrol engines were felt for the first time inLos Angeles. Pollution due to auto-exhaust remainsa serious environmental issue in manydeveloped and developing countries includingIndia.

The Air Pollution Control Act in India was passedin 1981 and the Motor Vehicle Act for controllingthe air pollution, very recently. These lawsare intended to prevent air from being polluted.The greatest industrial disaster leading to seriousair pollution took place in Bhopal whereextremely poisonous methyl isocyanide gas wasaccidentally released from the Union Carbide'spesticide manufacturing plant on the night ofDecember 3rd 1984. The effects of this disasteron human health and the soil are felt even today.

SOURCES AND COMMON EFFECTS OF COMMON AIR POLLUTANTS

Carbonmonoxide:

It is a colourless, odourless gas that is poisonous animals. It is formed by incomplete combustion of carbon containing fuels.Source of carbonmonoxide is cigarette smoking and incomplete combustion of fossil fuels (more than 77% comes from motor vehicle exhaust)

Health effects

include reduced ability of red blood cells to carry oxygen to body cells and tissues. This leads to headache and anemia. At high levels it causes coma, irreversible brain damage and death.

Nitrogen Dioxide:

It is a reddish-brown irritating gas that causes photochemical smog. In the atmosphere, it gets converted into nitric acid (HNO3). It is caused by burning fossil fuels in industries and power plants.

Health effects include lung irritation and damage. Environmental effects involve acid deposition leading to damage of trees, lakes, soil and ancient monuments. NO2 can damage fabrics.

Sulphur Dioxide:

It is a colourless and irritating gas that is formed by combustion of sulphur containing fossil fuels such as coal and oil. In the atmosphere it is converted into Sulphuric acid which is a major component of acid deposition.

Health effects involve breathing problems for healthy people.**Environmental effects** involve reduced visibility and acid deposition on trees, lakes, soils and monuments leading to their deterioration and adverse effect on aquatic life.

Suspended Particulate Matter (SPM):

Includes a variety of particles and droplets (aerosols) that can be suspended in atmosphere for short to long periods. Human sources for SPM include burning coal in power and industrial units, burning diesel and other fuels in vehicles, agriculture, unpaved roads, construction, etc.

Health effects include nose and throat irritation, ling damage, bronchitis, asthama, reproductive problems and cancer.**Environmental Effects** include reduced visibility and acid deposition. Acid deposition may lead to damaged trees, soils and aquatic life in lakes.

Ozone

It is a highly reactive gas with an unpleasant odour occurring in the stratosphere where it protects mankind fro the harmful ultra-violet rays from the Sun. However on earth, it is a pollutant. It occurs on earth due to reaction between Volatile Organic Compounds (VOCs) and Nitrogen Oxides. It moderates the climate

Photochemical smog

It is a browinsh smoke that frequently forms on clear, sunny days over large cities with significant amounts of automobile traffic.It is mainly due to chemical reactions among nitrogen oxides and hydrocarbons in the presence of sunlight.

Health effects include breathing problems, cough, eye, nose and throat irritation, heart diseases, reduced resistance to colds and pneumonia.

Environmental effects involve damage to plants and trees. Additionally, Smog reduces visibility.

Lead

It is a solid and highly toxic metal. Its compounds are emitted into the atmosphere as particulate matter.

Human Sources: Paint, Smelters (metal refineries), lead manufacture, storage batteries, leaded petrol, etc

Health effects: Lead accumulates in the body and brain leading to nervous system damage and mental retardation (especially in children), digestive and other health problems. Lead containing chemicals are known to cause cancer in test animals.

Environmental Effects: It can harm wildlife.

Hydrocarbons

Lower haydrocarbons accumulate due to decay of vegetable matter.**Human effects**: They are carcinogenic

Chromium:

It is a solid toxic metal emitted into the atmosphere as particulate matter.

Human sources: Paint, Smelters, Chromium manufacture, Chromium plating.

Health Effects: Perforation of nasal septum, chrome holes, etc.

Pollutants and Their Fate

- Our environment contains a multitude of chemicals that affect many species and ecosystems.
- Ecotoxicology can be viewed as a study of the harmful effects of chemicals on ecosystems.
- It deals with the delivery, transport, transformation, and effects of pollutants on the physical environment and on the species that live here.
- Pollutants enter ecosystems through discharges into the atmosphere, contamination of land, and entry into water
- Chemical pollution has been linked to:
- Loss of parental attention in birds
- Decreased fertility in invertebrates, fish, reptiles, birds, and mammals
- Decreased hatching success in fish, turtles, and birds
- Abnormal thyroid function in birds and mammals
- Feminization in males and masculinization in females in birds, fish, and invertebrates
- Altered immune function in birds and mammals

Atmosphere

- composed of a number of layers:
- Troposphere
- Stratosphere
- Mesosphere
- Thermosphere
- principal concern is the troposphere, which extends from the surface of the earth to an altitude of about 10 km
- This atmospheric air contains:
- "trace amounts" of chemicals such as carbon monoxide, sulfur dioxide, hydrogen sulfide, nitric acid, ammonia, formaldehyde, lead, oxides of nitrogen, particulates, and many others pollutants
- most of the atmospheric water, and this is where we observe clouds, storms, and other conditions of weather
- gases necessary to maintain life

Fate of a Chemical in the Air

- TRI estimates that about half of atmospheric pollutants result from industrial discharges
- Pollutants reach the atmosphere directly or through transport from other compartments.
- Airborne chemicals can enter into chemical reactions, such as oxidation and photolysis.
- This ability to enter into chemical reactions results in the production of other pollutants
- The fate of a chemical in the air depends on:
- Input (e.g., rate, type of pollutant, source, etc.)
- Dispersion (e.g., mixing from the wind and turbulence)
- Transport (e.g., vertical and horizontal by wind)
- Reactions and formation of secondary pollutants (e.g., chemical reactions such as oxidation hydrolysis and photolysis and physical reactions such as absorption and adsorption)
- Removal (e.g., through precipitation)

Indicators of pollution

Human activities can pollute the air and water. The presence or absence of certain living organisms acts as an indicator of the amount of pollution.

Air pollution

The most common source of air pollution is the **combustion of fossil fuels**. This usually happens in vehicle engines and power stations. Sulfur dioxide is released if the fuel contains sulfur compounds. This gas contributes to acid rain. Lichens can be used as air pollution indicators, especially of the concentration of sulfur dioxide in the atmosphere.

Lichens are plants that grow in exposed places such as rocks or tree bark. They need to be very good at absorbing water and nutrients to grow there. Rainwater contains just enough nutrients to keep them alive. Air pollutants dissolved in rainwater, especially sulfur dioxide, can damage lichens, and prevent them from growing. This makes lichens natural indicators of air pollution. For example:

- Bushy lichens need really clean air
- Leafy lichens can survive a small amount of air pollution
- Crusty lichens can survive in more polluted air.

In places where no lichens are growing, it is often a sign that the air is heavily polluted with sulfur dioxide.

AIRPOLLUTION EPISODES

Some Air Pollution Episodes

Bhopal Disaster, 1984

In the mid night of 2nd – 3 rdDecember 1984, in a densely populated area of Bhopal, CentralIndia, a poisonous vapor burst from the tall stacks of the Union Carbide pesticide plant. Aboutforty tons of toxic gases had leaked from the Carbides Bhopal plant and spread throughout thecity. The cause was the contamination of Methyl Isocyanate (MIC) storage tank with watercarrying catalytic materi

Mexico, Poza Rica, 1950

A catastrophic exposure episode involving the release of large quantities of hydrogen sulfideoccurred in Poza Rica, Mexico in November 1950. In an early morning malfunction of the wastegas flare resulted in the release of large quantities of unburned hydrogen sulfide into theatmosphere. The unburned gas, aided by a low-level temperature inversion and light earlymorning breezes, was carried to the residential area adjacent to the plant area. Residents of thearea succumbed while attempting to leave the area and assisting stricken neighbors. Within amatter of 3 hours, 320 persons were hospitalized and 22 were killed.

Los Angeles Smog, July 1973

A classical photochemical smog episode. A average of 63 pphm oxidants were observed. Ozoneconcentrations were observed to be highest in the studies of air quality. Brown haze over thetown was observed for 3 days. It was PAN smog which affected the city. Vehicular emission ismainly responsible and also NOx from combustion of fuels. And it caused the loss of visibilityand respiratory problems.

Kuwait oil firesCaused by Iraqi military forces setting fire to 700 oil wells as part of a scorched earth policyduring Gulf war in 1991. The fires started in January and February 1991 and the last one wasextinguished by November 1991. around 6 million barrels (950,000 m3) of oil were lost eachday. The byproducts of the petroleum burn caused pollution to the soil and air, known as Gulf war syndrome.Its consequences are as under:dramatic decrease in air quality, causing respiratory problems for many Kuwaitis.mixture of desert sand + unignited oil + soot formed layers of "tarcrete" which coverednearly 5% of the country.the oil has continued to sink into the sand affecting Kuwait's precious groundwaterresources.

Muhammad Imran Nawaz 2008-ENV-43Yokkaichi Asthma

In Japan in 1960 and 1972. Due to burning of petroleum and other fuels. Resulted in release of SO2. Causing chronic pulmonary diseases, chronic Bronchitis, pulmonary emphysema,bronchial asthma in people.

1983 Melbourne dust stormit was a meteorological phenomenon that occurred during the afternoon of 8February 1983.Red soil, dust and sand from Central and Southeastern Australia was swept up inhigh winds and carried southeast through Victoria.

2005 Malaysian haze:a week-long chokingsmog-like hazeoverMalaysiathat almost brought thecentral part of Peninsular Malaysiato a standstill

The 1997 Southeast Asian haze:□caused mainly byslash and burntechniques adopted by farmers in Indonesia.causes respiratory symptoms such asasthma,upper respiratory infectiondecreased lung function as well as eye and skin irritation due to particulate matter

Air Pollution Monitoring

The basic mission of the Office of Air Quality Planning and Standards is to preserve and improve the quality of our nation's air. To accomplish this, OAQPS must be able to evaluate the status of the atmosphere as compared to clean air standards and historical information. The following are some of the topics associated with monitoring air pollution:

- Air Quality Standards
- Air Quality Trends Report
- Ambient Monitoring Technology Information Center (AMTIC)
- Attainment/Nonattainment
- CICA (U.S. - Mexico Information Center on Air Pollution)

- Criteria Pollutants
- Databases
- Hazardous/Toxic Air Pollutants
- Monitor Types

Air Quality Standards

OAQPS manages EPA programs to improve air quality in areas where the current quality is unacceptable and to prevent deterioration in areas where the air is relatively free of contamination. To accomplish this task, OAQPS establishes the National Ambient Air Quality Standard (NAAQS) for each of the criteria pollutants.

There are two types of standards -- primary and secondary. Primary standards protect against adverse health effects; secondary standards protect against welfare effects, such as damage to farm crops and vegetation and damage to buildings. Because different pollutants have different effects, the NAAQS are also different. Some pollutants have standards for both long-term and short-term averaging times. The short-term standards are designed to protect against acute, or short-term, health effects, while the long-term standards were established to protect against chronic health effects.

This NAAQS table lists all criteria pollutants and standards.

Air Quality Trends Report

Each year EPA examines air pollution trends of each of the six principal pollutants in this country. A yearly EPA document titled National Air Quality and Emissions Trends Report gives a detailed analysis of changes in air pollution levels over the last 10 years time, plus a summary of the current air pollution status.

Ambient Monitoring Technology Information Center (AMTIC)

The Ambient Monitoring Technology Information Center (AMTIC) site is centeredaround the exchange of ambient monitoring related information. It contains all Federal Regulations pertaining to ambient monitoring, as well as ambient monitoring QA/QC related information and some information on ambient monitoring related publications. There is also available information on ambient monitoring news, field and laboratory studies of interest and available related training.

Attainment/Nonattainment

A geographic area that meets or does better than the national ambient air quality standard is called an **attainment area**; an area that doesn't meet this standard is called a **nonattainment area**.The Green Book site provides a look at nonattainment areas.

CICA (U.S. - Mexico Information Center on Air Pollution)

Bienvenido a CICA! The U.S. - Mexico Information Center on Air Pollution (CICA) is an organization that provides technical support and assistance in evaluating air pollution problems along the Mexico - U. S. Border. Our bilingual CICAsite discusses the organization and its projects in detail.

Criteria Pollutants

A few air pollutants, called **criteria air pollutants**, are common throughout the United The current criteria pollutants are:

- Carbon Monoxide (CO)
- Lead (Pb)
- Nitrogen Dioxide (NO_2)
- Ozone (O_3)
- Particulate matter (PM)
- Sulfur Dioxide (SO_2)

A good overview discussion of criteria pollutants can be found in the Plain English Guide To The Clean Air Act. Look to The Green Book site for a more detailed discussion. The EPA Air Trends Website also discusses the Six Principal Pollutants.

Databases

The Air Quality System (AQS), contains air pollution and meteorological measurements from EPA, state and local agency monitoring stations. The Air Facility System (AFS) contains compliance and permit data for stationary sources regulated by EPA, state and local air pollution agencies. The AirData Web site gives you access to air pollution data (annual summary data) for the entire United States by allowing you to produce reports and maps of air pollution data based on criteria that you specify.

Hazardous/Toxic Air Pollutants

Toxic air pollutants are substances in the air that are toxic or hazardous to humans. These pollutants can increase your chances of experiencing health problems, and cause ecological impacts.

The Air Toxics Website is a good source for further information, including detailed information about all 189 hazardous air pollutants at the The Health Effects Notebook for Hazardous Air Pollutants site.

Monitor Types

The Clean Air Act requires every state to establish a network of air monitoring stations for criteria pollutants, using criteria set by OAQPS for their location and operation. The monitoring stations in this network are called the State and Local Air Monitoring Stations (SLAMS). The states must provide OAQPS with an annual summary of monitoring results at each SLAMS monitor, and detailed results must be available to OAQPS upon request. To obtain more timely and detailed information about air quality in strategic locations across the nation, OAQPS established an additional network of monitors: the National Air Monitoring Stations (NAMS). NAMS sites, which are part of the SLAMS network, must meet more stringent monitor siting, equipment type, and quality assurance criteria. NAMS monitors also must submit detailed quarterly and annual monitoring results to OAQPS.

A third type of monitor, the Special Purpose Monitor (SPMS), is used by State and local agencies to fulfill very specific or short-term monitoring goals.

The 1990 Amendments to the Clean Air Act also requires a fourth category of a monitoring station, the Photochemical Assessment Monitoring Station (PAMS), which measures ozone precursors (approximately 60 volatile hydrocarbons and carbonyl).

The Ambient Monitoring Technology Information Center (AMTIC) site is centeredaround the exchange of ambient monitoring related information.

CONTROL MEASURES

The atmosphere has several built-in self cleaning processes such as dispersion, gravitational settling, flocculation, absorption, rain-washout, etc to cleanse the atmosphere. However, control of contaminants at their source level is a desirable and effective method through preventive or control technologies.

Source control: Some measures that can be adopted in this direction are:

1. Using unleaded petrol
2. Using fuels with low sulphur and ash content
3. Encouraging people to use public transport, walk or use a cycle as opposed to private vehicles
4. Ensure that houses, schools, restaurants and playgrounds are not located on busy streets
5. Plant trees along busy streets as they remove particulates, carbon dioxide and absorb noise

6. Industries and waste disposal sites should be situated outsdide the city preferably on the downwind of the city.
7. Catalytic converters should be used to help control emissions of carbon monoxide and hydrocarbons

Control measures in industrial centers

1. Emission rates should be restricted to permissible levels by each and every industry
2. Incorporation of air pollution control equipment in design of plant layout must be made mandatory
3. Continuous monitoring of the atmosphere for pollutants should be carried out to know the emission levels.

Equipment Used To Control Air Pollution

The following equipment is used to control air pollution:

1. Control of SPM by gravitation

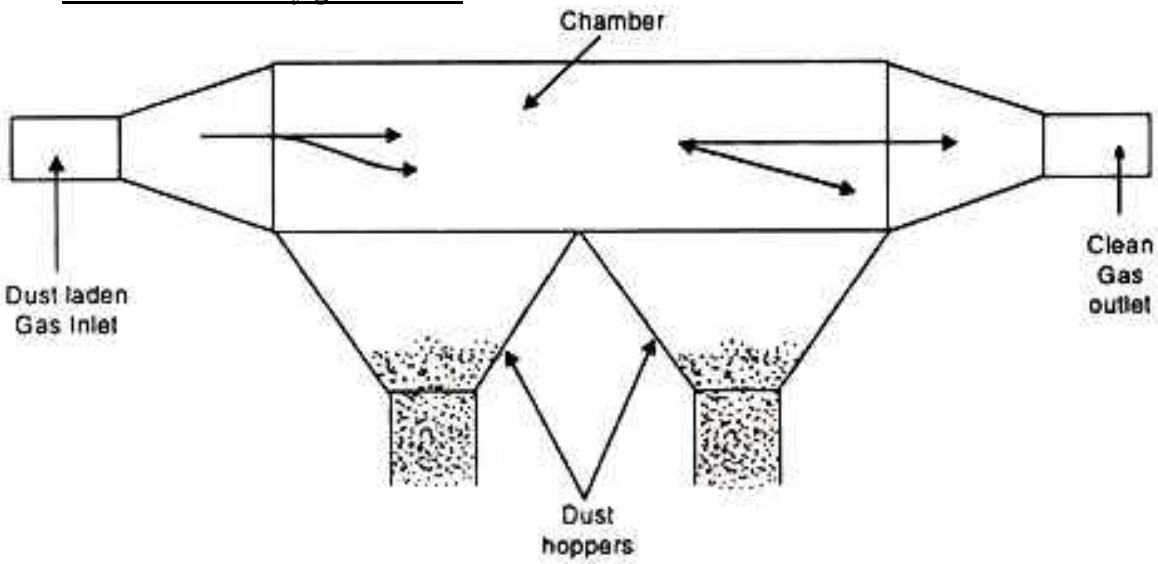
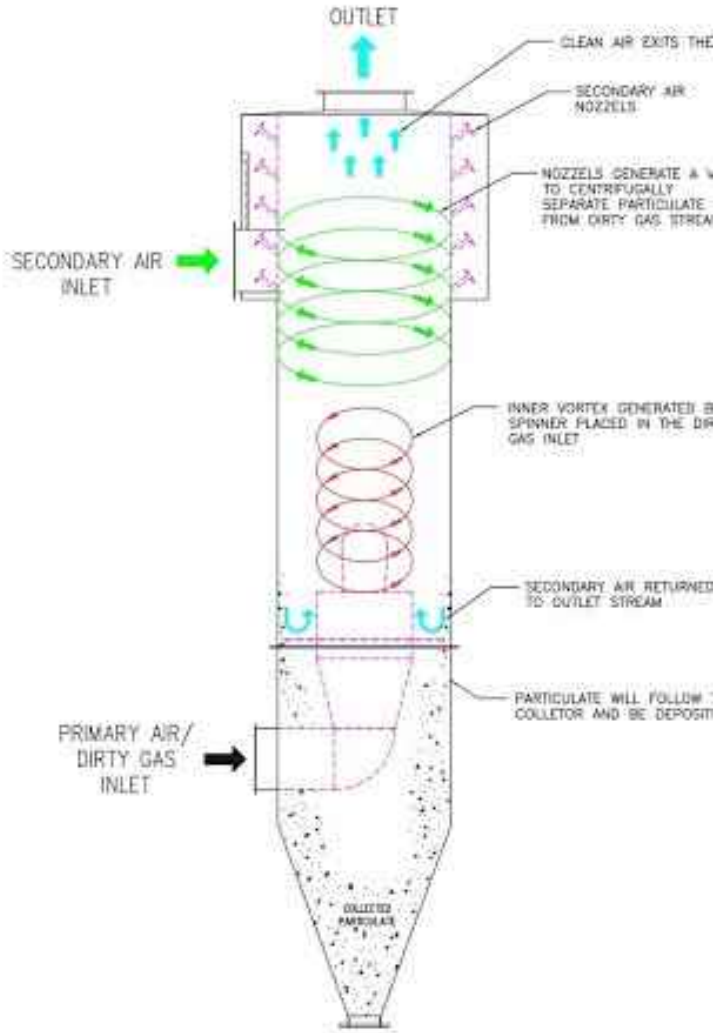


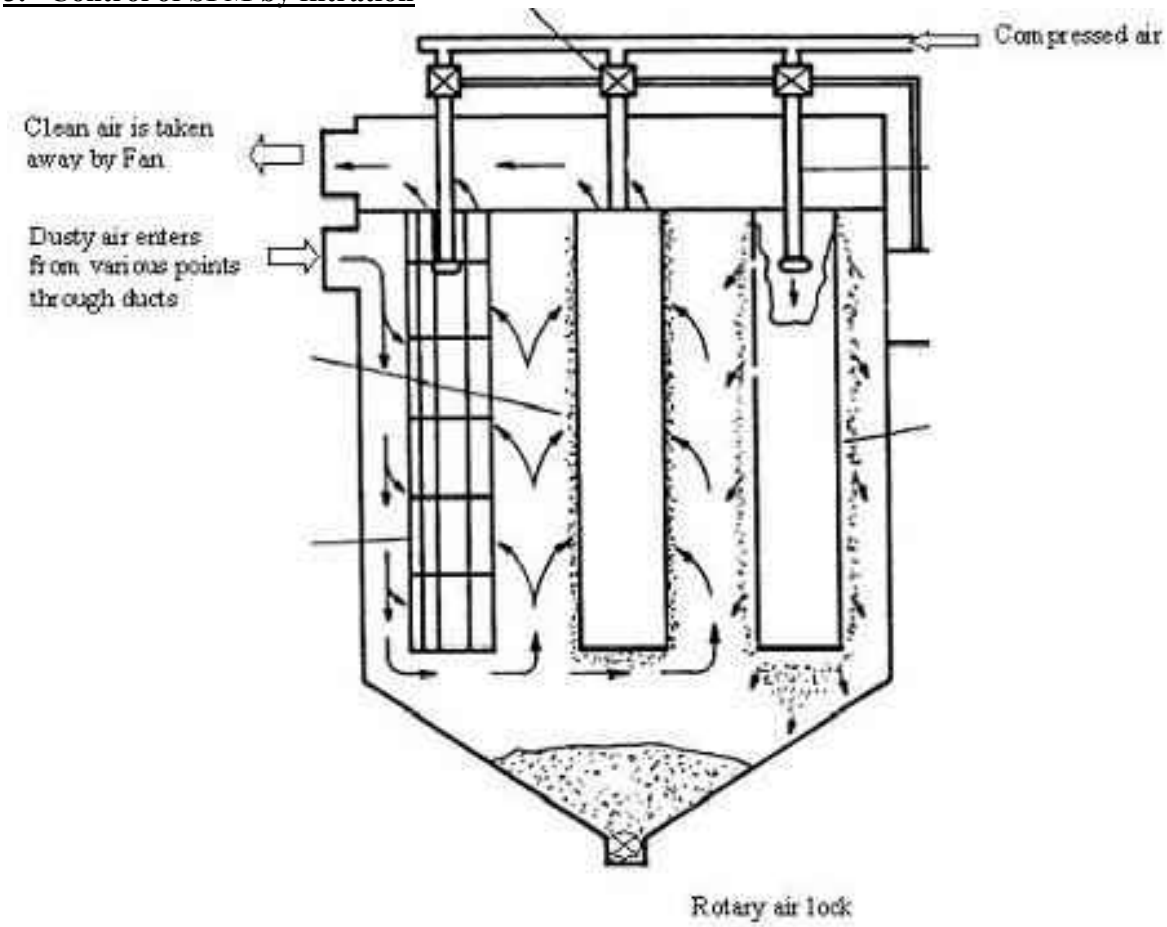
Fig. 6.4. Horizontal Flow Settling Chamber.



Control of SPM by centrifugation

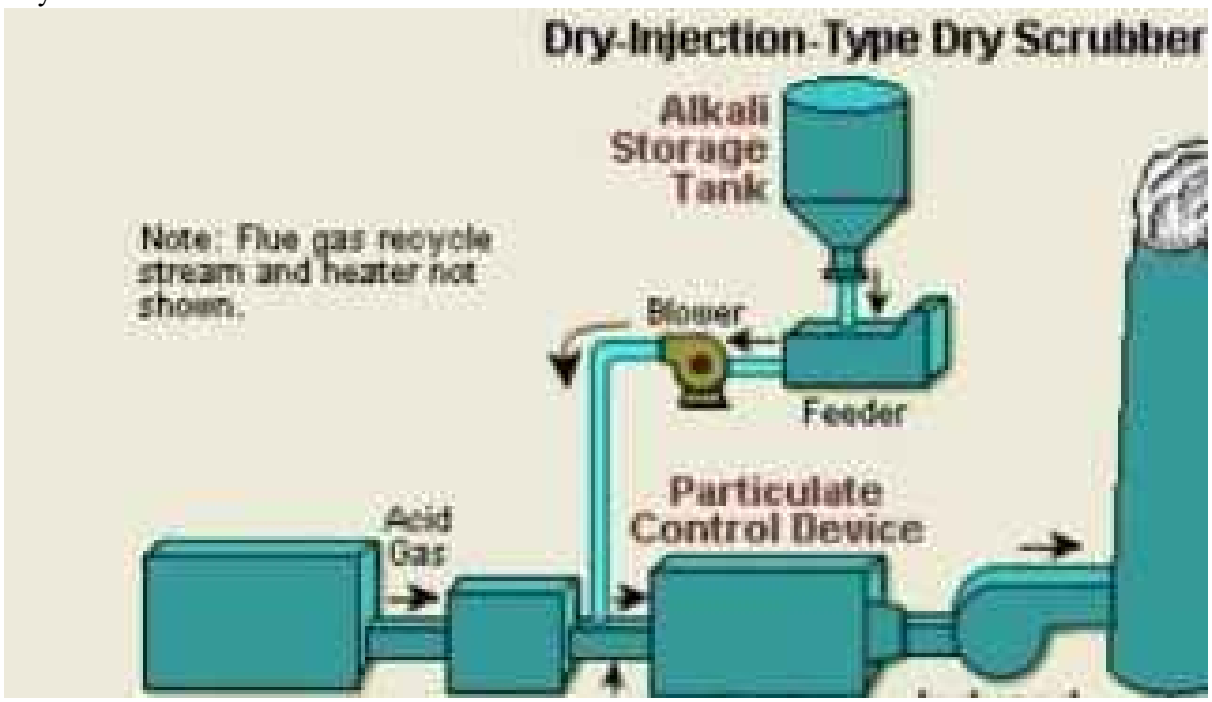
Equipment used: Cyclonic separator

3. Control of SPM by filtration

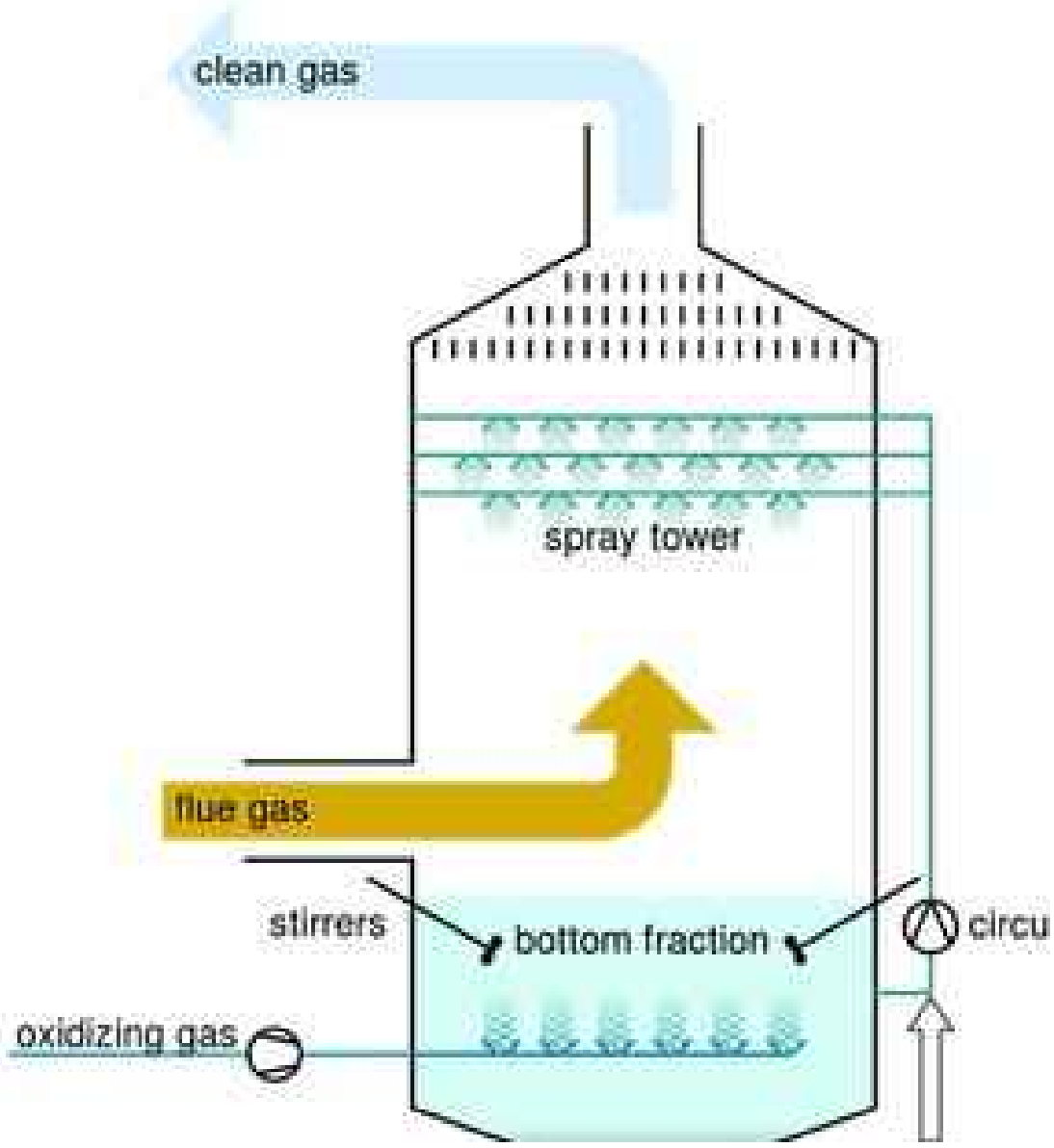


4. Control of SPM by scrubbing

A scrubber is a system used to remove harmful materials from industrial exhaust gases before they are released into the environment.



Wet scrubbing involves removal of harmful components from exhaust by spraying a liquid substance through the gas.


clean gas
spray tower
flue gas
stirrers
bottom fraction
circu
oxidizing gas

5. Control of SPM by Electrostatic precipitator

An Electrostatic precipitator is mainly used to control particulate matter.

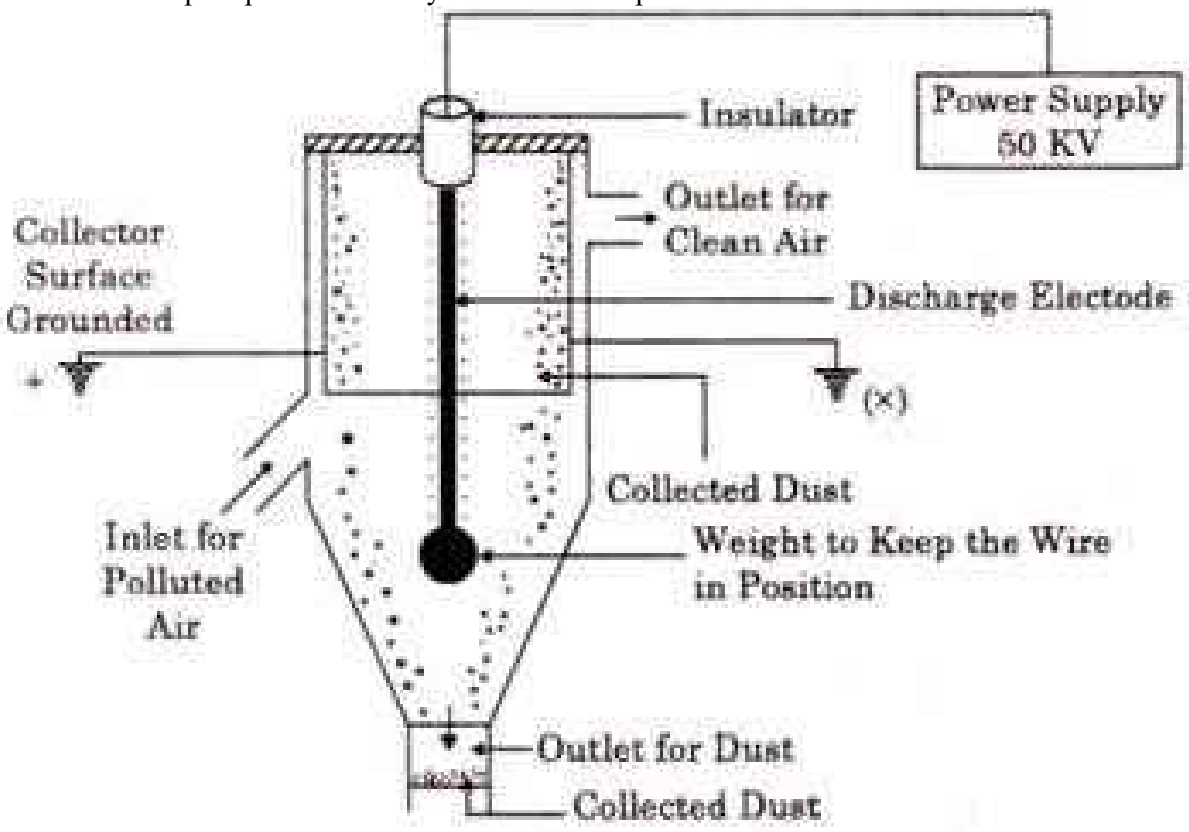


Fig. 5.4 Electrostatic Precipitator

Types of particulates

Term Meaning Examples

Aerosol General term for particles suspended in air Sprays from pressurized cans

Mist Aerosol consisting of liquid droplets Sulfuric acid mistDust Aerosol consisting of solid particles that are blown into Dust stormthe air or are produced from larger particles by grindingthem downSmoke Aerosol consisting of solid particles or a mixture of solid Cigarette smoke, smokeand liquid particles produced by chemical reaction such from burning garbageas firesFume Generally means the same as smoke but often applies Zinc/lead fumesspecifically to aerosols produced by condensation of hotvapors of metals.

Plume Geometrical shape or form of the smoke coming out ofa chimneyFog Aerosol consisting of water dropletsSmog Term used to describe a mixture of smoke

Air Pollution Episodes And Disasters

1. Air Pollution Episodes and DisastersKetanWadodkarM.Tech. 1st Year Environmental EngineeringUnder guidance of B.R. Gurjar
2. Meuse Valley, Belgium, 1930, Dec 2-5Meuse valley contains farms, villages, steel mills and chemical plant. Thermal inversion trapped fog over a stretch of 20kms of Meuse valley.
3. Effects:Nausea, short breath, stinging eyes, and burning throats.60 people died, and thousand more were ill due to a unknown disease.
4. Donora, Pennsylvania, 1948, Oct 30-31

5. Donora, Pennsylvania, 1948Location:Western, Pennsylvania on the Monongahela River50 km south of Pittsburgh. It's a valley.
6. Well known for Steel and Coal projects.Vast amount of natural resources.Major industrial location having prime manufacturing and shipping location causing industrial boom in America after world wars.About 14000 people lives on west bank.1000 people lives on Webster village on another bank.
7. On October 26, a east coast storm was replaced by a cloud anticyclone advancing from south west.This created inversion as high pressure zone was stagnated over western Pennsylvania for 5 days.The poor ventilation aggravated by local conditions of meteorology and pollutant emission.SO2 from US steel and temperature inversion was main reasons.
8. Reasons for the episodeRadiative inversion was observed.Also the temperature gradient was recorded very high.the pollutants in the air mixing with fog to form a thick, yellowish, acrid smog that hung over Donora for five days.PM = 4mg/m3, SO2= 0.5 ppm.
9. During the episode 43% of population experienced the effect from smog. 17% were moderately affected and 10% were severely affected.20 humans and nearly 800 animals killed during the incidentautopsy results showing fluorine levels in victims in the lethal range, as much as 20 times higher than normal.Fluorine gas generated in the zinc smelting process became trapped by the stagnant air and was the primary cause of the deaths.
10. The "GREAT" London smog ,December 1952, Lasted 5 days.A period of cold weather, anticyclone, also pollutant from use of coal.
11. London smog, Consequences of london smog.
12. ReasonsLow grade coal use, sulphurous variety post-war.numerous coal-fired power stations in the Greater London area,including Battersea, Bankside, and Kingston upon Thamessmoke from vehicle exhaust, especially diesel-fuelled busesPrevailing winds blown heavily polluted air across the English Channel from industrial areas of Europe .
13. Medical reports estimated that 4,000 had died prematurely100,000 more were ill because of the smog's effects on the human respiratory tract. fatalities was at about 12,000.
14. Los Angeles Smog, July 1973. A classical photochemical smog episode. A average of 63 pphm oxidants were observed. Ozone concentrations were observed to be highest in the studies of air quality. Brown haze over the town was observed for 3 days. It was PAN smog which affected the city.
15. Reason and effects:Vehicular emission is mainly responsible.Also Los Angeles is valley so it contributed inversion. NOx from combustion of fuels.The loss of visibilityAnd respiratory problems were common.
16. Bhopal gas tragedy Night of December 2–3, 1984 at the Union Carbide India Limited (UCIL) pesticide plant in Bhopal, Madhya Pradesh, India. MIC: Clear, colorless, B.P. 39deg C, odor threshold 2.1ppm.
17. ReasonA leak of methyl isocyanate gas and other chemicals from the plant. water entered a tank containing 42 tons of MIC. The resulting exothermic reaction increased the temperature inside the tank to over 200 °C (392 °F) and raised the pressure. The gases were blown by northwesterly winds over Bhopal40000kg of MIC was released in Bhopal on 3rd Dec.
18. Effects of 0.4 ppmCoughingChest painBreathing pain (dyspnea) Asthma Eye irritation Nose, throat, skin damage Effects of 21 ppm Lung oedema Emphysema (damage of lung tissue) hemorrhaginbronchial pneumoniadeath

19. 170,000 people were treated at hospitals and temporary dispensaries. 2,000 buffalo, goats, and other animals were collected and buried. leaves on trees yellowed and fell off. Fishing was prohibited 520,000 people were affected, 200,000 were below 15yrs. 3,928 deaths had been certified. But it is considered around 20000 people were died.
20. 22. Kuwait oil fires Caused by Iraqi military forces setting fire to 700 oil wells as part of a scorched earth policy during Gulf war in 1991. The fires started in January and February 1991 and the last one was extinguished by November 1991. around 6 million barrels (950,000 m3) of oil were lost each day. The byproducts of the petroleum burn caused pollution to the soil and air, known as Gulf war syndrome.
21. Gulf War Consequencesdramatic decrease in air quality, causing respiratory problems for many Kuwaitis.mixture of desert sand + unignited oil + soot formed layers of "tarcrete" which covered nearly 5% of the country.the oil has continued to sink into the sand affecting Kuwait's precious groundwater resources.
22. Yokkaichi AsthmaIn Japan in 1960 and 1972.Due to burning of petroleum and other fuels.Resulted in release of SO2.Causing chronic pulmonary diseases, chronic Bronchitis,pulmonary emphysema, bronchial asthma in people.
23. Other air pollution disasters through the world1983 Melbourne dust storm:was a meteorological phenomenon that occurred during the afternoon of 8 February 1983.Red soil, dust and sand from Central and Southeastern Australia was swept up in high winds and carried southeast through Victoria. Was seen as precursor to the Ash Wednesday bushfires.
24. 2005 Malaysian haze: a week-long choking smog-like haze over Malaysia that almost brought the central part of Peninsular Malaysia to a standstill. The 1997 Southeast Asian haze: caused mainly by slash and burn techniques adopted by farmers in Indonesia. causes respiratory symptoms such as asthma, upper respiratory infection decreased lung function as well as eye and skin irritation due to particulate matter.
25. Control Initiatives due to these episodes.World recognized that exposure to large amounts of pollution in a short period of time can result in injuries and fatalities.The event is credited for helping to trigger the clean-air movement in the United States,.Clean Air Act of 1970 was employed, which required the United States Environmental Protection Agency to develop and enforce regulations to protect the general public from exposure to hazardous airborne contaminants.Legislation such as the Clean Air Acts of 1956 and 1968, and the City of London (Various Powers) Act 1954, restricted air pollution greatly.
26. References Wikipedia. Google.US EPA website.J Black, Intussusceptions and the great smog of London, December 1952 Arch Dis Child 2003;88:1040–1042

Lecture 8

Role of plants in controlling air pollutants-legislation and air quality standards

There are various types of pollution e.g. air, water, soil, sound and mental pollution. Earth was abeautiful landscape but man has ruthlessly exploited for his greed specially, in the last century. Withrapid industrialization and random urbanization environmental pollution has become a seriousproblem. Over exploitation of open spaces, ever-increasing number of automobiles and demographicpressure has further aggravated the problem. There are various ways and means to mitigate the urbanenvironmental pollution. Plan-ting of trees and shrubs for abatement of pollution and improvement ofenvironment is an effective way and well recognized throughout the world.

Earlier, the purpose ofplanting trees in urban areas was purely aesthetic [1]. The incessant increase of urban environmentalpollution has necessitated to reconsider the whole approach of urban landscaping and its orientation inorder to achieve duel effect i.e. bio-aesthetics and mitigation of pollution. Proper planning and plantingscheme depending upon the magnitude and type of pollution, selection of pollution- tolerant and dustscavenging trees and shrubs should be done for bioremediation of urban environmental pollution.Pollution, the major problem in cities, is compounded by the fact that there is no exhaust for thepolluted air to escape. Landscape architects can solve the pollution problems related to urban landscapeby creating a micro-climate.

Pollution

Pollution is defined as ‘an undesirable change in physical, chemical and biological characteristics of air,water and land that may be harmful to living organisms, living conditions and cultural assets. Thepollution control board defined pollution as unfavourable alteration of our surrounding, largely as a byproductof human activities. The pollution may be due to human activities or natural ecosystems [3].

Natural pollution contaminates the air by storms, forest fire, volcanoes and natural processes (methanefrom marshy lands). Nature by and large treats, recycles and makes good use of the pollutants andrenders them less harmful, whereas man-made pollutants threaten the integrity of the nature.

Pollutants

The substances, which cause pollution, are called pollutants. Pollutant is defined as any substance thatis released intentionally or inadvertently by man into the environment in such a concentration that mayhave adverse effect on environmental health [4]. Environment Protection Act, 1986 EPA, 1986) definespollutant, as any solid, liquid or gaseous substance present in such a concentration as may be, or tend tobe, injurious to environment.

Air Pollution

Air is necessary for the survival of all higher forms of life on earth. On an average, a person needs atleast 30 lb of air every day to live, but only about 3 lb of water and 1.5 lb of food. A person can liveabout 5 weeks without food and about 5 days without water, but only 5 minutes without air. Naturally,every one likes to breathe fresh, clean air. But the atmosphere, that invisible yet essential ocean ofdifferent gases called air, is as susceptible to pollution from human activities as are water and landenvironments [5]. According to the WHO report, about 10 to 15 % of the total population of India is
suffering from common cold, bronchitis, asthma, hay fever etc. These diseases are no doubt airborneand spread the infection from several hundred kilometers under favourable atmospheric conditions.

Dust and soot in the air contribute to between 20 and 200 deaths each day in America’s biggest cities. Illhealth from microscopic particulates with tiny specks smaller than the width of a human hair can lodgedeep in the lungs and are associated with respiratory

diseases, heart attacks and premature deaths. Thenew research indicates elderly people suffer the most harm. In the United States the Environmental

Protection Agency (EPA) currently sets the maximum allowable concentration of microscopic particlesat 150 μm/m3 of air. The air is being continuously polluted in urban areas through heavy traffic,industry, domestic fuel combustion, stone quarries, coalmines and various agricultural activities fromthe adjoining areas. These particulates are no doubt dangerous to human health and environmentcausing various diseases to plants and animals, damage to properties including our cultural heritage,national monuments, archives etc. Dust concentration varies from place to place and hour to hour,diurnally depending upon traffic, type of industry etc. The highest dust concentration tends to be insummer, reaching maximum during mid-day and late–afternoon. In some large cities where wind andtemperature fall more steadily, the concentration of dust also reduces accordingly.

Criteria air pollutants

The five primary criteria pollutants include the gases- sulfur dioxide (SO2), nitrogen oxides (NOx) andcarbon monoxide (CO), solid or liquid particulates (smaller than 10 μm), and particulate lead.

EFFECTS OF DIFFERENT TYPES OF AIR POLLUTANTS

According to Agarwal [6], air pollution is broad term, which actually covers lots of different types ofproblems. They are, acid rain, domestic and industrial smoke, smog, greenhouse effect, particulates,radionuclides and ozone layer depletion.

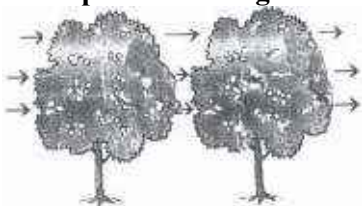
Plant species for pollution control

While selecting the species for pollution control the following are the important characteristics could beconsidered. Plants should be evergreen, large leaved, rough bark, indigenous, ecologically compatible,low water requirement, minimum care, high absorption of pollutants, resistant pollutants, agro-climaticsuitability, height and spread, Canopy architecture, Growth rate and habit (straight undivided trunk),Aesthetic effect (foliage, conspicuous and attractive flower colour), Pollution tolerance and dustscavenging capacity.

Thick plantations – small filtering effects



Loose plantations – good filtering effects



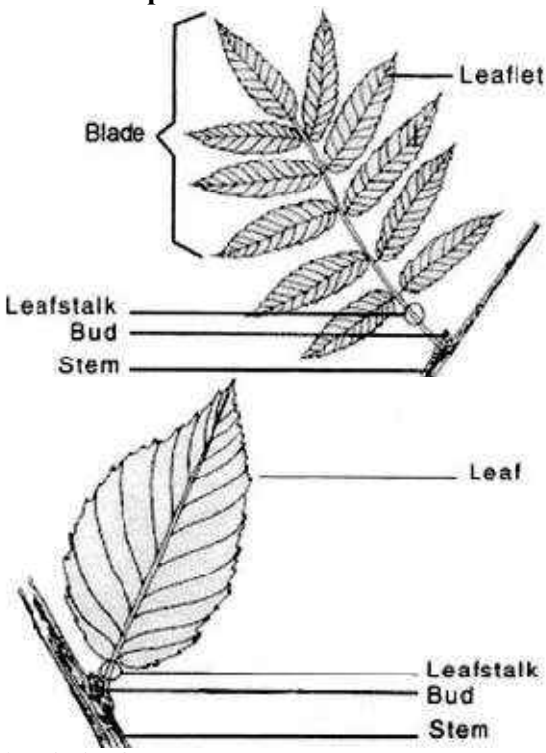
Morphological feature of plant leaves for dust capture efficiency

Different types of leaves tend to have differences in several aspects of their surfaces. Some types ofleaves have greater surface rigidity or roughness than other leaves, which may affect their stickiness orparticle solubility. Stickier leaves are better for collecting particles because more particles would stickto their surface. Therefore, certain plant leaves may be more useful for efficient dust capturing thanother plants. The various morphological features are also major factors for dust capturing by leaves.

The crown area of plants is depending upon the morphological features of the leaf [7]. The various typesof Morphological features viz. shape, size and surface texture of leaf are discussed below: Leaves can beof many different shapes. Primarily, leaves are divided into

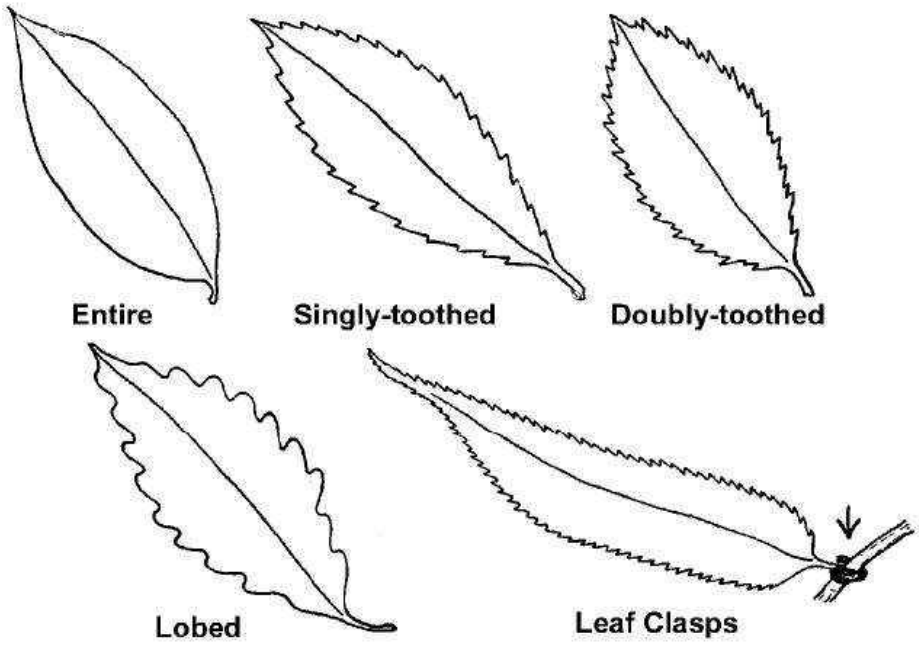
simple – a single leaf blade with a bud at thebase of the leaf stems; or compound leaf - a leaf with more than one blade. All blades are attached to asingle leaf stem. Where the leaf stems attaches to the twig with an axial bud.

Compound leaf

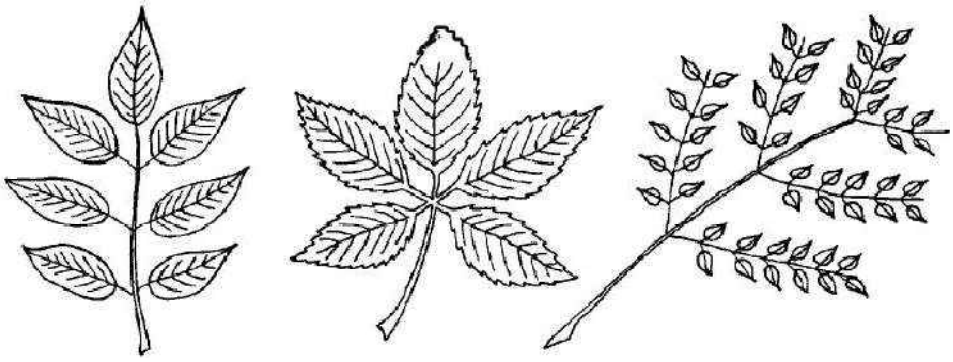


Simple leaf

The lamina or leaf blade (the edge of a leaf) may be entire, singly-toothed, do toothed, or lobed.



SIMPLE LEAVES - MARGIN STRUCTURE



COMPOUND LEAVES

Compound leaves may be **palmate** - having the leaflets arranged round a single point like fingers on thepalm of a hand; or **pinnate** - when the leaves are joined on the two sides of the

stalk, like the vanes of afeather. The form of leaves is related with all their functions and their environment. In addition tophotosynthesis, the leaf also carries out other exchanges with the atmosphere. It is through the leaf thatthe plant "breathes" (absorbs oxygen and releases carbon dioxide and generate energy) and transpires.

Epidermic tissues in the leaf contain **stomata** - microscopic openings like valves which regulate openingor closing, permitting or preventing transpiration, through which the plant loses the major part of thewater it absorbs so as to allow further absorption by the roots. In most plants the stomata are locatedon the underside of the leaves. Their function is regulated so that plants living in dry climates have asubstantially smaller number of stomata than those in humid climates, where stomata are numerousand prominent. Where humidity is low the stomata may actually be recessed or partly protected by softhairs which can prevent excessive transpiration.

Choice of eco-friendly plant species in urban environment to mitigate airborne particulate pollution

During tree plantation in an urban environment little or no attention has been paid to evaluate theeffect of trees on filtering the particulate matter. New housing developments offer an opportunity tocontrol atmospheric particulate pollution through tree plantations. Trees such as Tamarind(*Tamarindusindicus*) having smaller compound leaves are generally more efficient particle collectorsthan larger leaves.

Particle deposition is heaviest at the leaf tip and along leaf margin. In thepreliminary survey of dust fall on common roadside trees in Mumbai, carried out by Shetye andChaphekar [8] reported that the shape of leaves of Mango (*Mangiferaindica*), Ashoka (*Polyalthealongifolia*), Pongamia (*Derris indica*) and Umbrella (*Thespepsiapopulnea*) trees captured higheramounts of dust as compared to other neighboring plants.

Dochinger [9], a plant pathologist of USDAForest Service, Ohio, reported that the filtering effects of evergreen trees are better than the deciduoustrees. In Singapore; it has been noted that a single row of trees planted with or without shrubs canreduce particulate matter by 25% and each hectare (2.471 acres) of plantation can produce enoughoxygen to keep about 45 persons alive [1]. The value of trees in urban environment is now generallyrecognized not only aesthetically but also functionally in helping to make cities and towns agreeableplaces to live and work in. The first choice should be, therefore, to select easily propagated and readilyavailable, medium growing, ecologically much suitable, pest and disease resistant tree species and alsorequire less maintenance should be given top priority. Columnar and medium-sized trees are preferred.

Ingold [10] reported that the leaves with complex shapes and large circumference area reported to becollected particles more efficiently.

Many trees like Neem (*Azadirchtaindica*), Silk cotton (*Bombaxceiba*), Indian laburnum (*Cassia fistula* and *C. siamea*), Gulmohar (*Delonixregia*), Pipal (*Ficusreligiosa*), Jacaranda (*Jacaranda mimosifolia*), Indian lilac (*Lagerstroemia indica*), Temple or Pagoda tree (*Plumeriarubra*and *P. alba*), Java plum (*Syzygiumcumini*) and several other roadside and street trees have foundmore suitable in urban environment [11-14]. If such trees are to be planted, their local ecologicalrelationship with human environment has to be studied properly. It should be borne in mind that thesetrees may cause allergic disorders such as hay fever; asthma and toxemia due to airborne pollen grains, which can also contribute to atmospheric pollution significantly.

Chakre [15] has suggested that theinsect-pollinated trees with short flowering periods and also with less pollen productivity should beselected. It is also recommended that wind-pollinated tree species those, flowering during rainy seasoncan also be planted, as rains will wash out extra pollens. A tree should be relatively free of insects anddiseases and there

should not be dropping of messy fruits (*Muntingiacalabura*, *Cerberaodolam*), seedpods (*Acacia auriculaeformis*), twigs and leaves (*Dyeracostulata*). Trees with a tendency to drop largeand heavy fruits (*Durio*spp.) and emit bad smell (*Sterculiafoetida*) must be considered a seriousdrawback.

Plant species

Albizzialebbeck ,Cassia fistula ,Zizyphusjujuuba,Azadirachtaindica ,Ficusreligiosa Psidiumguajava,Phyllanthusemblica,Tamaridusindica,Moringaolifera,Delaonixregia Tectonagrandis,Morus alba, Bougainvillea spectabilis,CalotropisgigantesPoinsettia sp. Ricinuscommunis,Citrus lemon ,Lantana indica,Rosa indica.

Phytoremediation

Phytoremediation is a way to mitigate environment pollutions, such as in air, water and soilpollution in virtue of plants, more often than not, combined with their associated microorganisms. This concept has been widely applied to treat pollutants in soil and water (Cunningham & Ow, 1996; Schröder et al, 2002).

Principles of phytoremediation

Briefly, for phytoremediation, roles from roots to shoots are played in phytostabilization,rhyzodegradation, phytoextraction, phytodegradation, phytovolatilization and phytofiltration(Morikawa&Erkin, 2003) (see Fig. 1.). It is the process for phytostabilization thatcontaminants were sequestrated, but not degraded, in the root zones. The narrow region of1-3 mm from root surface is rhyzosphere, where there are much more profusemicroorganisms than in bulk soil. Root secretions and rhyzospheric microorganisms arejoint-agents to degrade the pollutants in soil in the process of rhyzodegradation.

Phytoextraction is with regard to extracting pollutants from soil and translocating themfrom the roots to shoots by plants. Phytodegradation is to transform and/or degrade pollutants, which enter the plants from soil, water or air. Phytovolatilization means the pollutants are volatile out of the stoma of the plants and degraded by hydroxyl radicals, otherwise they will become air pollutants. It is phytofiltration that particulate matters in air are reduced through the surface of plants (Singh &Tripathi, 2007).

Characteristics of phytoremediation

Main advantages

Phytoremediation is proved to be helpful to reduce pollutants in soil, water and air; it isenergy-economizing and cost-effective compared with those physical or chemical methods since it is solar-energy driven cleanup technology; it is a preference for people as its aesthetic value and less environment disruption, which letting nature take its course.

Primary disadvantages

Plants are also living things, and there exists tolerance limits for the toxicity of pollutants; itis a kind of “leave it to chance” thing since it is too natural to disrupt the environment:optimum concentrations of pollutants for remediation varying with species of plants,intermediates possibly leading to second pollutants, time-consuming when consideringgrowth period of plants.

Examples of phytoremediation of air pollutants

Particulate matters

Vegetation has been used to shield dust in many countries, and the process is called phytofiltration. A 8-m wide green belt may reduce 2 to 3 times of dust fall (Novoderzhikina et al., 1966 as cited in Singh &Tripathi, 2007). For phytofiltration, morphological characteristics, such as orientation of leaf on the main axis, size and shape, surface nature, the presence or absence of trichomes and wax deposition, are factors to trap or capture dust from ambient air.

Inorganic pollutants

NOX

NOX are air pollutants, and they are one of precursors of photochemical reaction. They canreach plant system through wet/dry deposition to foliar or roots. Plant species, plant age,NOX concentration and other environmental conditions are factors which influence the leafpenetration of NOX. After NO2 enter into the plant, most of them are metabolized to organiccompounds, e.g. amino acid, through nitrate assimilation pathway. The enzymes in plants,such as nitrate reductase, nitrite reductase or glutamine synthetase may play an importantrole in this process. Genetic engineering could turn plants into magic sink for NOx bymaking the enzymes overexpressed.

SO2

Nearly 70% SO2 in the atmosphere originates from fossil fuel combustion. SO2 enters intoplants mainly through stoma, and can be utilized in a reductive sulfur cycle in plants. Theyare changed into SO42- or SO32- in cell walls. Adenyl-5-phosphosulphate, carrier protein,carrier protein with bound sulphite and carrier protein with bound sulphide are theintermediates in the reduction pathway. The final products are cysteine or other organic compounds.

Organic pollutants

Formaldehyde

Formaldehyde, a ubiquitous air pollutant, is so harmful that it is classified as a mutagen andcarcinogen. In the 1980's, an NASA's research in the USA revealed that low level offormaldehyde in air could be removed by plant leaves alone, while higher concentrations ofthe toxic chemical can be filtered by activated carbon firstly and the plant roots andassociated microorganisms degrade and assimilate remained chemicals (Wolverton,1988). Incell-culture experiments (Giese,1994), when the concentration of formaldehyde is lowenough (8.5mg· m-3), spider plants (*Chlorophytumcomosum*L.) shoot can metabolize it toorganic acids, amino acids, free sugars, lipids and cell-wall components. In soybean (*Glycinemax* L.) cell cultures, serine and phosphatidylcholine are the major metabolic products forformaldehyde. NAS+-dependent formaldehyde dehydrogenase and glutathione-dependentformaldehyde dehydrogenases have been isolated from cell-suspension cultures of someplants. It appears that formaldehyde can be oxidized and then degraded in C1 metabolism
inphytodegradation.

Benzene and toluene

As members of VOCs (Volatile Organic Compounds, VOCs), benzene is genotoxicity and carcinogenicity while toluene is a neurotoxic chemical (Pariselli et al., 2009). Benzene and toluene have been reported to be removed from air and be assimilated by plants (Porter, 1994; Ugrekhelidze, 1997). Porter (1994) suggested at the lower toluene exposure, a fairlymodest amount light change was credited with significant increase or decrease of removalrate and the plant, the only photoresponsive organisms in the system, may be responsiblefor the phenomenon. Furthurmore, hydroxylation is considered to be the first step of thearomatic ring cleavage of benzene and toluene in higher plants (Ugrekhelidze, 1997).Besides, a substantial role of microorganisms in the growth medium in removing benzeneand toluene was also reported by previous researches (Orwell et al., 2004; Orwell et al., 2006;Wood et al., 2002; Wood et al., 2006).

Indoor air pollution

Indoor air pollution in urban environments has become a significant health concern, as citydwellers often spend over 90% of their time indoors (USEPA, 1987; Abbritti&Muzi, 1995;American Lung Assoc., 2001). "Sick building syndrome" (SBS) or "building-related illness"is partly attributed to the indoor chemicals, particularly in air-conditioned buildings

(Burgeet al., 1987; Mendell and Smith, 1990; Carpenter, 1998; Brasche et al., 1999; Carrer et al., 1999;Jones, 1999; Sundell, 2004).
As indoor pollutants, volatile organic compounds (VOCs) with300 species having been detected, should not be negligible (Orwell et al., 2004), after thecocktail of VOCs might lead to additive effects on human health (Wolkoff, 1995; Weschlerand Shields, 1997; Pariselli*et al.*, 2009). Thus, removing these chemical mixtures from indoorair merits serious consideration. Many experimental studies have investigated the effects ofplants on VOCs (Cape, 2003). This part presents a study on applicability ofphytoremediation of benzene and toluene from indoor air.

LEGISLATION AND AIR QUALITY STANDARDS

How is air pollution defined under the Air Act of 1981?
Air pollution means the presence in the atmosphere of any air pollutant.
Air pollutant means any solid, liquid or gaseous substance (including noise) present in the atmosphere in such concentration as may be or tend to be injurious to human beings or other living creatures or plants or property or environment.

What are objectives of the Air (Prevention and Control of Pollution) Act 1981?
The objective of this Act is to provide for the prevention, control and abatement of air pollution, for the establishment, with a view to carrying out the aforesaid purposes, of Boards, for conferring on and assigning to such Boards powers and functions relating thereto and for matters connected therewith.
Decisions were taken at the United Nations Conference on the Human Environment held in Stockholm in June 1972, in which India participated, to take appropriate steps for the preservation of the natural resources of the earth which, among other things, includes the preservation of the quality of air and control of air pollution.Therefore it is considered necessary to implement the decisions aforesaid insofar as they relate to the preservation of the quality of air and control of
air pollution.

What are the functions of Central Board under the Air Act?
The main functions of the Central Board, as specified in Section 16 of the Act.
To improve the quality of air and to prevent, control or abate air pollution in the country; and in particular, and without prejudice to the generality of the foregoing functions, the Central Board, may

- Advise the Central Government on any matter concerning the improvement of the quality of air and the prevention, control or abatement of air pollution
- Plan and cause to be executed a nation wide programme for the prevention, control or abatement of air pollution
- Coordinate the activities of the State Boards and resolve disputes among them
- Provide technical assistance and guidance to the State Boards, carry out and sponsor investigations and research relating to problems of air pollution and prevention, control or abatement of air pollution
- Plan and organize the training of persons engaged or to be engaged in programmes for the Can prevention, control or abatement of air pollution on such terms and conditions as the Central Board may specify
- Organise through mass media a comprehensive programme regarding the prevention, control or abatement of air pollution
- Collect, compile and publish technical and statistical data relating to air pollution and the measures devised for its effective prevention, control or abatement and prepare manuals, codes, or guides relating to prevention, control or abatement of air pollution
- Lay down standards for the quality of air

- Collect and disseminate information in respect of matters relating to air pollution
- Perform such other functions as may be prescribed, under Rules or under an Order.

In addition to the above functions, the Central Board may establish or recognize a laboratory or laboratories to enable the Central Board to perform its functions under this Section efficiently, and it may

a) delegate any of its functions under the Act generally or specially to any of the Committees appointed by it; and

b) do such other things and perform such other acts as it may think necessary for the proper discharge of its functions and generally for the purpose of carrying into effect the purposes of the Act.

What are the functions of the State Boards under the Air Act 1981?

The functions of the State Board, as specified in Section 17, shall be:

- To plan a comprehensive programme for the prevention, control or abatement of air pollution and to secure the execution thereof
- To advise the State Government on any matter concerning the prevention, control or abatement of air pollution
- To collect and disseminate information relating to air pollution
- To collaborate with the Central Board in organizing the training of persons engaged or to be engaged in programmes relating to prevention, control or abatement of air pollution and to organize mass education programme relating thereto
- To inspect, at all reasonable times, any control equipment, industrial plant or manufacturing process and to give by order, such directions to such persons as it may consider necessary to take steps for the prevention, control or abatement of air pollution
- To inspect air pollution control areas at such intervals as it may think necessary, assess the quality of air therein and take steps for the prevention, control or abatement of air pollution in such areas
- To lay down, in consultation with the Central Board and having regard to the standards for the quality of air laid down by the Central Board, standards for emission of air pollutants into the atmosphere from industrial plants and automobiles or for the discharge of any air pollutant into the atmosphere from any other source whatsoever not being a ship or an aircraft
- Provided that different standards for emission may be laid down under this clause for different industrial plants having regard to the quantity and composition of emission of air pollutants into the atmosphere from such industrial plants
- To advise the State Government with respect to the suitability of any premises or location for carrying on any industry which is likely to cause air pollution
- To perform such other functions as may be prescribed or as may, from time to time, be entrusted to it by the Central Board or the State Government
- To do such other things and to perform such other acts as it may think necessary for the proper discharge of its functions and generally for the purpose of carrying into effect the purposes of the Act.
- In addition to the above functions, the State Board may establish or recognize a laboratory or laboratories to enable the State Board to perform its above functions efficiently.

What is the importance of Sections 19, 20, 21 and 22 of the Air Act?

Section 19. Declaration of air pollution control area.

The Act has provided for measures which are

- ❖ Preventive in nature, in the case of industries to be established
- ❖ In the case of industries already established they are remedial

The primary responsibility of controlling air pollution is on the Board. The very first measure to be adopted in this respect is the declaration of any area or areas within the State as air pollution control area. The sub-section thus provides that the State Government may, after consultation with the State Board, by notification in the Official Gazette, declare in such manner as may be prescribed, any area or areas within the State as air pollution control area or areas for the purposes of the Act.

As regards power to give instructions for ensuring standards for emission from automobiles, Section 20 of the Act lays down that with a view to ensuring that the standards for emission of air pollutants from automobiles laid down by the State Board under clause (g) of sub-section(1) of Section 17 are complied with the State Government shall, in consultation with the State Board, give such instructions as may be deemed necessary to the concerned authority in charge of registration of motor vehicles under the Motor Vehicles Act, 1988, and such authority shall notwithstanding anything contained in that Act or the rules made thereunder be bound to comply with such instructions.

What are the penalties for violation of various provisions the Air Act 1981?

Section 37. Failure to comply with the provisions of section 21 or section 22 or with the directions issued under section 31-A.

a) Whoever fails to comply with the provisions of section 21 or section 22 or directions issued under section 31-A, shall, in respect of each such failure, be punishable with imprisonment for a term which shall not be less than one year and six months but which may extend to six years and with fine, and in case the failure continues, with an additional fine which may extend to five thousand rupees for every day during which such failure continues after the conviction for the first such failure.
b) If the failure referred to in sub-section(1) continues beyond a period of one year after the date of conviction, the offender shall be punishable with imprisonment with a term which shall not be less than two years but which may extend to seven years and with fine.

Section 38. Penalties for certain acts:

Whoever

- Destroys, pulls down, removes, injures or defaces any pillar, post or stake fixed in the ground or any notice or other matter put up, inscribed or placed, by or under the authority of the Board, or
- Obstructs any person acting under the orders or directions of the Board from exercising his powers and performing his functions under this Act, or
- Damages any works or property belonging to the Board, or
- Fails to furnish to the Board or any officer or other employee of the Board any information required by the Board or such officer or other employee for the purpose of this Act, or
- Fails to intimate the occurrence of the emission of air pollutants into the atmosphere in excess of the standards laid down by the State Board or the apprehension of such occurrence, to the State Board and other prescribed authorities or agencies as required under sub-section (1) of Section 23, or
- In giving any information which he is required to give under this Act, makes a statement which is false in any material particular, or
- For the purpose of obtaining any consent under Section 21, makes a statement which is false in any material particular,
- shall be punishable with imprisonment for a term which may extend to three months or with fine which may extend to [ten thousand rupees] or with both.
- **Section 39. Penalty for contravention of certain provisions of the Act.** Whoever contravenes any of the provisions of this Act or any order or direction issued

thereunder, for which no penalty has been elsewhere provided in this Act, shall be punishable with imprisonment for a term which may extend to three months or with fine which may extend to ten thousand rupees or with both, and in the case of continuing contravention, with an additional fine which may extend to five thousand rupees for every day during which such contravention continues after conviction for the first such contravention.

Can companies and government departments be prosecuted under the Air Act? Yes. This is provided under Section 40 and 41.

Section 40.

Offences by companies

1. Where an offence under this Act has been committed by a company, every person who, at the time the offence was committed, was directly in charge of, and was responsible to, the company for the conduct of the business of the company, as well as the company, shall be deemed to be guilty of the offence and shall be liable to be

Proceeded against and punished accordingly:

Provided that nothing contained in this sub-section shall render any such person liable to any punishment provided in this Act, if he proves that the offence was committed without his knowledge or that he exercised all due diligence to prevent the commission of such offence.

2. Notwithstanding anything contained in sub section (1), where an offence under this Act has been committed by a company and it is proved that the offence has been committed with the consent or connivance of, or is attributable to any neglect on the part of any director, manager, secretary or other officer of the company, such director, manager, secretary or other officer shall also be deemed to be guilty of that offence and shall be liable to be proceeded against and punished accordingly.

Explanation:

a) "Company" means any body corporate, and includes a firm or other association of individuals; and
b) "Director", in relation to a firm, means a partner in the firm.

Section 41. Offences by Government departments:

1 Where an offence under this act has been committed by any Department of Government, the Head of the Department shall be deemed to be guilty of the offence and shall be liable to be proceeded against and punished accordingly: Provided that nothing contained in this section shall render such Head of the Department liable to any punishment if he proves that the offence was committed without his knowledge or that he exercised all due diligence to prevent the commission of such offence.

2 Notwithstanding anything contained in sub section (1), where an offence under this Act has been committed by a Department of Government and it is proved that the offence has been committed with the consent or connivance of, or is attributable to any neglect on the part of, any officer, other than the Head of the Department, such officer shall also be deemed to be guilty of that offence and shall be liable to be proceeded against and punished accordingly.

What restriction does the Air Act impose on private citizens with respect to courts taking offences under the Air Act?

No court shall take cognizance of any offence except on a complaint made by any person who has given notice of not less than sixty days, in the manner prescribed, of the alleged offence and of his intention to make a complaint to the Board or officer authorized by the Board.

1. No Court inferior to that of a Metropolitan Magistrate or a Judicial Magistrate of the first class shall try any offence punishable under this Act.

2. Where a complaint has been made by any private citizen the Board shall, on demand by such person, make available the relevant reports in its possession to that person. The Board may refuse to make any such report available to such person if the same is, in its opinion, against the public interest.

What are the legal provisions under Indian Law to control automobile pollution? "Automobile" means any vehicle powered either by internal combustion engine or by any method of generating power to drive such vehicle by burning fuel.

Power to give instructions for ensuring standards for emission from automobiles. As regards power to give instructions for ensuring standards for emission from automobiles, Section 20 of the Act lays down that with a view to ensuring that the standards for emission of air pollutants from automobiles laid down by the State Board under clause (g) of sub-section(1) of Section 17 are complied with, the State Government shall, in consultation with the State Board, give such instructions as may be deemed necessary to the concerned authority in charge of registration of motor vehicles under the Motor Vehicles Act, 1988, and such authority shall notwithstanding anything contained in that Act or the rules made there under be bound to comply with such instructions.

Lecture 10

Noise Pollution, Sources, Effect and Control Measures, Indoor air pollutants and control measures

The word noise is derived from the Latin word nausea. Unwanted sound that causes discomfort to the listener is called noise. Wrong sound in wrong place at the wrong time is also called as noise.

The root cause:

- Modern mechanization and Technological innovation that have greatly increased the source of noise forced to consider it a major component of environmental pollution.
- Formerly it was limited with the industrial environment only where heavy machines used to work round the clock. But with growing population heavy traffic and urban crowd and electrical entertainment it has become new irritant and some of environmental assistance.

Source:

Natural - Thunder
Man made - Industrial and Non-industrial

Industrial:

Industries of various kinds such as textile, iron, steel, utensils, automobiles, fertilizers, paper, ceramics and thermal power stations are the major contribution of noise.

- Machines in connection with threshing, grinding, drilling, bunching, weaving, boiler making, forging, pressing and blasting operations.

Non-industrial:

(a) Domestic noise (b) Loud speaker (c) Construction work
(d) Traffic (e) Crowded markets (f) Theatres
(g) Religious function (h) Cultural festivals (i) Trains
(j) Air crafts (k) Projection of satellites (l) Atomic explosions.

Measurement of Noise:

Sound consists of repeated alternate compressions and expansions of air. The pitch of sound is determined by frequency and intensity of sound. The intensity of noise is measured by 'decibels' dB. (deci = 10 and the name of scientist Alfred Graham Bell).

$$dB = 10 \times \log_{10} \frac{I}{I_0}$$

I = Sound Intensity
I_0 = Softest audible sound intensity

Human ear is known to be sensitive to an extremely wide intensity from 0 to 190 dB. Here, '0' dB is threshold of hearing while 180 dB is threshold of pain. Some people will feel discomfort with 85 db and most people may not feel discomfort even with sound of 115 fB. Pain is usually felt at 140 dB. The sound intensity is measured by instrument Larnharometer.

$$\text{Frequency} = \frac{\text{The effect of sound as man depends upon its frequency}}{\text{Number of vibrations per second}}$$

It is denoted by Hertz. One Hz = 1 vibration / second.

People can hear sound from 16 to 20000 hz, but this range is reduced with age. Why the sound which at one time gives soothing touch to the ear becomes noise at another occasion. To understand this:

Ambient and quality standards for noise

	Limits dB	
	Day time	Night time
Industrial area	75	70
Commercial area	65	55
Residential area	55	45
Silent zone	50	40

Effect on human beings

- Affects physiological and mental health
- Reduces job efficiency
- Reduces enjoyment of life at home.
- Prolonged exposure to noise beyond 90 dB may cause hearing loss.
- In some cases, it causes auditory fatigue, nervousness, irritability, increased blood pressure, cardio-vascular, respiratory, glandular, deafness and neurological disorders. Physiological disorder : Neurosis, anxiety, insannia, hypertension

India is a noise living country; on every occasion from birth to death loud noise and dance is the part of our culture.

Effects on wildlife:

Decline is migratory birds to a habitat if it becomes noisy animals become dull and inactive.

Effects on non living things:

High intensity of noise, such as vibrations emanating from heavy machineries cause shattering of window glasses, loosen the plaster of house walls, cracks in walls.

- Some time even cause the shattering of the foundation of the building.
- Noise may also cause depreciation of the residential property located near by air path, highways, industrial areas and other noise prone areas.

Control of noise pollution:

(1) Control of noise at source:

(i) Designing and fabricating silencing devices and then use in air craft, motor cycles, industrial machines and home appliances.

1. Segregating the noisy equipments.
2. Introducing less noisy machines in plan of noisy areas.
3. Conducting noisy operations in open spaces far away from residential zones.
4. Setting up industries 5 km away from residential zones.

(2) Control of transmission:

This is achieved by covering the room walls with sound absorbing materials constrained layer damping technique.

(3) Protection of exposed person:

Using weaning devices such as ear plugs and ear muffs.

(4) Creation of vegetation and buffer zones:

Growing of noise absorbing plants like Ashoka, Banyan, Neem, Legumine, Tamarind, Coconut etc. in the road side and in front of buildings.

(5) Environmental education and awareness:

People may be made aware of health hazards of noise pollution through news papers, Television, Articles, Radio, News reels in cinema halls etc.

(6) Legislation:

Strict laws and enforcement; Appointing Noise Inspectors.

Noise Source	Intensity
Breathing	10 dB
Soft whisper	20 dB
Library	30 dB
Low volume radio (Walkman)	35 dB
Normal conversation	35 to 60 dB
Telephone	60 dB
Alarm clock	70 dB
Traffic	70 to 90 dB
Lions roar (at 12 feet)	105 dB
Thunder, Jet, Train whistle (at 50 feet)	110 dB
Air craft Jetlakoft (at 100 feet)	120 dB
Siren	150 dB
Space rocket	170 dB

Lecture 11

Solid waste –Sources – Sludge from Industry and farm waste-Characteristics – Environmental problems

Types & Source of Solid Wastes:

Basically solid waste can be classified into different types depending on their source:

Source	*Typical waste generators*	*Types of solid wastes*
Residential	Single and multifamily dwellings.	Food wastes, paper, cardboard, plastics, textiles, leather, yard wastes, wood, glass, metals, ashes, special wastes (*e.g.*, bulky items, consumer electronics, white goods, batteries, oil, tires), and household hazardous wastes.).
Industrial	Light and heavy manufacturing, fabrication, construction sites, power and chemical plants.	Housekeeping wastes, packaging, food wastes, construction and demolition materials, hazardous wastes, ashes, special wastes.
Commercial	Stores, hotels, restaurants, markets, office buildings, etc.	Paper, cardboard, plastics, wood, food wastes, glass, metals, special wastes, hazardous wastes.
Institutional	Schools, hospitals, prisons, government centers.	Same as commercial.
Construction and demolition	New construction sites, road repair, renovation sites, demolition of buildings.	Wood, steel, concrete, dirt, etc.
Municipal services	Street cleaning, landscaping, parks, beaches, other recreational areas, water and wastewater treatment plants.	Street sweepings; landscape and tree trimmings; general wastes from parks, beaches, and other recreational areas; sludge.
Process (manufacturing, etc.)	Heavy and light manufacturing, refineries, chemical plants, power plants, mineral extraction and processing.	Industrial process wastes, scrap materials, off-specification products, slay, tailings.
Agriculture	Crops, orchards, vineyards, dairies, feedlots, farms.	Spoiled food wastes, agricultural wastes, hazardous wastes (*e.g.*, pesticides).

Effects of Solid Waste Pollution:

Municipal solid wastes heap up on the roads due to improper disposal system. People clean their own houses and litter their immediate surroundings which affects the community including themselves.

This type of dumping allows biodegradable materials to decompose under uncontrolled and unhygienic conditions. This produces foul smell and breeds various types of insects and infectious organisms besides spoiling the aesthetics of the site. Industrial solid wastes are sources of toxic metals and hazardous wastes, which may spread on land and can cause changes in physicochemical and biological characteristics thereby affecting productivity of soils.

Toxic substances may leach or percolate to contaminate the ground water. In refuse mixing, the hazardous wastes are mixed with garbage and other combustible wastes. This makes segregation and disposal all the more difficult and risky.

Various types of wastes like cans, pesticides, cleaning solvents, batteries (zinc, lead or mercury), radioactive materials, plastics and e-waste are mixed up with paper, scraps and other non-toxic materials which could be recycled. Burning of some of these materials produces dioxins, furans and polychlorinated biphenyls, which have the potential to cause various types of ailments including cancer.

Lecture 12

Management of solid waste, Disposal methods, Sanitary land fills, Incineration, Pyrolysis

Methods of Solid Wastes Disposal:

i. Sanitary Landfill
ii Incineration
iii. Composting
iv. Pyrolysis

i. Sanitary Land Filling:

In a sanitary landfill, garbage is spread out in thin layers, compacted and covered with clay or plastic foam. In the modern landfills the bottom is covered with an impermeable liner, usually several layers of clay, thick plastic and sand. The liner protects the ground water from being contaminated due to percolation of leachate.

Leachate from bottom is pumped and sent for treatment. When landfill is full it is covered with clay, sand, gravel and top soil to prevent seepage of water. Several wells are drilled near the landfill site to monitor if any leakage is contaminating ground water. Methane produced by anaerobic decomposition is collected and burnt to produce electricity or heat. Sanitary Landfills Site Selection:

i. Should be above the water table, to minimize interaction with groundwater.
ii. Preferably located in clay or silt.
iii. Do not want to place in a rock quarry, as water can leech through the cracks inherent in rocks into a water fracture system.
iv. Do not want to locate in sand or gravel pits, as these have high leeching. Unfortunately, most of Long Island is sand or gravel, and many landfills are located in gravel pits, after they were no longer being used.
v. Do not want to locate in a flood plain. Most garbage tends to be less dense than water, so if the area of the landfill floods, the garbage will float to the top and wash away downstream.

A large number of adverse impacts may occur from landfill operations. These impacts can vary:

i. Fatal accidents (e.g., scavengers buried under waste piles).
ii. Infrastructure damage (e.g., damage to access roads by heavy vehicles).
iii. Pollution of the local environment (such as contamination of groundwater and/or aquifers by leakage and residual soil contamination during landfill usage, as well as after landfill closure).
iv. Off gassing of methane generated by decaying organic wastes (methane is a greenhouse gas many times more potent than carbon dioxide, and can itself be a danger to inhabitants of an area).
v. Harbouring of disease vectors such as rats and flies, particularly from improperly operated landfills.

ii. Incineration:

The term incinerates means to burn something until nothing is left but ashes. An incinerator is a unit or facility used to burn trash and other types of waste until it is reduced to ash. An incinerator is constructed of heavy, well-insulated materials, so that it does not give off extreme amounts of external heat.

The high levels of heat are kept inside the furnace or unit so that the waste is burned quickly and efficiently. If the heat were allowed to escape, the waste would not burn as completely or as rapidly. Incineration is a disposal method in which solid organic wastes are subjected to combustion so as to convert them into residue and gaseous products. This method is useful for disposal of residue of both solid waste management and solid residue from waste water

management. This process reduces the volumes of solid waste to 20 to 30 per cent of the original volume.

Incineration and other high temperature waste treatment systems are sometimes described as "thermal treatment". Incinerators convert waste materials into heat, gas, steam and ash. Incineration is carried out both on a small scale by individuals and on a large scale by industry. It is used to dispose of solid, liquid and gaseous waste. It is recognized as a practical method of disposing of certain hazardous waste materials. Incineration is a controversial method of waste disposal, due to issues such as emission of gaseous pollutants.

iii. Composting:

Due to shortage of space for landfill in bigger cities, the biodegradable yard waste (kept separate from the municipal waste) is allowed to degrade or decompose in a medium. A good quality nutrient rich and environmental friendly manure is formed which improves the soil conditions and fertility.

Organic matter constitutes 35%-40% of the municipal solid waste generated in India. This waste can be recycled by the method of composting, one of the oldest forms of disposal. It is the natural process of decomposition of organic waste that yields manure or compost, which is very rich in nutrients.

Composting is a biological process in which micro-organisms, mainly fungi and bacteria, convert degradable organic waste into humus like substance. This finished product, which looks like soil, is high in carbon and nitrogen and is an excellent medium for growing plants.

The process of composting ensures the waste that is produced in the kitchens is not carelessly thrown and left to rot. It recycles the nutrients and returns them to the soil as nutrients. Apart from being clean, cheap, and safe, composting can significantly reduce the amount of disposable garbage.

The organic fertilizer can be used instead of chemical fertilizers and is better specially when used for vegetables. It increases the soil's ability to hold water and makes the soil easier to cultivate. It helped the soil retain more of the plant nutrients.

Vermi-composting has become very popular in the last few years. In this method, worms are added to the compost. These help to break the waste and the added excreta of the worms makes the compost very rich in nutrients. In the activity section of this web site you can learn how to make a compost pit or a vermi-compost pit in your school or in the garden at home.

To make a compost pit, you have to select a cool, shaded corner of the garden or the school compound and dig a pit, which ideally should be 3 feet deep. This depth is convenient for aerobic composting as the compost has to be turned at regular intervals in this process.

Preferably the pit should be lined with granite or brick to prevent nitrite pollution of the subsoil water, which is known to be highly toxic. Each time organic matter is added to the pit it should be covered with a layer of dried leaves or a thin layer of soil which allows air to enter the pit thereby preventing bad odour. At the end of 45 days, the rich pure organic matter is ready to be used. Composting: some benefits

i. Compost allows the soil to retain more plant nutrients over a longer period.

ii. It supplies part of the 16 essential elements needed by the plants.

iii. It helps reduce the adverse effects of excessive alkalinity, acidity, or the excessive use of chemical fertilizer.

iv. It makes soil easier to cultivate.

v. It helps keep the soil cool in summer and warm in winter.

vi. It aids in preventing soil erosion by keeping the soil covered.

vii. It helps in controlling the growth of weeds in the garden.

iv. Pyrolysis:

Pyrolysis is a form of incineration that chemically decomposes organic materials by heat in the absence of oxygen. Pyrolysis typically occurs under pressure and at operating temperatures above 430 °C (800 °F).

In practice, it is not possible to achieve a completely oxygen-free atmosphere. Because some oxygen is present in any pyrolysis system, a small amount of oxidation occurs. If volatile or semi-volatile materials are present in the waste, thermal desorption will also occur.

Organic materials are transformed into gases, small quantities of liquid, and a solid residue containing carbon and ash. The off-gases may also be treated in a secondary thermal oxidation unit. Particulate removal equipment is also required. Several types of pyrolysis units are available, including the rotary kiln, rotary hearth furnace, and fluidized bed furnace. These units are similar to incinerators except that they operate at lower temperatures and with less air supply.

Limitations and Concerns:

i. The technology requires drying of soil prior to treatment.

ii. Limited performance data are available for systems treating hazardous wastes containing polychlorinated biphenyls (PCBs), dioxins, and other organics. There is concern that systems that destroy chlorinated organic molecules by heat have the potential to create products of incomplete combustion, including dioxins and furans. These compounds are extremely toxic in the parts per trillion ranges. The MSO process reportedly does not produce dioxins and furans.

iii. The molten salt is usually recycled in the reactor chamber. However, depending on the waste treated (especially inorganics) and the amount of ash, spent molten salt may be hazardous and require special care in disposal.

iv. pyrolysis is not effective in either destroying or physically separating in organics from the contaminated medium. Volatile metals may be removed as a result of the higher temperatures associated with the process, but they are not destroyed. By-products containing heavy metals may require stabilization before final disposal.

v. When the off-gases are cooled, liquids condense, producing an oil/tar residue and contaminated water. These oils and tars may be hazardous wastes, requiring proper treatment, storage, and disposal.

Lecture 13

Recycling –Energy recovery –Composting – Vermicomposting – Maturity indices assessment-Standards and Legislation

Compost :

Compost is a key ingredient in organic farming. Compost is organic matter that has been decomposed and recycled as a fertilizer and soil amendment

Benefits of Compost

Compost improves the quality of soil, and for this reason it is considered as a soil conditioner. It contains a variety of the basic nutrients required for healthy growth of plants.

In addition to, nitrogen, phosphorous, and potassium, certain micronutrients *viz.* manganese, copper, iron, and zinc also found in compost which helps them to control diseases and insects.

Compost improves the structure and texture of the soil enable them to retain nutrients, moisture, and air for the betterment of growth of plants.

Mechanism of Composting

Composting is a biochemical process in which aerobic and anaerobic microorganism decomposes organic matter into valuable manure called as compost.

Microorganism Involved in Composting

Two type of microbes which help in composting process are:

Aerobes

Which decompose organic matter in presence of oxygen

Anaerobes

Which decompose organic matter in absence of oxygen

Methods of Preparation of Compost

Indore Method

This method was developed by A. Howard and Y. D. Wad at the Institute of Plant industry, Indore, India

Bangalore Method

This method was worked out by L. N. Acharya at Indian Institute of Science, Bangalore.

NADEP Method

Demonstration of this method at large scale was initiated at J. N. Krishi Vidyalaya, Indore

Compost :

Indore method

Size of the pit

Breadth - 6-8 feet

Depth - 2-3 feet (not more than 3 feet)

Length - 10 feet or more as per requirement

Bangalore method

Spread the moist farm refuse at the bottom of the pit up to one inch.

Then, spread two inch of cattle dung and urinated mud followed by 1 or 2 inch layer of soil

This heap is made up to 1.5-2.0 feet above the ground level following above process.

Finally the heap is covered with 1 inch thick mud

After 8-9 months all material decomposes and compost becomes ready for the application

NADEP Method

This method facilitates a lot of composting through minimum use of cattle dung.

In this method, the decomposition process takes place aerobically

Selection of site

The tank should be located near cattle shed or farm site.

Size of pit

The tank should be 10' ×6' ×3'in size and are prepared with 9' inch thick wall

Proper blocks and holes of 7 inches should be left on all the four side of the tank wall for the circulation of air.

Plastering of inner wall and floor of the tank should be done by mixture of dung and mud.

Method of Filling Tank

Slurry made of cow dung and water should be sprinkled on the floor and the walls of tank. The filling of tank follows these steps:

First layer

Plant residues are spread evenly in layer up to 6 inches (10-100 Kg) in tank.

Second layer:

4-5 Kg Cattle dung of *gobber* gas-slurry in 125 to 150 litres of water should be apply on the first layer.

Third layer:

50-60 Kg sieved soil added on the second layer of tank

In this way, the tank is filled layer by layer up to 1.5 feet above the brick level of tank.

Filled tank should be covered and sealed by 3 inch layer of soil (300-400Kg)

It should also be pasted with a mixture of dung and soil.

Second Filling

At this stage, the process of the first filling is repeated and again sealed with paste of mud & dung.

After 20 days, the plant residue contracts and goes down in the tank by 20-25 inches.

Periodically the paste of cattle dung and water should be sprinkle to maintain 15-20% moisture.

windrow composting

windrow composting is the production of **compost** by piling organic matter or biodegradable waste, such as animal manure and crop residues, in long rows (**windrows**). ... **Windrow composting** is a commonly used farm scale **composting method**.

Factors Affecting Composting

- **Carbon and Nitrogen Content in Materials**
- The ideal C:N ratio for composting is between 25:1 and 30:1, with carbon being the higher number. **High-**
- **carbon materials** are almost always plant materials such as straw, cornstalks, sawdust and leaves. Nitrogenous materials more often include animal by-products
- Examples of **high-nitrogen materials** are grass clippings, alfalfa meal, blood meal and poultry manure. A few materials, such as fresh clover, most kitchen garbage and manure mixed with bedding,
- **Moisture & Aeration**
- The optimal moisture level for composting is 40 to 60 percent
- necessary to ensure that oxygen is supplied throughout the mass and aerobic activity is maintained. during the decomposition,
- **Temperature**
- Microorganisms generate heat as they decompose organic material. A compost pile with temperatures between 90= and 140=F (32=-60=C) is composting efficiently.
- Temperatures higher than 140=F (60=C) inhibit the activity of many of the most important and active organisms

Organisms:

- Aerobic composting is a dynamic system where in bacteria actinomycetes, fungi and other biological forms are actively involved.

- Facultative and obligate forms of bacteria, actinomycetes and fungi are most active in this process.
- In the initial stages mesiphilic forms predominate and thermophilic bacteria and fungi soon take over except in the final stage of composting
- Thermophilic bacteria are mainly responsible for the breakdown of proteins and other readily biodegradable organic matter.

Vermicomposting production

- **Materials Required**
- Animal excreta, kitchen waste, farm residues, forest litter, animal dung, mostly cow dung and dried chopped crop residues, leguminous and non-leguminous crop residues
- Types of earth warm
- African earthworm (*Eudrillus engenial),* Red worms (*Eisenia foetida)* and composting worm (*Peronyx excavatus)*
- African worm (*Eudrillus eugenial)* is preferred over other two types, because it produces higher production of vermicompost
- **Following steps are followed for vermicomposting preparation**
- **Pit method**: Composting is done in the cemented pits of size 5x5x3 feetor silpaulin sheet .The unit is covered with thatch grass or any other locally available materials.
- Vermicomposting unit should be in a cool, moist and shady site
- Cow dung and chopped dried leafy materials are mixed in the proportion of 3: 1 and are kept for partial decomposition for 15 – 20 days.
- A layer of 15-20cm of chopped dried leaves/grasses/ saw dust / coir pith waste should be kept as bedding material at the bottom of the bed. To maintain moisture level
- Beds of partially decomposed material of size 6x2x2 feet should be made
- Earthworm (1500-2000 nos) should be released on the upper layer of bed
- Water should be sprinkled with can immediately after the release of worms
- Beds should be kept moist by sprinkling of water (daily) and by covering with
- gunny bags/polythene – Moisture maintain 60% level
- Bed should be turned once after 30 days for maintaining aeration and for proper
- decomposition.
- Compost gets ready in 45-50 days 3 days before the harvest watering of the tank are discontinued to allow migration of worms towards the bottom of the bed,

Nutrient	Conten
Organic carbon	20-25%
Nitrogen	1.5-2.0
Phosphorus	0.5-1.5
Potassium	0.5-1.0
Calcium	0.4-0.8
Magnesium	0.3-0.6
Sulphur	100-50
Iron	6.7-9.3
Copper	2.0-9.5

Doses

- Field crops- 5- 6 t / ha
- Fruit crops- 3- 5 kg/ ha
- Pot- 100-200 g / poy
- 1 kg vermicompost- Rs 10
- 1 kg of earth warm- Rs 500

Vermiwash is a liquid that is collected after the passage of water through a column of worm action and is very useful as a foliar spray.

Lecture 14

Environmental standards-CPCB Normsfor discharging industrial effluents to public sewers

Pollutant	Time weighted Average	Concentration in Ambient Air		
		New Coal Mines (commenced after 25.09.2000)	Existing Coal Mines (commenced prior to 25.09.2000)	Work Zone (Power House)
Suspended Particulates Matter (SPM)	Annual Average	360 $\mu g/m^3$	430 $\mu g/m^3$	360 $\mu g/m^3$
	24 hours	500 $\mu g/m^3$	600 $\mu g/m^3$	500 $\mu g/m^3$
Respirable Particulate Matter (size less than 10 µm) (RPM)	Annual Average	180 $\mu g/m^3$	215 $\mu g/m^3$	120 $\mu g/m^3$
	24 hours	250 $\mu g/m^3$	300 $\mu g/m^3$	150 $\mu g/m^3$
Sulphur Dioxide (SO_2)	Annual Average	80 $\mu g/m^3$	80 $\mu g/m^3$	80 $\mu g/m^3$
	24 hours	120 $\mu g/m^3$	120 $\mu g/m^3$	120 $\mu g/m^3$
Oxides of Nitrogen as NO_2	Annual Average	80 $\mu g/m^3$	80 $\mu g/m^3$	80 $\mu g/m^3$
	24 hours	120 $\mu g/m^3$	120 $\mu g/m^3$	120 $\mu g/m^3$

Micro Gram / Cubic meter ($\mu g/m^3$)

Note : -(i) Annual arithmetic mean of 24- hourly / 8- hourly values shall be met 92% of the time in a year. However, 8% of the time it may exceed but not on two consecutive days.

(ii) In case of residential or commercial or industrial place falls within 500 metres of any dust generating sources, the National Ambient Air Quality Standards shall be made applicable.

(b) Frequency

(1) Air quality monitoring at a frequency of once in a fortnight (24 hourly sampling) at the identified locations near the dust generating sources.

(2) As a result of monthly monitoring, if it is found that the concentration of the pollutants is less than the 50% of the specified standards for three consecutive months, then the sampling frequency may be shifted to two days in a quarter year.

(3) In case the value has exceeded the specified standards, the air quality sampling shall be done twice in a week. If the results of four consecutive weeks indicate that the concentration of pollutants is within the specified standards, then fortnightly monitoring may be reverted to.

II. EFFLUENT WATER QUALITY STANDARDS :

(a) Standards

pH -- 5.5 to 9.0

Chemical Oxygen Demand (COD) -- 250 mg/l

Total Suspended Solids (TSS) -- 100 mg/l

-- 200 mg/l (Land for irrigation)

Oil & Grease (O & G) -- 10 mg/l

Note: Milligrams / litre (mg/l)

(b) Frequency

Monitoring of these parameters shall be done at a frequency of once in a fortnight.

All the 33 Parameters as given in Part-A of General Standards for discharge of Environmental Pollutants, GSR 801 (E) EPA 1993, prescribed by CPCB shall be monitored once in a year.

III. NOISE LEVEL STANDARDS:

(a) Standards

Time duration:	6.00 A.M.- 10.00 P.M.	10.00 P.M. - 6.00 A.M
Noise Level:	Leq 75 dB(A)	Leq 70 dB(A)

Note : decibel (dB)

(b) Frequency

Monitoring frequency for noise levels shall be once in a fortnight.

Occupational exposure limit of noise specified by Director General of Mines Safety (DGMS) shall be complied with by the coalmines.

B. **NATIONAL AMBIENT AIR QUALITY STANDARDS**

Pollutants	Time-weighted average	Concentration in ambient air			Method of measurement
		Industrial Areas	Residential, Rural & other Areas	Sensitive Areas	
Sulphur Dioxide (SO_2)	Annual Average*	80 µg/m^3	60 µg/m^3	15 µg/m^3	-Improved West and Geake Method - Ultraviolet Fluorescence
	24 hours**	120 µg/m^3	80 µg/m^3	30 µg/m^3	
Oxides of Nitrogen as (NO_2)	Annual Average*	80 µg/m^3	60 µg/m^3	15 µg/m^3	-Jacob & Hochheiser Modified (Na-Arsenite) Method -Gas Phase Chemiluminescence
	24 hours**	120 µg/m^3	80 µg/m^3	30 µg/m^3	
Suspended Particulate Matter (SPM)	Annual Average*	360 µg/m^3	140 µg/m^3	70 µg/m^3	- High Volume Sampling, (Average flow rate not less than 1.1 m3/minute).
	24 hours**	500 µg/m^3	200 µg/m^3	100 µg/m^3	
RespirableParticulate Matter (RPM) (size less than 10 microns)	Annual Average*	120 µg/m^3	60 µg/m^3	50 µg/m^3	-Respirable particulate matter sampler
	24 hours**	150 µg/m^3	100 µg/m^3	75 µg/m^3	
Lead (Pb)	Annual Average*	1.0 µg/m^3	0.75 µg/m^3	0.50 µg/m^3	-ASS Method after Sampling using EPM 2000 or equivalent Filter paper
	24 hours**	1.5 µg/m^3	1.00 µg/m^3	0.75 µg/m^3	
Ammonia1	Annual Average*	0.1 mg/ m^3	0.1 mg/ m^3	0.1 mg/m^3	
	24 hours**	0.4 mg/ m^3	0.4 mg/m^3	0.4 mg/m^3	

CarbonMonoxide(CO)	8 hours**	5.0 mg/m^3	2.0 mg/m^3	1.0 mg/m^3	- Non Dispersive Infra Red (NDIR) Spectroscopy
	1 hour	10.0 mg/m^3	4.0 mg/m^3	2.0 mg/m^3	

Note:
* Annual Arithmetic mean of minimum 104 measurements in a year taken twice a week 24 hourly at uniform interval.
**
24 hourly/8 hourly values should be met 98% of the time in a year. However, 2% of the time, it may exceed but not on two consecutive days.

NOTE:

1. National Ambient Air Quality Standard: The levels of air quality with an adequate margin of safety, to protect the public health,, vegetation and property.
2. Whenever and wherever two consecutive values exceeds the limit specified above for the respective category, it would be considered adequate reason to institute regular/continuous monitoring and further investigations.

[EPA Notification: GSR 176 (E), April 02, 1996]

C. GENERAL STANDARDS* FOR DISCHARGE OF ENVIRONMENTAL POLLUTANTS

Part - A : Effluents

S. No.	Parameter	Inland surface water	Public sewers	Land for irrigation	Marine/coastalareas
.	2		3	.	.
.	.	(a)	(b)	(c)	(d)
1	Colour and odour	See 6 of Annexure-1I		See 6 of Annexure-1I	See 6 of Annexure-1I
2	Suspended solids mg/l, max.	100	600	200	(a) For process waste water (b) For cooling water effluent 10 per cent above total suspended matter of influent.
3	Particle size of suspended solids	shall pass 850 micron IS Sieve	-	-	(a) Floatable solids, solidsmax. 3 mm (b) Settleable solids, max 856 microns
4	pH value	5.5 to 9.0	5.5 to 9.0	5.5 to 9.0	5.5 to 9.0
5	Temperature	shall not exceed 5°C above the receiving water temperature			shall not exceed 5oCabove the receiving water temperature
6	Oil and grease, mg/l max,	10	20	10	20

7	Total residual chlorine, mg/l max	1.0	-	-	1.0
8	Ammonicalnitrogen (as N),mg/l, max.	50	50	-	50
9	Total kjeldahlnitrogen (as N);mg/l, max. mg/l, max.	100	-	-	100
10	Free ammonia (as NH3), mg/l,max.	5.0	-	-	5.0
11	Biochemical oxygen demand (3 days at 27°C), mg/l, max.	30	350	100	100
12	Chemical oxygen demand, mg/l, max.	250	-	-	250
13	Arsenic(as As).	0.2	0.2	0.2	0.2
14	Mercury (As Hg), mg/l, max.	0.01	0.01	-	0.01
15	Lead (as Pb) mg/l, max	0.1	1.0	-	2.0
16	Cadmium (as Cd) mg/l, max	2.0	1.0	-	2.0
17	Hexavalent chro-mium (as Cr + 6),mg/l, max.	0.1	2.0	-	1.0
18	Total chromium (as Cr) mg/l, max.	2.0	2.0	-	2.0
19	Copper (as Cu)mg/l, max.	3.0	3.0	-	3.0
20	Zinc (as Zn) mg/l, max.	5.0	15	-	15
21	Selenium (as Se)	0.05	0.05	-	0.05
22	Nickel (as Ni) mg/l, max.	3.0	3.0	-	5.0
23	Cyanide (as CN) mg/l, max.	0.2	2.0	0.2	0.2
24	Fluoride (as F) mg/l, max.	2.0	15	-	15
25	Dissolved phos-phates (as P),mg/l, max.	5.0	-	-	-
26	Sulphide (as S) mg/l, max.	2.0	-	-	5.0
27	Phenoliccompounds (as C6H50H)mg/l, max.	1.0	5.0	-	5.0
28	Radioactive materials: (a) Alpha emitters micro curie mg/l, max. (b)Beta emittersmicrocurie mg/l	10^{-7} 10^{-6}	10^{-7} 10^{-6}	10^{-8} 10^{-7}	10^{-7} 10^{-6}
29	Bio-assay test	90% suivivalof fish after 96 hours in 100% effluent	90% suivival of fish after 96 hours in 100% effluen	90% suivivalof fish after 96 hours in 100% effluen	90% suivival of fish after 96 hours in 100% effluen
30	Manganese	2 mg/l	2 mg/l	-	2 mg/l
31	Iron (as Fe)	3mg/l	3mg/l	-	3mg/l

32	Vanadium (as V)	0.2mg/l	0.2mg/l	-	0.2mg/l
33	Nitrate Nitrogen	10 mg/l	-	-	20 mg/l

Lecture 15

Environment Impact Assessment, Introduction, Stages of EIA, -Monitoring and Auditing

Definitions of EIA:

Environmental Impact Assessment is defined as an activity designed to identify the impact on the biogeophysical environment, on man and well-being of legislative proposals, projects, policies, operational procedures and to interpret and communicate information.

EIA is a systematic process of identifying future consequences of a current or proposed action.

Objective of EIA:

The objective of EIA is (i) to identify, predict and evaluate the economic, environmental and social impact of development activities (ii) to provide information on the environmental consequences for decision making and (iii) to promote environmentally sound and sustainable development through the identification of appropriate alternatives and mitigation measures.

First Environmental Legislation:

The first comprehensive environmental legislation (Section 102) in United States came into force on 1st January 1970 in the form of National Environmental Policy Act (NEPA). In India, the Central Ministry of Environment and Forests issued a Notification on 27th January, 1994 making EIA statutory for 29 specified activities falling under sectors such as industries, mining, irrigation, power and transport etc.

This Notification was amended on 4th May, 1994 and the amended version includes a self-explanatory note detailing the procedure for obtaining environmental clearance, technical information, documents required to be submitted for getting environmental clearance from the Ministry of Environment and Forests.

Environmental Impact Statement (EIS) should contain the following information's/data:

1. Description of proposed action (construction, operation and shut down phase) and selection of alternatives to the proposed action.
2. Nature and magnitude of the likely environmental effects.
3. Possibility of earthquakes and cyclones.
4. Possible effects on surface and ground water quality, soil and air quality.
5. Effects on vegetation, wild life and endangered species.
6. Economic and demographic factors.
7. Identification of relevant human concerns.
8. Noise pollution. Efficient use of inputs.

Recycling and reduction of waste.

10. Risk analysis and disaster management.

EIA Methodology:

Whenever a new development project is planned which is likely to affect environmental quality, it is necessary to carry out EIA.

1. The first step in EIA method is to determine whether the project under consideration follows the jurisdiction of the relevant acts and regulations and if so, whether it is likely to create a significant environmental disruption.
2. If so, an EIA is undertaken and the environmental impact statement (EIS) is prepared.
3. In many countries, EIS is open to public scrutiny and is reviewed at public hearings.
4. Finally, a political decision is taken. The development project may be (i) accepted or (ii) accepted with amendments or (iii) an alternative proposal is accepted or (iv) rejected.

Environment Impact Assessment Process:

In EIA system, there are sequence of activities implemented in a project in logical manner termed as EIA process.

Guiding Principles:

The entire process of EIA is governed by eight guiding principles.

1. Participation:

An appropriate and timely access to the process for all interested parties.

2. Transparency:

All assessment decisions and their basis should be open and accessible.

3. Certainty:

The process and timing of the assessment should be agreed by all participants in advance.

4. Accountability:

The decision makers of all parties are responsible for their action and decisions under the assessment process.

5. Credibility:

Assessment is undertaken with professionalism and objectivity.

6. Cost effectiveness:

The assessment process and its outcomes will ensure environmental protection at the least cost to the society.

7. Flexibility:

The assessment process should be able to deal efficiently with any proposal and decision making situation.

8. Practicality:

The information and outputs provided by the assessment process are readily usable in decision making and planning.

Participants in EIA Process:

1. Proponent:

Government or Private Agency which initiates the project.

2. Decision maker:

Designated individual or group.

3. Assessor:

Agency responsible for the preparation of EIS.

4. Reviewer:

Individual/Agency/Board.

5. Expert advisers, Media and Public, Environmental organisations etc.

EIA Process in Sequence of Application:

1. Stakeholder's Involvement:

Stakeholders' involvement occurs in various stages of EIA to ensure quality, efficiency and effectiveness.

2. Project Screening and Scoping:

(i) Determine necessity for EIA requirement.
(ii) Describe various screening criteria.
(iii) Scoping determines coverage or scope of EIA.

3. Project Design and Construction:

(i) Type of project under consideration.
(ii) Physical dimensions of the area being considered.
(iii) Whether the resources will be used optically?
(iv) Whether there is an irretrievable commitment of land?
(v) Whether the project is a critical phase of a larger development?
(vi) Whether there will be serious environmental disruptions during construction?

(vii) What are the long-term plans of the proponent?

4. Project Operation:

(i) What provisions have been made to check the safety equipment regularly?
(ii) How will the hazardous waste products be handled?
(iii) What are the contingency plans developed to cope up with the possible accidents?
(iv) What provisions have been made for training the employees for environmental protection?
(v) What plans have been made for environmental monitoring?

5. Site Characteristics:

(i) Whether the site is susceptible to floods, earth quakes and other natural disasters?
(ii) Whether the terrain is creating problems in predicting ground water characteristics and air pollution etc.?
(iii) Whether the local environment is conductive for the success of the project?
(iv) How many people are likely to be displaced because of the project?
(v) What are the main attributes (e.g., protein content, calorie content, weed or pest status, carnivorousness, rarity of species, etc.) of the local fauna and flora?
(vi) Whether the project will interfere with the movements of fish population and important migratory animals?
(vii) Whether historic sites are likely to be endangered because of the project?

6. Possible Environmental Impacts:

(i) What are the possible short-term and long-term environmental impacts from the projects during construction and after construction?
(ii) Who would be effected because of these impacts?

7. Mitigation Measures:

(i) Design system to avoid, reduce and minimize adverse impacts.
(ii) Enhance beneficial outcomes.

8. Monitoring and auditing measures:

(i) Identify impacts that require monitoring and auditing.

9. Socio-Economic Factors:

(i) Who are the expected gainers and losers by the projects?
(ii) Where are the expected trade-offs?
(iii) Will the project interfere (blend, increase or reduce) with the existing inequalities between occupational, ethnic and age groups?
(iv) Will it effect the patterns of local/regional/national culture?

10. Availability of Information and Resources:

(i) Whether local and outside experts are available to consult specific impacts of the project?
(ii) Whether the relevant guidelines, technical information and other publications are available to identify the possible impacts of similar projects?
(iii) Whether relevant environmental standards, by-laws etc. are considered?
(iv) Whether the sources of relevant environmental data are identified and whether they are accessible?
(v) Whether the views of the specialist groups and general public regarding the project have been considered?
(vi) Whether the competent technical manpower is available to handle the project?

11. EIA Report and Review:

Complete information in report including non-technical summary, methodologies used, results, interpretation and conclusions. Review assesses adequacy of issues and facilitate decision making process.

12. Decision Making:

The project may be accepted, accepted with alterations or rejected.

Importance of EIA:

1. EIA is potentially a useful component of good environmental management.
2. It is the Government policy that any industrial project has to obtain EIA clearance from the Ministry of Environment before approval by the planning commission.

Lecture 16

CDM and Carbon foot print

The Clean Development Mechanism (CDM),is one among the three flexible Kyoto Protocol mechanisms facilitated implementation of Green House Gas (GHG) emission reductions in developing countries. Such GHG emissions are procured by the developed countries to meet their emission reduction targets under the Kyoto Protocol.

What is a carbon footprint?

carbon footprint is the amount of greenhouse gases—primarily carbon dioxide—released into the atmosphere by a particular human activity. A carbon footprint can be a broad meaasure or be applied to the actions of an individual, a family, an event, an organization, or even an entire nation. It is usually measured as tons of CO_2 emitted per year, a number that can be supplemented by tons of CO_2-equivalent gases, including methane, nitrous oxide, and other greenhouse gases.

How is a carbon footprint calculated?

When calculating a carbon footprint, a lot of factors are taken into consideration. For example, driving to the grocery store burns a certain amount of fuel, and fossil fuels are the primary sources of greenhouses gases. But that grocery store is powered by electricity, and its employees probably drove to work, so the store has its own carbon footprint. In addition, the products that the store sells were all shipped there, so that must also be factored into the total carbon footprint. Beyond that, the fruits, vegetables, and meats that the store sells were all grown or raised on farms, a process that produces methane, which has a greenhouse effect 25 times greater than CO_2. All of those elements must be combined to understand the full carbon footprint of a given activity.

How can I calculate my carbon footprint?

Although adding up one's individual carbon footprint can be difficult, online calculators can do some of the work for you, giving a rough estimate of your carbon footprint based on the size of your household, the efficiency of your appliances, how much you drive or fly, what you eat, and how much you recycle.

Fossil fuel use and electricity consumption play a major role in your carbon footprint.
(Photo: Getty Images)

This isn't perfect, but it is a good way to measure your activities so you can understand roughly how much CO_2 they generate and take steps to reduce your carbon footprint.

How can I reduce my carbon footprint?

Methods of reducing your carbon footprint include driving more-efficient vehicles (or making sure that your current vehicles are properly maintained), taking public transportation, using energy-efficient appliances, insulating your home to reduce heating and air conditioning costs, consuming food that doesn't require as much transportation, and eating less meat, which has a higher carbon footprint than fruits and vegetables. Individuals and companies can also offset some of their CO_2 emissions by purchasing carbon credits, the money from which can go into projects such as planting trees or investing in renewable energy.

What flashcards would you like to see? Email us or let us know in the comments below.

Lecture 17

Environmental Clearance - The Process

the environmental clearance process is required for 39 types of projects and covers aspects like screening, scoping and evaluation of the upcoming project. The main purpose is to assess impact of the planned project on the environment and people and to try to abate/minimise the same.

The process consists of following steps:

• Project proponent identifies the location of proposed plant after ensuring compliance with existing siting guidelines. If project site does not agree with the siting guideline, the proponent has to identify other alternative site for the project

• The project proponent then assesses if the proposed activity/project falls under the purview of environmental clearance. If it is mentioned in schedule of the notification, the proponent conducts an EIA study either directly or through a consultant. If the project falls in B category, the project goes to state government for clearance which further categorise into B1 and B2 projects. B2 projects doe not require preparation of EIA reports.

• After the EIA report is ready, the investor approaches the concerned State Pollution Control Board (SPCB) and the State Forest Department (if the location involves use of forestland). The SPCB evaluates and assesses the quantity and quality of effluents likely to be generated by the proposed unit as well as the efficacy of the control measures proposed by the investor to meet the prescribed standards. If the SPCB is satisfied that the proposed unit will meet all the prescribed effluent and emissions standards, it issues consent to establish (popularly known as NOC), which is valid for 15 years.

• The public hearing is a mandatory step in the process of environmental clearance for certain developmental projects. This provides a legal space for people of an area to come face-to-face with the project proponent and the government and express their concerns. The process of public hearing is conducted prior to the issue of NOC from SPCB. The District Collector is the chairperson of the public hearing committee. Other members of the committee includes the official from the district development body, SPCB, Department of Environment and Forest, Taluka and Gram Panchayat representative, and senior citizen of the district, etc. The hearing committee hears the objections/suggestions from the public and after inserting certain clauses it is passed on to the next stage of approval (Ministry of Forest and Environment).

• The project proponent submits an application for environmental clearance with the MoEF if it falls under Project A category or the state government if it falls under project B category. The application form is submitted with EIA report, EMP, details of public hearing and NOC granted by the state regulators**.**

• Environmental appraisal: The documents submitted by an investor are first scrutinised by a multi-disciplinary staff functioning in the Ministry of Environment and Forests who may also undertake site-visits wherever required, interact with the investors and hold consultations with experts on specific issues as and when necessary. After this preliminary scrutiny, the proposals are placed before specially constituted committees of experts whose composition is specified in the EIA Notification. Such committees, known as Environmental Appraisal Committees have been constituted for each sector such as River Valley, Industries, Mining etc. and these committees meet regularly to appraise the proposals received in the Ministry. In case of certain very special/controversial projects, which have aroused considerable public interest, the committee may also decide to arrange for public hearings on those projects to ensure public participation in developmental decisions. Announcements for such public hearing shall be made atleast 30 days before through newspapers. On the basis of the exercise described in the foregoing paragraphs, the Appraisal Committees make their recommendations for approval or rejection of particular projects. The recommendations of

the Committees are then processed in the Ministry of Environment and Forests for approval or rejection.

• Issues of clearance or rejection letter: When a project requires both environmental clearance as well as approval under the Forest (Conservation) Act, 1980. Proposals for both are required to be given simultaneously to the concerned divisions of the ministry. The processing is done simultaneously for clearance/rejection, although separate letters may be issued. If the project does not involve diversion of forest land, the case is processed only for environmental clearance.

Once all the requisite documents and data from the project authorities are received and public hearings (where required) have been held, assessment and evaluation of the project from the environment angle is completed within 90 days and the decision of the ministry shall be conveyed within 30 days thereafter. The clearance granted shall be valid for a period of five years for commencements of the construction or operation of the project.

The process is summarised in Figure 2.1: Environmental clearance

